

In what follows we will discuss “aggregation”. What do we mean with this term? We say that an economy admits aggregation if the behavior of the aggregate equilibrium quantities (e.g., aggregate consumption, investment, wealth, etc.) and prices (e.g., wage, interest rate, etc.) does not depend on the distribution of the individual quantities across agents. In other words, we can aggregate whenever we can define a fictitious “representative agent” that behaves, in equilibrium, as the sum of all individual consumers.

1 Two elementary results on aggregation

We begin with two elementary results on aggregation of technologies and preferences.

1.1 Aggregating firms with the same technology

Consider an economy with M firms, indexed by $i = 1, 2, \dots, M$ which produce a homogeneous good with the same technology $zF(k^i, n^i)$ where z is aggregate productivity. Assume that F is strictly increasing, strictly concave, differentiable in both arguments and constant returns to scale. Can we aggregate these individual firms into a representative firm?

Suppose inputs markets are competitive, so that every firm faces the same prices. Then, each price-taking firm of type i solves

$$\max_{\{k^i, n^i\}} zF(k^i, n^i) - wn^i - (r + \delta)k^i,$$

with first-order conditions

$$\begin{aligned} zF_k(k^i, n^i) &= r + \delta, \\ zF_n(k^i, n^i) &= w, \end{aligned} \tag{1}$$

Recall that by CRS, F_k and F_n are homogenous of degree zero, hence:

$$\frac{F_k(k^i, n^i)}{F_n(k^i, n^i)} = \frac{F_k\left(\frac{k^i}{n^i}, 1\right)}{F_n\left(1, \left(\frac{k^i}{n^i}\right)^{-1}\right)} = \frac{f_k(k^i/n^i)}{f_n(k^i/n^i)}.$$

Dividing through the two first-order conditions, we obtain

$$g\left(\frac{k^i}{n^i}\right) \equiv \frac{f_k(k^i/n^i)}{f_n(k^i/n^i)} = \frac{r + \delta}{w},$$

and using the fact that the left-hand side is a strictly decreasing function of (k^i/n^i) , we obtain

$$\frac{k^i}{n^i} = g^{-1} \left(\frac{r + \delta}{w} \right) = \frac{K}{N}, \text{ for every } i = 1, 2, \dots, M$$

where $K = \sum_{i=1}^M k_i$ and $N = \sum_{i=1}^M n_i$: every firm chooses the same capital-labor ratio.

Aggregate production across all firms:

$$\begin{aligned} z \sum_{i=1}^M F(k^i, n^i) &= z \sum_{i=1}^M [F_k(k^i, n^i) k_i + F_n(k^i, n^i) n_i] \\ &= z \sum_{i=1}^M [f_k(K/N) k_i + f_n(K/N) n_i] \\ &= z f_k(K/N) K + z f_n(K/N) N \\ &= z F_k(K, N) K + z F_n(K, N) N \\ &= z F(K, N) \end{aligned}$$

where the first and last lines use the CRS property of F . This derivation proves the existence of a “representative” firm with technology $zF(K, N)$. Note that z is the same across firms. With CRS, if one firm is more productive than all the others, it gets all the inputs.

1.2 Aggregating consumers with the same preferences

Consider a version of the neoclassical growth model with mass 1 of consumers indexed by $i \in [0, 1]$ with the same endowments of capital $k_0^i = \kappa$ for all i 's and same preferences

$$U(c_0^i, c_1^i, \dots) = \sum_{t=0}^{\infty} \beta^t u(c_t^i).$$

Assume that markets are competitive, so that every consumer faces the same prices. Then, as long as the utility function u is strictly concave, all the consumers make the same decisions and we can aggregate them into a representative agent.

Result 1.0: If every firm has the same CRS production function, consumers have the same initial endowments and same preferences and their utility function is strictly concave, then the neoclassical growth model admits a formulation with one representative firm and one representative household.

2 Gorman aggregation

That aggregation obtains when firms and consumers are identical on so many dimensions is good to know, but quite unexciting. We now provide a more interesting and useful theory of aggregation for the consumer side of the economy with heterogeneity in preferences and endowments.

Consider a static economy populated by N agents indexed by $i = 1, \dots, N$. Assume utility of each agent is u_i strictly increasing and concave over the M consumption goods $\{c^1, \dots, c^M\}$ whose price vector is p . Assume also that each consumer is endowed with a_i units of wealth. Consider a particular good c . Aggregate demand of good c is given by the sum of all individual demands, or

$$C(p, \{a_i\}) = \sum_{i=1}^N c_i(p, a_i),$$

which makes it clear that, in general, to determine aggregate quantities you need to know the entire distribution of assets across agents.¹

When can we, instead, write aggregate demand as $C(p, A)$ where $A = \sum_{i=1}^N a_i$? In order for this representation to be true, it must be that if we reallocate one dollar of wealth from consumer i to j aggregate demand does not change, or

$$\frac{\partial c_i(p, a_i)}{\partial a_i} = \frac{\partial c_j(p, a_j)}{\partial a_j} \Big|_{da_j = -da_i} \text{ for all } (i, j).$$

This condition is true if all agents have the same marginal propensity to consume out of wealth, or if the individual decision rule for consumption (recall, an outcome of optimization) can be written as

$$c_i(p, a_i) = \kappa_i^0(p) + \kappa(p) a_i. \quad (2)$$

Condition (2) means that Engel curves (expenditures as a function of wealth) are linear. Then aggregate consumption of good c is

$$C(p, A) = \sum_{i=1}^I \kappa_i^0(p) + \kappa(p) A.$$

¹The individual demand vector for agent i is obtained from the indirect utility function v_i according to Roy's identity:

$$c_i(p, a_i) = - \frac{\nabla_p v_i(p, a_i)}{\nabla_a v_i(p, a_i)}$$

This demand aggregation result is due to Gorman (1961), who stated it in terms of indirect utility as follows:

Result 1.0.1: If agents' indirect utility functions can be represented as $v_i(p, a_i) = \alpha_i(p) + \beta(p) a_i$, then aggregate consumption can be expressed as the choice of a representative agent with indirect utility $v(p, A) = \alpha(p) + \beta(p) A$ where $\alpha(p) = \sum_{i=1}^I \alpha_i(p)$.

That is, for Gorman aggregation what we want is an indirect utility function that can be separated into a term that depends on prices and the consumer's identity but not on her wealth, and a term that depends on a function of prices that is common to all consumers that is multiplied by that consumer's wealth. This indirect utility is said to be of the Gorman form.

Gorman aggregation (or demand aggregation) is a very powerful result because it explains how to construct the preferences of the representative agent. Moreover, it requires only (somewhat strong) assumptions on the demand side, such as restrictions on u , but no restriction on markets or technology. For example, we don't need complete markets (but later on, we'll see that constraints like borrowing constraints will make it fail).

Quasilinear utility: We now consider an example of Gorman aggregation: agent i with quasilinear utility over two goods (c_1, c_2) solves

$$\begin{aligned} \max_{c_{i1}, c_{i2}} \quad & u_i(c_{i1}) + \beta c_{i2} \\ \text{s.t.} \quad & \\ & p_1 c_{i1} + c_{i2} = a_i \end{aligned}$$

which implies FOCs

$$\begin{aligned} u_{i,c}(c_{i1}) &= \lambda_i p_1, \\ \beta &= \lambda_i, \end{aligned}$$

where λ_i is the multiplier on the budget constraint. Note that $c_{i1} = u_{i,c}^{-1}(\beta p_1)$ and from the budget constraint $c_{i2} = a_i - p_1 u_{i,c}^{-1}(\beta p_1)$, hence both individual demand functions show constant marginal propensities (respectively zero and one) out of wealth. Gorman aggregation holds. Note that the indirect utility function becomes

$$\begin{aligned} v(p_1; a_i) &= u(u_{i,c}^{-1}(\beta p_1)) + \beta [a_i - p_1 u_{i,c}^{-1}(\beta p_1)] \\ &= \alpha_i(p_1) + \beta a_i \end{aligned}$$

which is linear in a and has common coefficient β , so we show again that we satisfy Gorman aggregation. Another example of utility for which Gorman aggregation holds is when preferences are homothetic. A preference relation is said to be homothetic if the slope of the indifference curves remains constant along any ray from the origin. This implies that demand and utility scale linearly with wealth (income). Given that the slope of an indifference curve is also the marginal rate of substitution, it follows that for homothetic preferences the marginal rate of substitution is constant and depends only on the relative quantity of the goods and not on the absolute quantity (i.e. how far from the origin the indifference curve is).

2.1 Neoclassical growth model with heterogeneity in endowments and complete markets

We study a version of the neoclassical growth model with heterogeneity where consumers are different only in terms of their initial endowments of wealth. There is no individual or aggregate uncertainty.

2.1.1 The economy

Demographics— The economy is inhabited by N types of infinitely lived agents, indexed by $i = 1, 2, \dots, N$. Denote by μ^i the number of agents i and normalize the total number of agents to one, so that averages and aggregates are the same.

Preferences— Preferences are time separable, defined over streams of consumption, given by

$$U = \sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where the period utility function u belongs to one of the following three classes: log, power, exponential, i.e.

$$u(c) = \begin{cases} \ln(\bar{c} + c) & \text{with } \bar{c} + c > 0, \quad \bar{c} \leq 0 \\ \frac{(\bar{c} + c)^{1-\sigma}}{1-\sigma} & \text{with } \bar{c} + c > 0, \quad \bar{c} \leq 0 \\ -\bar{c} \exp(-\sigma c) & \text{with } \bar{c} > 0 \end{cases} \quad (3)$$

We impose $\bar{c} \leq 0$ for log and power utility to allow for a subsistence level for consumption, and we impose $\bar{c} > 0$ for the exponential case. Note that utility is the same for each type (what happens is $\bar{c}$ is indexed by i ?).

Markets and property rights— There are spot markets for the final good (which can be used for both consumption and investment) and complete financial markets, i.e. there are no constraints on transfers of income across periods. We follow Chatterjee (1994) in assigning the property rights on capital to the firm and the ownership of the firm to the household.² This is a different arrangement of property rights from the one you are used to seeing. Here households own shares of the firm.

We will describe a time 0 trading arrangement in which all claims are traded at time 0. See L.S Chapter 8 for equivalence of allocations under this arrangement with that under which agents trade one-period securities each period. We will let the initial level of wealth, at date 0, differ across agents. Let a_0^i be the individual wealth of type i at date 0. Then, $a_0^i = s_0^i A_0$, where s_0^i is the share of the firm-value owned by consumer i at date 0. By summing both sides of this equation over i and exploiting the fact that $\sum_{i=1}^N \mu^i s_t^i = 1$ for every t , we obtain that aggregate household wealth equals the value of the firm. Besides intertemporal trading, there will be no other securities traded among agents in equilibrium, since there is no risk.

Technology and firm's problem— The aggregate production technology is $Y_t = f(K_t)$ with f strictly increasing, strictly concave and differentiable. The representative firm owns physical capital and makes the investment decision by solving the problem

$$A_0 = \max_{\{I_t\}} \sum_{t=0}^{\infty} \left(\frac{p_t}{p_0} \right) [f(K_t) - I_t] \quad (4)$$

s.t.

$$K_{t+1} = (1 - \delta) K_t + I_t,$$

where p_t is the time t price of the final good. Let's define real profits $\pi_t \equiv f(K_t) - I_t$. Then, it is easy to see that A_0 is the value of the firm, i.e. the present value of future profits discounted at rate (p_t/p_0) , the relative price of consumption between time t and date 0. Recall that $p_{t+1}/p_t = (1 + r_{t+1})^{-1}$ where r is the interest rate.

It is useful to compute the first-order condition (FOC) of the firm problem with respect

²If we had chosen to model the firm's problem as static (i.e. the firm rents capital services from households), every argument in this lecture would still hold. You should check this, as well as every other claim I make without proving it!

to K_{t+1} by substituting the law of motion for capital into (4). The problem becomes:

$$\max_{\{K_{t+1}\}} \sum_{t=0}^{\infty} \left(\frac{p_t}{p_0} \right) [f(K_t) - K_{t+1} + (1 - \delta) K_t]$$

with FOC:

$$1 = \frac{p_{t+1}}{p_t} [f'(K_{t+1}) + (1 - \delta)] \quad (5)$$

which, incidentally, is exactly the same FOC of a “static” firm who rents capital from households at every date.

Household’s problem— Given complete markets, the maximization problem of household i at date 0 can therefore be stated as:

$$\begin{aligned} \max_{\{c_t^i\}} \sum_{t=0}^{\infty} \beta^t u(c_t^i) \\ \text{s.t.} \\ \sum_{t=0}^{\infty} p_t c_t^i \leq p_0 a_0^i \end{aligned} \quad (6)$$

where a_0^i is the wealth of agent i in term of consumption units at date 0. Let λ^i be the Lagrange multiplier on the individual i ’s Arrow-Debreu budget constraint.

Solution— Consider the log-preferences case. From the FOC of the household problem with respect to consumption at time t , we have:

$$\beta^t u'(c_t^i) = \lambda^i p_t \quad \Rightarrow \quad \beta^t \left(\frac{1}{\bar{c} + c_t^i} \right) = \lambda^i p_t \quad \Rightarrow \quad c_t^i = \frac{\beta^t}{\lambda^i p_t} - \bar{c}. \quad (7)$$

Substituting this FOC into the budget constraint of (6), we can derive an expression for the multiplier λ^i :

$$\begin{aligned} \sum_{t=0}^{\infty} p_t \left(\frac{\beta^t}{\lambda^i p_t} - \bar{c} \right) &= p_0 a_0^i \\ \frac{1}{\lambda^i (1 - \beta)} - \bar{c} \sum_{t=0}^{\infty} p_t &= p_0 a_0^i \\ \frac{1}{\lambda^i} &= (1 - \beta) \left[p_0 a_0^i + \bar{c} \sum_{t=0}^{\infty} p_t \right] \end{aligned} \quad (8)$$

Let’s now substitute the expression on the last line into equation (7), i.e.

$$c_t^i = \frac{\beta^t}{\lambda_t^i p_t} - \bar{c},$$

in order to solve explicitly for c_t^i :

$$\begin{aligned} c_t^i &= \frac{\beta^t}{p_t} \left[(1 - \beta) p_0 a_0^i + (1 - \beta) \bar{c} \sum_{\tau=0}^{\infty} p_{\tau} \right] - \bar{c} \\ &= \bar{c} \left[(1 - \beta) \beta^t \sum_{\tau=0}^{\infty} \left(\frac{p_{\tau}}{p_t} \right) - 1 \right] + (1 - \beta) \beta^t \frac{p_0}{p_t} a_0^i \end{aligned} \quad (9)$$

Finally, let a_t^i denote the period t wealth of consumer i expressed in units of date t consumption. Then the household's budget constraint at date t is

$$p_t c_t^i + \sum_{\tau=t+1}^{\infty} p_{\tau} c_{\tau}^i = p_t a_t^i.$$

Plugging in the budget constraint at date $t + 1$, which is $\sum_{\tau=t+1}^{\infty} p_{\tau} c_{\tau}^i = p_{t+1} a_{t+1}^i$, we can rewrite the date t budget constraint as $c_t^i + \frac{p_{t+1}}{p_t} a_{t+1}^i = a_t^i$. Using this and that the FOC imply $\frac{u'(c_t^i)}{\beta u'(c_{t+1}^i)} = \frac{p_t}{p_{t+1}}$, one can show that³

$$\begin{aligned} c_t^i &= \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right] + (1 - \beta) a_t^i \\ &= \Theta(p^t, \bar{c}) + (1 - \beta) a_t^i, \end{aligned} \quad (10)$$

where

$$\Theta(p^t, \bar{c}) \equiv \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \left(\frac{p_{\tau}}{p_t} \right) - 1 \right] \quad (11)$$

is a function of the subsistence level and of the whole price sequence $p^t = \{p_t, p_{t+1}, \dots\}$.

Thus, we have the optimal individual consumption rule

$$c_t^i = \Theta(p^t, \bar{c}) + (1 - \beta) a_t^i, \quad (12)$$

which is an *affine function* of asset holdings at time t for each type i . More in general, when period utility belongs to the families in (3), then preferences share a common property. They are *quasi-homothetic*, i.e. they have affine Engel curves in wealth: the wealth-expansion path is linear.⁴ We know that (12) implies that we can Gorman-aggregate demand functions. Even though we have only derived it for the log-case, it is easy to check that this representation of the consumption function holds also for the other two classes of preferences (power and exponential utility).

³The proof is by induction. See Appendix.

⁴When $\bar{c} = 0$, preferences are homothetic because the constant in the consumption function becomes zero and Engel curves start at the origin, i.e. they are linear. However, linearity of the wealth-expansion path is not affected by the constant $\bar{c}$.

2.1.2 Equilibrium aggregate dynamics

Denote aggregate variables with capital letters. From (12), we derive easily that aggregate consumption only depends on aggregate variables (prices and aggregate wealth), but it is independent of the distribution of wealth. By summing over i on the LHS and RHS of (12) with weights μ^i we arrive at:

$$C_t = \Theta(p^t, \bar{c}) + (1 - \beta) A_t. \quad (13)$$

Since equilibrium aggregate consumption C_t has the same form as individual optimal consumption choice c_t^i , it is clear that (13) can be obtained as the solution to the following representative agent problem:

$$\begin{aligned} \max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ s.t. \end{aligned} \quad (PP)$$

$$\sum_{t=0}^{\infty} p_t C_t \leq p_0 A_0$$

which is exactly as in (6) but we have replaced small letters with capital letters. Let's make some further progress on the solution. From the FOCs

$$\frac{u'(C_t)}{\beta u'(C_{t+1})} = \frac{p_t}{p_{t+1}}. \quad (14)$$

From (14) and the FOC for the firm's problem (5), we obtain the familiar Euler equation of the neoclassical growth model

$$u'(C_t) = \beta u'(C_{t+1}) [f'(K_{t+1}) + (1 - \delta)]. \quad (15)$$

We can state our first important result:

Result 1.1: If preferences are quasi-homothetic and agents are heterogeneous in initial endowments, the neoclassical growth model with complete markets admits a single-agent representation. The dynamics of aggregate quantities and prices are exactly the same as in the standard neoclassical growth model with representative agent.

Two remarks are in order. First, equation (15) governs the dynamics of capital in the representative agent growth model where firms rent capital from households, instead of owning it. Therefore, we have discovered that in complete markets it is irrelevant

whether we attribute property rights on capital to firms (and let households own shares of the firms) or to workers (and let firms rent capital from households). Second, equation (15) also governs the dynamics of capital in the social-planner problem. We are still in complete markets, and the Welfare Theorems hold.

Steady-state— The dynamics of the economy will converge to the steady-state values of capital stock satisfying the modified golden rule $f'(K^*) = 1/\beta - (1 - \delta)$. Note now that in steady-state $p_{t+1}/p_t = \beta$ for all t , hence from the definition of $\Theta(p^t, \bar{c})$ in (11) we conclude that $c^i = (1 - \beta) a^i$. In other words, in steady-state, the average propensity to save is β , independently of wealth, for every type of household.

To conclude, in the neoclassical growth model with complete markets and where agents have heterogeneous wealth endowments, the dynamics of the aggregate variables do not depend on the evolution of the wealth distribution. But is the inverse statement true? Does the evolution of the wealth distribution across households (i.e., wealth inequality) depend on the dynamics of aggregate variables (prices and quantities)? We show below that the answer is: yes, it does.

2.1.3 Equilibrium dynamics of the wealth distribution

To characterize the dynamics of the wealth distribution, we must characterize the evolution of wealth of each agent i . In particular, we want to establish conditions under which an individual's share of total wealth will grow over time, i.e. $s_{t+1}^i > s_t^i$. First of all, note that:

$$\frac{s_{t+1}^i}{s_t^i} > 1 \quad \Leftrightarrow \quad \frac{a_{t+1}^i}{a_t^i} > \frac{A_{t+1}}{A_t} \quad (16)$$

We can characterize the evolution of wealth for an agent i from the lifetime budget constraint the agent at time t

$$p_t c_t^i + \sum_{\tau=t+1}^{\infty} p_{\tau} c_{\tau}^i = p_t a_t^i \quad \Rightarrow \quad c_t^i + \left(\frac{p_{t+1}}{p_t} \right) a_{t+1}^i = a_t^i \quad (17)$$

$$\frac{c_t^i}{a_t^i} + \frac{p_{t+1} a_{t+1}^i}{p_t a_t^i} = 1 \quad \Rightarrow \quad \frac{a_{t+1}^i}{a_t^i} = \left(\frac{p_t}{p_{t+1}} \right) \left(1 - \frac{c_t^i}{a_t^i} \right), \quad (18)$$

which expresses the growth rate of wealth for type i as a function of her consumption-wealth ratio.

By aggregating over types in equation (17), we can obtain an equivalent equation at the aggregate level:

$$\begin{aligned}\sum_i \mu^i c_t^i + \left(\frac{p_{t+1}}{p_t}\right) \sum_i \mu^i a_{t+1}^i &= \sum_i \mu^i a_t^i \\ C_t + \left(\frac{p_{t+1}}{p_t}\right) A_{t+1} &= A_t \\ \frac{A_{t+1}}{A_t} &= \left(\frac{p_t}{p_{t+1}}\right) \left(1 - \frac{C_t}{A_t}\right)\end{aligned}$$

Therefore, the condition under which the agent's share of total wealth will grow over time becomes

$$\frac{a_{t+1}^i}{a_t^i} > \frac{A_{t+1}}{A_t} \Leftrightarrow \frac{c_t^i}{a_t^i} < \frac{C_t}{A_t} \quad (19)$$

Moreover, from equations (12) and (13),

$$\frac{c_t^i}{a_t^i} = \frac{\Theta(p^t, \bar{c})}{a_t^i} + (1 - \beta), \quad \text{and} \quad \frac{C_t^i}{A_t} = \frac{\Theta(p^t, \bar{c})}{A_t} + (1 - \beta)$$

and therefore

$$\frac{c_t^i}{a_t^i} < \frac{C_t}{A_t} \Leftrightarrow \frac{\Theta(p^t, \bar{c})}{a_t^i} < \frac{\Theta(p^t, \bar{c})}{A_t} \Leftrightarrow \Theta(p^t, \bar{c}) (a_t^i - A_t) > 0$$

and, thus, summarizing we have the following equivalence (i.e., “if and only if”) condition:

$$\frac{s_{t+1}^i}{s_t^i} > 1 \Leftrightarrow \Theta(p^t, \bar{c}) (a_t^i - A_t) > 0,$$

which means that whether consumer's i wealth share is increasing or decreasing over time depends on 1) the sign of the constant Θ (equal for everyone) and 2) on her relative position in the distribution. For example, if $\Theta > 0$ then for a consumer whose initial wealth is above average, her share will grow, whereas for a consumer whose initial wealth is below average, her share will fall. And hence the distribution will become more unequal over time. Note that the dynamics of the wealth distribution depend on the entire sequence of prices, hence on the dynamics of aggregate variables in equilibrium.

First of all, in absence of subsistence level, $\bar{c} = 0$ we have $\Theta = 0$, and therefore the neoclassical growth model with heterogeneous endowments has a sharp prediction for the evolution of inequality.

Result 1.2: In the neoclassical growth model with complete markets, homothetic preferences, heterogeneous endowments, but without subsistence level ($\bar{c} = 0$), the wealth distribution remains unchanged along the transition path, i.e. initial conditions in endowments (and inequality) persist forever.

The intuition is that if $\bar{c} = 0$ then $c^i = (1 - \beta) a^i$, so the average propensity to consume, and save, is the same for every agent. Every agent accumulates wealth at the same rate.

In presence of a subsistence level, the dynamics are more interesting. We now determine the sign of Θ , through:

Lemma 1.1 (Chatterjee, 1994): $\Theta(p^t, \bar{c})$ is greater (less) than zero if and only if the economy is converging from below (above) to the steady-state, i.e. if $K_\tau < (>) K^*$.

Proof: Suppose the economy grows towards the steady-state, i.e. $K_\tau < K^*$. Then the sequence $\{f'(K_\tau)\}$ is decreasing and, from equation (15), the sequence $\{p_{\tau+1}/p_\tau\}$ is increasing towards β , i.e., $p_{\tau+1}/p_\tau \leq \beta$ for all $\tau \geq t$ where the strict inequality holds at least for some τ . It follows that

$$p_\tau/p_t = (p_\tau/p_{\tau-1})(p_{\tau-1}/p_{\tau-2}) \dots (p_{t+2}/p_{t+1})(p_{t+1}/p_t) < \beta^{\tau-t}.$$

From the definition of $\Theta(p^t, \bar{c})$ in (10), use the above equation to obtain

$$\Theta(p^t, \bar{c}) = \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \left(\frac{p_\tau}{p_t} \right) - 1 \right] > \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \beta^{\tau-t} - 1 \right] = 0$$

where the inequality follows from $\bar{c} < 0$. **QED**

The implications for the evolution of the wealth distribution in an economy growing towards the steady-state (the empirically interesting case) are easy to determine, at this point. In the presence of a subsistence level ($\bar{c} < 0$), $\Theta > 0$. From equation (19) this implies that the average propensity to consume (save) declines (increases) with wealth: poor agents must consume proportionately more out of their wealth to satisfy the subsistence level, so wealth inequality increases along the transition. In other words:

Result 1.3: In the neoclassical growth model with complete markets, homothetic preferences, heterogeneous endowments and subsistence level $\bar{c} < 0$, as the economy grows towards the steady-state: (i) the wealth distribution becomes more unequal, as rich agents accumulate more than poor agents along the transition path, and (ii) there is no change in the ranking of households in the wealth distribution, i.e., initial conditions in wealth (and consumption) ranking persist forever.

The main conclusion of this lecture is that in this model there is no economic or social mobility. This is not a good model to understand why some individuals are born poor

and make it in life, while other are born rich and end up poor as rats. This is just a model of castes.

Robustness— We now discuss how robust this result is to three of the key assumptions made so far in the analysis: 1) all agents have same discount factor β , 2) all agents are equally productive, 3) markets are complete.

1. When agents have different discount factors, then none of the results hold any longer. Suppose that $\bar{c} = 0$ to simplify the analysis. Then, from (12)

$$c_t^i = (1 - \beta^i) a_t^i,$$

therefore the average propensity to save out of wealth is higher the more patient is the individual and from (19), wealth grows faster for the more patient individuals. In the limit, in steady-state, the most patient type holds all the wealth, and the distribution becomes degenerate.

2. Caselli and Ventura (2000) extend the Gorman aggregation result to a version of the neoclassical growth model where agents also differ in their endowments of efficiency units of labor.
3. In absence of markets and trade (autarky), every consumer has access to her own technology. Each agent i will solve his own maximization problem in isolation

$$\begin{aligned} & \max_{\{c_\tau^i\}} \sum_{\tau=t}^{\infty} \beta^{\tau-t} u(c_\tau^i) \\ & s.t. \\ & k_{\tau+1}^i = (1 - \delta) k_\tau^i + f(k_\tau^i) - c_\tau^i \\ & k_t^i \text{ given} \end{aligned}$$

with different initial conditions k_t^i . It is easy to see that, independently of initial conditions, each agent will converge to the same capital stock k^* , hence in the long-run the distribution of wealth is perfectly equal. Interestingly, we conclude that less developed financial markets induce less wealth inequality, in the long-run.

2.1.4 Indeterminacy of the wealth distribution in steady-state

One very important implication of the aggregation Result 1.1 is that in steady-state the wealth distribution is indeterminate. From (15), (4) and (12), the set of equations

characterizing the steady-state is:

$$\begin{aligned} f'(K^*) &= 1/\beta - (1 - \delta), \\ A^* &= \frac{1}{1 - \beta} [f(K^*) - \delta K^*] \\ \sum_{i=1}^N \mu^i a^i &= A^*, \\ c^i &= (1 - \beta) a^i, i = 1, 2, \dots, N \end{aligned}$$

We therefore have $(N + 3)$ equations and $(2N + 2)$ unknowns $(\{c^i, a^i\}_{i=1}^N, K^*, A^*)$. In other words, the multiplicity of the steady-state wealth distributions is of order $N - 1$.⁵

However, suppose we start from a given wealth distribution at date $t = 0$ when the economy has not yet reached its steady-state, then the dynamics of the model are uniquely determined by Results 1.2 and 1.3 and the final steady-state distribution is determined as well. So, let's restate our finding in:

Result 1.4: In the steady-state of the neoclassical growth model with N agents, heterogeneous initial endowments and homothetic preferences, there is a continuum with dimension $(N - 1)$ of steady-state wealth distributions. However, given an initial wealth distribution $\{a_0^i\}_{i=1}^N$ at $t = 0$, the equilibrium wealth distribution $\{a_t^i\}_{i=1}^N$ in every period t is uniquely determined, and so is the final steady-state distribution.

Finally, in terms of language, this whole section shows that it is important to distinguish “steady-state” from “equilibrium path”. In this economy, the equilibrium path is always unique (given initial conditions), but the steady-state is not.

3 The Negishi Approach

What can we do when no aggregation theorem applies? Solving for competitive equilibria of economies with heterogeneous agents can be a nightmare: you must compute the demand for each individual, for every good, in every period, and find the right infinite-dimensional vector of prices such that markets clear...

Negishi (1960) suggested a method to calculate the competitive equilibrium (CE) prices and allocations of complete markets economies (in particular, economies for which

⁵This means that, if $N = 1$ (representative agent), the steady-state is unique.

the first welfare theorem holds) with heterogeneous households. This method is particularly useful for those economies where Gorman aggregation does not go through and, hence we do not know how to write the preferences of the representative agent.⁶

From the first welfare theorem, we know that any CE is a Pareto optimum (PO), hence it can be found as the solution to a social planner problem with “some” Pareto weights given to each agent. Suppose we want to compute a particular CE of an economy where agents are initially endowed with heterogeneous shares $\{s_0^i\}_{i=1}^N$ of the aggregate wealth. Can we use the planner problem for this purpose? Negishi showed that the key is to search for the “right” weights given to each type of agent in the social welfare function of the planner. Each set of weights corresponds to a Pareto efficient allocation, the key is to find the set of weights which correspond to our desired CE in the original economy.⁷

3.1 An Example

Consider our neoclassical growth model of section (2.1) with two consumers ($N = 2$). The agent’s i problem in the decentralized Arrow-Debreu equilibrium can be written as

$$\begin{aligned} \max_{\{c_t^i\}} & \sum_{t=0}^{\infty} \beta^t u(c_t^i) \\ \text{s.t.} & \\ & \sum_{t=0}^{\infty} p_t c_t^i \leq p_0 a_0^i \end{aligned}$$

where $a_0^i = s_0^i A_0$ is the initial wealth endowment, given at $t = 0$. Let’s assign the property rights on capital to the firm, so the firm’s problem is exactly the one of the previous section.

From the FOC of the agent of type i , we obtain

$$FOC(c_t^i) \longrightarrow \beta^t u'(c_t^i) = \lambda^i p_t,$$

where λ^i is the multiplier on the time zero budget constraint. Thus, putting together the FOC’s for the two types:

$$\frac{u'(c_t^1)}{u'(c_t^2)} = \frac{\lambda^1}{\lambda^2}. \quad (20)$$

⁶In his original paper, Negishi (1960) used this equivalence result to propose a simple way to show existence of competitive equilibria.

⁷Incidentally, the notion of social welfare was introduced by Samuelson (1956).

Now, write down the following Negishi planner problem (NP) for our economy

$$\begin{aligned}
& \max_{\{c_t^1, c_t^2, K_{t+1}\}} \sum_{t=0}^{\infty} \beta^t [\alpha^1 u(c_t^1) + \alpha^2 u(c_t^2)] \quad (\text{NP}) \\
& s.t. \\
& c_t^1 + c_t^2 + K_{t+1} \leq f(K_t) + (1 - \delta) K_t \\
& K_0 \text{ given}
\end{aligned}$$

where (α^1, α^2) are the planner's weights for each type of household in the social welfare function.

The FOC's for this problem are

$$FOC(c_t^i) \longrightarrow \alpha^i \beta^t u'(c_t^i) = \theta_t, \quad (21)$$

$$FOC(K_{t+1}) \longrightarrow \theta_t = \theta_{t+1} [f'(K_{t+1}) + (1 - \delta)] \quad (22)$$

where θ_t is the Lagrange multiplier on the planner's resource constraint at time t . Note that putting together the first-order conditions for consumption for the two agents we arrive at

$$\frac{u'(c_t^1)}{u'(c_t^2)} = \frac{\alpha^2}{\alpha^1}, \quad (23)$$

which tells us that the planner allocates consumption proportionately to the weight it gives to each consumer (with strictly concave utility).⁸

If we want the NP to deliver the same solution as the CE, we need the PO allocations and the CE allocations to be the same. Given strict concavity of preferences, this implies that, putting together (23) and (20):

$$\frac{\alpha^2}{\alpha^1} = \frac{\lambda^1}{\lambda^2}$$

Hence, the relative weights of the planner must correspond to the inverse of the ratio of the Lagrange multipliers on the time-zero Arrow-Debreu budget constraint for the two agents in the CE.

In particular, for log preferences $u(c_t^i) = \log c_t^i$ and $\bar{c} = 0$, we derived in equation (8) that

$$\left(\frac{1}{\lambda^i} \right) = (1 - \beta) p_0 a_0^i \Rightarrow \lambda^i = \left[\frac{1}{(1 - \beta) p_0 A_0} \right] \frac{1}{s_0^i},$$

⁸Note also that the ratio of marginal utility across agents is kept constant in every period (a key features of complete markets allocations, also called *full insurance*).

therefore we obtain that

$$\frac{\alpha^2}{\alpha^1} = \frac{\lambda^1}{\lambda^2} = \frac{s_0^2}{s_0^1}.$$

Imposing the natural and innocuous normalization $\alpha^1 + \alpha^2 = 1$, we can solve explicitly for the two weights: $\alpha^1 = s_0^1$ and $\alpha^2 = s_0^2$, i.e., the weights are exactly equal to the initial wealth shares (“exactly” is because of log utility, in general weights will be proportional to the initial shares). The higher is the initial wealth share s_0^i , the lower is the multiplier λ^i and the larger is the Pareto weight α^i on the Negishi problem: the planner must deliver more to consumption to the agent who has a large initial share of wealth in the decentralized equilibrium.

Result 1.5: Consider an economy with agents heterogeneous in endowments where the First Welfare Theorem holds. Then, the competitive equilibrium allocations can be computed through an appropriate planner’s problem where the relative weights on each agent in the social welfare function are proportional to the relative individual endowments: those agents who initially have more wealth will get a higher weight in the planner’s problem.

Now, note that (22) implies a relationship between the sequence of capital stocks and the sequence of Lagrange multipliers on the planner’s resource constraint

$$\frac{\theta_t}{\theta_{t+1}} = f'(K_{t+1}) + (1 - \delta).$$

Recall that equation (15) dictating the optimal choice of capital for the firm in the CE stated that

$$\frac{p_t}{p_{t+1}} = f'(K_{t+1}) + (1 - \delta).$$

Hence, we have

$$\frac{\theta_t}{\theta_{t+1}} = \frac{p_t}{p_{t+1}}, \tag{24}$$

in other words, the Arrow-Debreu equilibrium prices can be uncovered as the sequence of Lagrange multipliers in the Pareto problem: intuitively, the multipliers give us the shadow value of an extra unit of consumption and, in the CE, prices signal exactly this type of scarcity.⁹ Note that equation (24) is true for any well behaved u .

⁹Using p_0 as the numeraire and imposing the normalizations $p_0 = \theta_0$, the above relationship implies that $p_t = \theta_t$ so equilibrium prices are exactly equal to the shadow prices of consumption in the planner’s problem.

In conclusion, we have uncovered a tight relation between weights of the NP problem and initial endowments in the CE and an equivalence between Lagrange multiplier on the resource constraint of the NP problem and prices in the CE. This strict relationship, that we have uncovered for the log utility case, is true more in general.

3.2 General application of the Negishi method

In general, without specific restrictions on preferences (e.g., CRRA utility), one may not have closed form solutions for the λ 's in the CE, so the algorithm is a little more involved. The objective is to compute the CE allocations for an economy with N types of agents and endowment distribution $\{a_0^i\}_{i=1}^N$. We can describe the algorithm in four steps:

1. In the social planner problem (NP), guess a vector of weights $\alpha = \{\alpha^1, \alpha^2, \dots, \alpha^N\}$. The normalization $\sum_{i=1}^N \alpha^i = 1$ means that α belongs to the N -dimensional simplex

$$\Delta_N = \{\alpha \in \mathbb{R}_+^N : \sum_{i=1}^N \alpha^i = 1\}$$

and the simplex traces out the entire set of Pareto-optimal allocations.

2. From the NP problem compute the sequence of allocations $\left\{\{c_t^i\}_{i=1}^N, K_t\right\}_{t=0}^\infty$ and the implied sequence of multipliers $\{\theta_t\}_{t=0}^\infty$ on the resource constraint in each period t . In practice, at every t , one needs to solve the $N + 2$ equations

$$\begin{aligned} \alpha^i \beta^t u'(c_t^i) &= \theta_t, \quad i = 1, \dots, N \\ \frac{\theta_t}{\theta_{t+1}} &= f'(K_{t+1}) + (1 - \delta) \\ \sum_{i=1}^N \mu^i c_t^i + K_{t+1} &= f(K_t) + (1 - \delta) K_t \end{aligned}$$

in $N + 2$ unknowns $(\{c_t^i\}_{i=1}^N, K_{t+1}, \theta_{t+1})$. At every t , (K_t, θ_t) are given, therefore the Negishi method simplifies enormously the computation of the equilibrium: the Negishi solution requires solving, for every time t , a small system of equations. Recall that, instead, to solve for the CE allocations, at every time t one must set the excess demand function to zero and the excess demand function depends on the entire price sequence—an infinitely dimensional object. To understand, take another look at the consumption allocation (12) where $\Theta(\cdot)$ depends on the entire

price sequence from t onward.

Instead of guessing (and iterating over) an infinite sequence of prices, one guesses and iterates over a finite set of weights. With a caveat: even though K_0 is given, θ_0 is not. So, one has also to guess a value for θ_0 . The reason is that, unless you have the right value for θ_0 , the system will not be on the saddle-path and capital will diverge. In other words, there is another condition that we need to satisfy in the growth model, the transversality condition.

3. Exploit the equivalence between prices p_t and multipliers θ_t to verify whether the time-zero Arrow-Debreu budget constraint of each agent holds exactly at the guessed vector of weight α . Specifically, for each agent, compute the implicit transfer function $\tau^i(\alpha)$ associated with the assumed vector of weights

$$\tau^i(\alpha) = \sum_{t=0}^{\infty} \theta_t c_t^i(\alpha) - \theta_0 a_0^i, \text{ for every } i = 1, 2, \dots, N \quad (25)$$

and if (25) holds for agent i with a “greater (smaller) than” sign, it means that the planner is giving too much (little) weight to agent i . So, in the next iteration reduce (increase) the weight α^i given to agent i . Note one property of the transfer functions:

$$\sum_{i=1}^N \mu^i \tau^i(\alpha) = \sum_{i=1}^N \left[\sum_{t=0}^{\infty} \theta_t \mu^i c_t^i(\alpha) - \theta_0 \mu^i a_0^i \right] = \sum_{t=0}^{\infty} p_t C_t - p_0 A_0 = 0$$

since the discounted present value of resources of the economy cannot be greater than its current wealth. Put differently, recall that from the representative firm problem

$$A_0 = \sum_{t=0}^{\infty} \left(\frac{p_t}{p_0} \right) [f(K_t) - I_t] = \sum_{t=0}^{\infty} \left(\frac{p_t}{p_0} \right) C_t$$

which establishes the same result. To conclude, the individual transfer functions $\tau^i(\alpha)$ must sum to zero.

4. Iterate over α until you find the vector of weights α^* that sets *every* individual transfer function $\tau^i(\alpha^*)$ to zero. This vector corresponds to the PO allocations that are affordable by each agent in the CE, given their initial endowment, without the need for any transfer across-agents. Thus, we are computing exactly the CE associated to initial conditions $\{a_0^i\}_{i=1}^N$.

See also Ljungqvist-Sargent, section 8.5.3, for a discussion of the Negishi algorithm.

4 Aggregation with complete markets

We just learned that the equilibrium allocations of a complete market economy with heterogeneity can be obtained as the solution of a Negishi planner problem. The Negishi approach can be used to prove a more general aggregation result for complete markets economies due to Constantinides (1982) and then refined by Ogaki (2003). Consider the economy described above where we know that the Welfare Theorems hold and therefore we can solve for the equilibrium allocations through the following Negishi planner problem

$$\begin{aligned} \max_{\{c_t^i, K_{t+1}\}} & \sum_{t=0}^{\infty} \beta^t \sum_{i=1}^N \alpha^i u(c_t^i) \mu_i \\ \text{s.t.} & \\ K_{t+1} + C_t &= f(K_t) + (1 - \delta) K_t \\ C_t &= \sum_{i=1}^N \mu_i c_t^i \end{aligned}$$

Now, split the problem in two stages. First, given a sequence of aggregate consumption $\{C_t\}$, consider the static problem of how to allocate C_t across agents at every t

$$\begin{aligned} U(C_t) &= \max_{\{c_t^i\}} \sum_{i=1}^N \alpha^i u(c_t^i) \mu_i \\ \text{s.t.} & \\ C_t &= \sum_{i=1}^N \mu_i c_t^i \text{ for all } t \end{aligned}$$

where $U(C_t)$ is the indirect utility function of the planner at date t . Second, consider the consumption/investment problem

$$\begin{aligned} \max_{\{C_t, K_{t+1}\}} & \sum_{t=0}^{\infty} \beta^t U(C_t) \\ \text{s.t.} & \\ K_{t+1} + C_t &= f(K_t) + (1 - \delta) K_t \\ K_0 &\text{ given} \end{aligned}$$

This second-stage problem shows that the aggregate dynamics of the model can be described by the solution to the problem of a representative agent (RA), but the RA's preferences are different from preferences of the individual consumer, i.e., $U \neq u$. Indeed,

one can even allow for different utility functions u^i across agents and show that this result still holds.

The Constantinides aggregation theorem is a very general existence (i.e., existence of a representative agent) result: note that the only restriction on preferences is strict concavity of u , homotheticity or quasilinearity are not required. But it requires complete markets, while Gorman aggregation does not. It also suggests an algorithm for constructing the preferences of the RA, but in most cases an analytical solution for U is not attainable, even though we know U exists. In what follows, we show an example with closed-form solution.

4.1 An Example

This example is taken from Maliar and Maliar (2001, 2003). In their model, agents have non-homothetic preferences in consumption and leisure, and are subject to idiosyncratic, but insurable, shocks to labor endowment (i.e., markets are still complete). They show how to recover preferences of a fictitious representative agent whose optimization problem yields, as a solution, the equilibrium aggregate dynamics of the original economy. This representative agent ends up having a *different* (but not so different) utility function from that of individuals populating the economy.

Demographics— The economy is inhabited by a continuum of infinitely lived agents, indexed by $i \in I \equiv [0, 1]$. Denote by μ^i the measure of agents i in the set I and normalize the total number of agents to one, $\int_I d\mu^i = 1$, so that averages and aggregates are the same. Initial heterogeneity is in the dimension of initial wealth endowments.

Uncertainty— Agents are subject to idiosyncratic productivity shocks to skills. Let ε_t^i be the shock of agent i , and suppose shocks are iid, with mean 1, and defined over the set E . This is not necessary, but it simplifies the notation.

Preferences— Preferences are time separable, defined over streams of consumption, given by

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i, 1 - h_t^i).$$

where period utility is given by

$$u(c_t, 1 - h_t) = \frac{c_t^{1-\gamma} - 1}{1 - \gamma} + \phi \frac{(1 - h_t)^{1-\sigma} - 1}{1 - \sigma} \quad (26)$$

and note that preferences are not quasi-homothetic, unless $\sigma = \gamma$.

Markets and property rights– There are spot markets for the final good (which can be used for both consumption and investment) whose price is normalized to one, and complete financial markets, i.e. agents can trade a full set of state-contingent claims. The agent's portfolio is composed by Arrow securities of the type $a_{t+1}^i(\varepsilon)$ which pay one unit of consumption at time $t + 1$ if the individual's shock is ε and zero otherwise. Let $p_t(\varepsilon)$ the price of this security and $\int_E p_t(\varepsilon) a_{t+1}^i(\varepsilon) d\varepsilon$ the value of such portfolio for agent i .

Technology and firm's problem– The aggregate production technology is $Y_t = Z_t f(K_t, N_t)$ with f strictly increasing and strictly concave in both arguments and differentiable. The representative firm rents capital from households. N_t is aggregate labor input in efficiency units, i.e. $N_t = \int_I \varepsilon_t^i h_t^i d\mu^i$.

Household problem– The sequential problem of an agent i :

$$\begin{aligned} \max_{c_t^i, h_t^i, k_{t+1}^i, a_{t+1}^i(\varepsilon)} \quad & \mathbb{E}_t \sum_{j=0}^{\infty} \beta^{t+j} u(c_{t+j}^i, 1 - h_{t+j}^i) \quad (27) \\ \text{s.t.} \quad & \\ c_t^i + k_{t+1}^i + \int_E p_t(\varepsilon) a_{t+1}^i(\varepsilon) d\varepsilon = & (1 - \delta + r_t) k_t^i + w_t \varepsilon_t^i h_t^i + a_t^i(\varepsilon_t^i) \\ & \{k_t^i, a_t^i\} \text{ given} \end{aligned}$$

Equilibrium– This is a complete markets economy. The First Welfare Theorem tells us that the equilibrium is Pareto optimal, so we can use a social planner problem to characterize the equilibrium by applying the Negishi method. The key, as usual, is to find the right weights that guarantee that allocations are affordable for each agent, given their initial endowments.

Aggregation?– Given that preferences are not homothetic, we know that Gorman's strong aggregation concept will not hold. But can we, nevertheless, obtain a RA whose choices describe the evolution of the aggregate economy? And how the preferences of the RA look like?

To characterize the preferences of the RA, we split the Negishi planner problem in two stages. Given a sequence of aggregate consumption C_t and aggregate labor input N_t

the problem of the planner is

$$\begin{aligned} & \max_{c_t^i, h_t^i} \int_i \alpha_i u(c_t^i, 1 - h_t^i) d\mu_i \\ & s.t. \\ & \int_i c_t^i d\mu_i = C_t \\ & \int_i \varepsilon_t^i h_t^i d\mu_i = N_t \end{aligned}$$

Letting θ_t^C and θ_t^N be the multipliers on the aggregate consumption and labor feasibility constraints and given the assumption on the utility function, the FOC with respect to individual i in the Negishi planner's static problem are:

$$\begin{aligned} \alpha^i (c_t^i)^{-\gamma} &= \theta_t^C \\ \alpha^i \phi (1 - h_t^i)^{-\sigma} &= \theta_t^N \varepsilon_t^i \end{aligned} \tag{28}$$

Rearranging gives

$$\begin{aligned} c_t^i &= \left(\frac{\alpha^i}{\theta_t^C} \right)^{\frac{1}{\gamma}} \\ (1 - h_t^i) &= \left(\frac{\phi \alpha^i}{\theta_t^N} \right)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma} \end{aligned}$$

and note that consumption of individual i is proportional to its weight α^i in the social welfare function. Leisure is directly proportional to its weight (a wealth effect) and inversely proportional to individual productivity: efficiency arguments induce the planner to make high-productivity individuals work harder.

Integrating the two FOCs across agents gives

$$\begin{aligned} C_t &= \int_I c_t^i d\mu^i = \int_I \left(\frac{\alpha^i}{\theta_t^C} \right)^{\frac{1}{\gamma}} d\mu^i \\ 1 - N_t &= 1 - \int_I \varepsilon_t^i h_t^i d\mu^i = \int_I \left(\frac{\phi \alpha^i}{\theta_t^N} \right)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{1-1/\sigma} d\mu^i \end{aligned} \tag{29}$$

Now, note that

$$\begin{aligned} c_t^i &= \frac{\left(\frac{\alpha^i}{\theta_t^C} \right)^{\frac{1}{\gamma}}}{\int_I \left(\frac{\alpha^i}{\theta_t^C} \right)^{\frac{1}{\gamma}} d\mu^i} C_t \Rightarrow c_t^i = \frac{(\alpha^i)^{\frac{1}{\gamma}}}{\int_I (\alpha^i)^{\frac{1}{\gamma}} d\mu^i} C_t \\ (1 - h_t^i) &= \frac{\left(\frac{\phi \alpha^i}{\theta_t^N} \right)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma}}{\int_I \left(\frac{\phi \alpha^i}{\theta_t^N} \right)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{1-1/\sigma} d\mu^i} (1 - N_t) \Rightarrow (1 - h_t^i) = \frac{(\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma}}{\int_I (\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{1-1/\sigma} d\mu^i} (1 - N_t) \end{aligned} \tag{30}$$

Now, consider the social welfare function for the Negishi planner who is using weights $\{\alpha^i\}$

$$\int_I \left[\frac{(c_t^i)^{1-\gamma} - 1}{1-\gamma} + \phi \frac{(1-h_t^i)^{1-\sigma} - 1}{1-\sigma} \right] \alpha^i d\mu^i$$

and substitute the two expressions in (30) into the social welfare function:

$$\begin{aligned} & \int_I \alpha^i \frac{\left[\frac{(\alpha^i)^{\frac{1}{\gamma}}}{\int_I (\alpha^i)^{\frac{1}{\gamma}} d\mu^i} C_t \right]^{1-\gamma} - 1}{1-\gamma} d\mu^i + \phi \int_I \alpha^i \frac{\left[\frac{(\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma}}{\int_I (\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma} d\mu^i} (1-N_t) \right]^{1-\sigma} - 1}{1-\sigma} d\mu^i \\ &= \frac{\frac{\int_I \alpha^i (\alpha^i)^{\frac{1-\gamma}{\gamma}} d\mu^i}{\left[\int_I (\alpha^i)^{\frac{1}{\gamma}} d\mu^i \right]^{1-\gamma}} C_t^{1-\gamma} - 1}{1-\gamma} + \phi \frac{\frac{\int_I \alpha^i (\alpha^i)^{\frac{1-\sigma}{\sigma}} (\varepsilon_t^i)^{1-1/\sigma} d\mu^i}{\left[\int_I (\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{-1/\sigma} d\mu^i \right]^{1-\sigma}} (1-N_t)^{1-\sigma} - 1}{1-\sigma} \end{aligned}$$

which yields the utility for the RA

$$\begin{aligned} & \frac{C_t^{1-\gamma} - 1}{1-\gamma} + \phi \Phi \frac{(1-N_t)^{1-\sigma} - 1}{1-\sigma} \\ & \text{where} \\ \Phi &= \frac{\left[\int_I (\alpha^i)^{\frac{1}{\sigma}} (\varepsilon_t^i)^{1-1/\sigma} d\mu^i \right]^\sigma}{\left[\int_I (\alpha^i)^{\frac{1}{\gamma}} d\mu^i \right]^\gamma} \end{aligned}$$

is independent of t because shocks are iid. Therefore the RA problem which describes the aggregate allocations for this economy becomes:

$$\max_{\{C_t, N_t, K_{t+1}\}} \mathbb{E}_t \sum_{j=0}^{\infty} \beta^{t+j} \frac{C_{t+j}^{1-\gamma} - 1}{1-\gamma} + \phi \Phi \frac{(1-N_{t+j})^{1-\sigma} - 1}{1-\sigma} \quad (31)$$

s.t.

$$C_t + K_{t+1} = (1-\delta) K_t + Z_t f(K_t, N_t)$$

K_t given

Some remarks are in order:

1. We have found a RA, but its preferences are not the ones of the individual agent. Note that preferences of the RA depend on N_t (aggregate efficiency-weighted hours) instead of H_t (aggregate hours), and note that they feature a new weight on leisure Φ . This is the first reason why this is not a Gorman aggregation type of result.

2. The preference shifter Φ , in general, *depends on the distribution of shocks and the initial distribution of endowments*, therefore aggregate quantities do depend on the distribution of exogenous shocks and initial wealth (but not on the time-varying distribution of wealth). This is the second reason why this is, technically, not Gorman's aggregation. Note, however, that because these distributions are exogenous, it is a simple problem. In particular, in the case of no initial wealth heterogeneity $\alpha^i = 1$ for all i , $\Phi = \left[\int_I (\varepsilon_t^i)^{1-1/\sigma} d\mu^i \right]^\sigma$. Suppose that $\log \varepsilon \sim N(-v_\varepsilon/2, v_\varepsilon)$, then

$$\Phi = \exp \left(\sigma \left(\frac{\sigma-1}{\sigma} \right) \left(\frac{\sigma-1}{\sigma} - 1 \right) \frac{v_\varepsilon}{2} \right) = \exp \left(\frac{1-\sigma}{\sigma} \frac{v_\varepsilon}{2} \right)$$

which shows how the variance of the shocks affects the taste for leisure of the fictitious representative agent.

3. Suppose $\gamma = \sigma$. Then utility is quasi-homothetic. If there are no skill shocks, but only differences in endowments, then $\Phi = 1$ and $H_t = N_t$ and the utility of the representative agent is the same as the individual agent. We are back to Gorman's aggregation. If there are idiosyncratic shocks, then $\Phi \neq 1$ and Gorman aggregation fails, which establishes that Gorman aggregation holds only if the unique source of heterogeneity across agents is in initial wealth.
4. Suppose $\gamma \neq \sigma$. Then utility is not quasi-homothetic. Even if there are no skill shocks, but only differences in endowments, then Φ depends on the distribution of endowments and Gorman's aggregation fails.

Finally, note that the assumption that agents can trade a full set of claims contingent on all possible realizations of idiosyncratic labor productivity shocks is not very realistic, as we will argue later in the course.

Appendices

Proof for:

$$\begin{aligned} c_t^i &= \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right] + (1 - \beta) a_t^i \\ &= \Theta(p^t, \bar{c}) + (1 - \beta) a_t^i, \end{aligned}$$

where

$$\Theta(p^t, \bar{c}) \equiv \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \left(\frac{p_{\tau}}{p_t} \right) - 1 \right]$$

Proof: The proof is by induction. Suppose $t = 0$. Then using (9), we have

$$c_0^i = \bar{c} \left[(1 - \beta) \sum_{\tau=0}^{\infty} \frac{p_{\tau}}{p_0} - 1 \right] + (1 - \beta) a_0^i$$

Suppose that $c_t^i = \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right] + (1 - \beta) a_t^i$. We need to show that this also holds at $t + 1$, i.e. that

$$c_{t+1}^i = \bar{c} \left[(1 - \beta) \sum_{\tau=t+1}^{\infty} \frac{p_{\tau}}{p_{t+1}} - 1 \right] + (1 - \beta) a_{t+1}^i$$

To show this, note first that using that $a_t^i = c_t^i + a_{t+1}^i \frac{p_{t+1}}{p_t}$, we can write c_{it} as

$$\beta c_t^i = \bar{c} \left[(1 - \beta) \sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right] + (1 - \beta) a_{t+1}^i \frac{p_{t+1}}{p_t}, \quad (32)$$

Next, from the FOC we have that

$$\beta c_t^i = (c_{t+1}^i + \bar{c}) \frac{p_{t+1}}{p_t} - \beta \bar{c}$$

which we can plug into (32) and manipulate it to obtain

$$c_{t+1}^i = -\bar{c} + \bar{c} \frac{p_t}{p_{t+1}} \left[(1 - \beta) \left(\sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right) \right] + (1 - \beta) a_{t+1}^i \quad (33)$$

Finally, note that

$$\sum_{\tau=t+1}^{\infty} \frac{p_{\tau}}{p_{t+1}} = \frac{p_t}{p_{t+1}} \left(\sum_{\tau=t}^{\infty} \frac{p_{\tau}}{p_t} - 1 \right) \quad (34)$$

Plugging (34) in (33), we obtain

$$\begin{aligned}
c_{t+1}^i &= -\bar{c} + \bar{c} \left((1 - \beta) \sum_{\tau=t+1}^{\infty} \frac{p_{\tau}}{p_{t+1}} \right) + (1 - \beta) a_{t+1}^i \\
&= \bar{c} \left[(1 - \beta) \sum_{\tau=t+1}^{\infty} \frac{p_{\tau}}{p_{t+1}} - 1 \right] + (1 - \beta) a_{t+1}^i
\end{aligned}$$

QED

1 Consumption Insurance

We will use the following notation to represent uncertainty. In each period $t \geq 0$, there is a realization of a stochastic event $s_t \in S_t$. We refer to this as the current state of the economy and let $s^t = \{s_0, s_1, \dots, s_t\}$ be the history up to time t , with $s^t \in S^t \equiv S_0 \times S_1 \times \dots \times S_t$. Let $\pi(s^t)$ the probability of this history occurring. Let $y_t^i(s^t)$ be the individual i realization of endowment/income upon the realization of history s^t , with $\sum_{i \in I} y_t^i(s^t) = Y_t(s^t)$ denoting the aggregate endowment/income.

1.1 Two Benchmarks: Autarky and Complete Markets

If we consider the spectrum of all possible market arrangements, at the two extremes we find autarky and complete markets. Let's analyze these two cases first.

1.1.1 No Risk Sharing in Autarky

The simplest starting point to analyze consumption is an endowment economy where insurance markets to trade across states s_t at a given point in time t are completely absent, and there is no storage technology to transfer resources across periods (e.g., the consumption good is perishable). In this economy, an individual i who receives a random stream of income shocks $\{\{y_t^i(s^t)\}_{s^t \in S^t}\}_{t=0}^{\infty}$ has no other choice than consuming her income every period, i.e.

$$c_t^i(s^t) = y_t^i(s^t). \quad (1)$$

1.1.2 Full Risk Sharing in Complete Markets

AD budget constraint— The diametrically opposite benchmark is an endowment economy with a full set of insurance and financial markets. In an Arrow-Debreu market structure we assume that trade takes place at $t = 0$, before any uncertainty has realized. Letting $p_t(s^t)$ denote the price of one unit of consumption quoted at $t = 0$ to be delivered in period t if history s^t is realized, the Arrow-Debreu budget constraint of agent i is¹

$$\sum_{t=0}^{\infty} \sum_{s^t \in S^t} p_t(s^t) [c_t^i(s^t) - y_t^i(s^t)] = 0, \text{ for all } i \in I. \quad (2)$$

¹Underlying the single AD budget constraint is the fact that multilateral trades are possible through a clearing operation that keeps track of net claims, i.e. a system with net settlements of trades. All trades occur at time 0. After time 0, trades that were agreed to at time 0 are executed, but no more trades occur. See LS, Chapter 8.

Hence, every possible transfer of income across states and time is possible, as long as discounted consumption expenditures equal discounted income for each individual. Agents maximize expected lifetime utility

$$\sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) u(c_t^i(s^t)).$$

SP solution– Using the First Welfare Theorem, we can characterize the complete markets competitive equilibrium allocations as the solution to the Pareto problem

$$\begin{aligned} & \max_{\{c_t^i(s^t)\}} \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) \sum_{i \in I} \alpha^i u(c_t^i(s^t)) \\ & s.t. \\ & \sum_{i \in I} c_t^i(s^t) = Y_t(s^t), \text{ for all } t, s^t \in S^t \end{aligned}$$

with solution

$$\beta^t \pi(s^t) \alpha^i u'(c_t^i(s^t)) = \theta_t(s^t),$$

which implies that for any pair of agents (i, j) the ratio of marginal utility is constant in every period and every state of the world, i.e.

$$\frac{u'(c_t^i(s^t))}{u'(c_t^j(s^t))} = \frac{\alpha^j}{\alpha^i}, \text{ for all } s^t, (i, j), \quad (3)$$

which is the definition of *full risk-sharing (or full-insurance)*.

For the CRRA case, where $u(c_t^i(s_t)) = \frac{c_t^i(s_t)^{1-\sigma}}{1-\sigma}$, it is easy to derive that

$$\frac{c_t^i(s^t)}{c_t^j(s^t)} = \left(\frac{\alpha^i}{\alpha^j} \right)^{\frac{1}{\sigma}},$$

hence the ratio of consumption allocations is constant. Summing over $i = 1, \dots, I$, this implies clearly that

$$c_t^j(s^t) = \left[\frac{(\alpha^j)^{\frac{1}{\sigma}}}{\sum_{i=1}^I (\alpha^i)^{\frac{1}{\sigma}}} \right] C_t(s^t), \quad (4)$$

so, individual consumption tracks perfectly aggregate consumption for every household and depends only on aggregate consumption and not on the individual's own income realization.²

²In general, full risk sharing does not mean constant consumption over time/across states. In the above model, individual consumption is constant over time and across states only if aggregate consumption is constant. Even if the aggregate endowment is constant, with flexible labor supply and non-separability between consumption and leisure in preferences, consumption may not be constant over time. The only statement that is always true in complete markets is that the ratio of marginal utility for any pair of agents is constant over time.

1.1.3 Empirical Test

The full risk sharing hypothesis can be tested empirically. Under CRRA preferences, for example, equation (4) implies that the log-change in individual consumption should equal the log-change in aggregate consumption, for every individual, in every period. If we estimate from microdata the relationship

$$\Delta \log c_t^i = \beta_0 + \beta_1 \Delta \log C_t + \beta_2 \Delta \log y_t^i + \varepsilon_t^i,$$

where y_t^i is current individual income, then the full risk-sharing hypothesis implies $(\beta_1 = 1, \beta_2 = 0)$. Contrast this prediction with the “autarky hypothesis” which implies $(\beta_1 = 0, \beta_2 = 1)$, i.e. consumption tracks perfectly current income. In general, they are both rejected, albeit the data seem to be much closer to full risk-sharing in many contexts.³

Result 2.0: A good model (empirically, at least) for consumption lies between autarky and full risk-sharing, i.e. it must be a model where agents have access to “partial” consumption insurance.

1.1.4 Modeling Partial Risk Sharing

How do we model partial consumption insurance? There are two approaches. The first one, which we can call the “endogenous incomplete markets” approach is to model explicitly the frictions that undermine full insurance (e.g., limited enforcement, moral hazard, adverse selection). The second is the “exogenous incomplete markets” approach which suggests to model only the contracts/assets that we observe in the data (e.g., stock, bond, housing, etc.). The first approach is deeper than the second, but it yields usually a lot of state-contingent contracts in equilibrium that we do not see in the data, so the second approach is more empirically useful.

Consider an example of the first approach. Suppose that the trades agreed upon at time $t = 0$ cannot be fully enforced. However, consumers who do not honor their contracts are excluded from financial markets forever and will leave in autarky thereon. Then, we can write “participation constraints” for agents, in every state s_t , of the form

$$\sum_{\tau=t+1}^{\infty} \sum_{s^{\tau} \in S^{\tau}} \beta^{\tau-t} \pi(s^{\tau}) u(c_{\tau}^i(s^{\tau})) \geq \sum_{\tau=t+1}^{\infty} \sum_{s^{\tau} \in S^{\tau}} \beta^{\tau-t} \pi(s^{\tau}) u(y_{\tau}^i(s^{\tau}))$$

³See Mace (1991), Cochrane (1991).

These constraints state that agents will always strictly prefer staying in the contract than defaulting. Clearly, these constraints limit the amount of insurance that is available in equilibrium relative to an A-D equilibrium with full enforcement. At the end of the course, we will study consumer default in economies with limited contract enforcement.

1.2 Exogenous Restrictions on Trade: The Bond Economy

We now consider an example of the “exogenous incomplete markets” approach.

Sequential formulation of complete markets– It is useful to start from the sequential formulation version of the Arrow-Debreu constraint (2), i.e., a sequence of budget constraints, for every period t and history s^t , of the form

$$c_t^i(s^t) + \sum_{s_{t+1} \in S_{t+1}} q_t(s_{t+1}, s^t) a_{t+1}^i(s_{t+1}, s^t) = y_t^i(s^t) + a_t^i(s^t), \quad (5)$$

and a no Ponzi scheme condition that rules out excessively high debt at every node.⁴ Here, $q_t(s_{t+1}, s^t)$ is the price at date t and state s^t of an Arrow security that pays one unit of consumption if state s_{t+1} occurs next period. Obviously, this sequential formulation of the complete markets model gives rise to the same conclusion, i.e. full risk sharing.⁵

The bond economy– The key difference between the bond economy and the complete markets model is in the set of securities that the household is allowed to trade. In the bond economy, agents are restricted to trade only a non state-contingent asset. The budget constraint (5) is replaced by the more restrictive constraint, for every history s^t ,

$$c_t^i(s^t) + q_t(s^t) a_{t+1}^i(s^t) = y_t^i(s^t) + a_t^i(s^{t-1}),$$

where $q_t(s^t)$ is the price at date t and state s^t of an asset that pays one unit of consumption next period, *independently* of the realization of the state s_{t+1} , i.e. it is a one-period bond. In other words, the agent is cut-off from every state-contingent insurance market and has

⁴The notation $a_t^i(s_t, s^{t-1})$ and the somewhat more concise formulation $a_t^i(s^t)$ are obviously equivalent, since $s^t = \{s_t, s^{t-1}\}$.

⁵It can be proved that the sequence of constraints in (5) and the no Ponzi scheme condition

$$\lim_{t \rightarrow \infty} \sum_{s^t \in S^t} q_t(s_t, s^{t-1}) a_t^i(s_t, s^{t-1}) \geq 0$$

is equivalent to the Arrow-Debreu constraint (2). See chapter 8 in LS.

only access to a simple financial instrument to transfer resources over time. The absence of insurance opportunities induces the consumer to hold a certain amount of the bond in order to smooth consumption.

Define

$$q_t(s^t) \equiv \frac{1}{1 + r_t(s^t)},$$

where $r_t(s^t)$ is the interest rate on a risk-free bond, and reformulate the budget constraint with lighter notation as

$$a_{t+1} = (1 + r_t)(y_t + a_t - c_t), \quad (6)$$

i.e., we omit the explicit dependence on histories and the i index.

No-Ponzi schemes— Absent any other constraints, agents would choose to borrow an infinite amount. We rule this out by imposing a No-Ponzi scheme condition stating that in the limit assets cannot be negative, i.e.

$$\lim_{J \rightarrow \infty} a_{t+J+1} \Pi_{j=0}^J \left(\frac{1}{1 + r_{t+j}} \right) \geq 0.$$

This constraint holds in the uncertain case, along all possible consumption sequences (technically, it holds almost surely).⁶ Optimality implies that the weak inequality above holds with the “=” sign.⁷

Borrowing constraints— In some cases, we would like to study the effect of a more restrictive borrowing constraint than the No-Ponzi condition. We thus impose the additional ad-hoc borrowing constraint that prevents borrowing from growing too much

$$a_{t+1} \geq -\Phi,$$

in which Φ is some parameter that we will sometimes set to 0 or to a strictly positive value. These are ad-hoc constraints that we impose to capture, in a reduced form, frictions that are not explicitly modeled. The type of frictions that these constraints are supposed to

⁶It is not sufficient that the No-Ponzi condition holds in expectation, that is $E_0 \lim_{J \rightarrow \infty} a_{t+J+1} \Pi_{j=0}^J \left(\frac{1}{1 + r_{t+j}} \right) \geq 0$. If this were the case, then there would be possible paths with non-zero measure where the No-Ponzi condition would be violated. Along these paths, lenders would have to agree to provide an infinite level of consumption to the household. Note also that if the No-Ponzi condition holds almost surely, then it holds in expectation, while the reverse is obviously not true.

⁷See the Appendix for a more exhaustive discussion of the no Ponzi scheme condition.

capture are asymmetric information, moral hazard, adverse selection, defaults, etc : all sorts of frictions that can limit trade in assets.

Household's problem—We can write the full consumption-saving problem as follows:

$$\begin{aligned} \max_{c_t, a_{t+1}} E_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right] \\ \text{s.t. } c_t + \frac{a_{t+1}}{1+r_t} = y_t + a_t \\ a_{t+1} \geq -\Phi \\ +\text{no-Ponzi condition} \end{aligned}$$

and write the corresponding Lagrangian

$$\mathcal{L} = E_0 \left\{ \sum_{t=0}^{\infty} \beta^t \left[u(c_t) - \lambda_t \left(c_t + \frac{a_{t+1}}{1+r_t} - y_t - a_t \right) + \mu_t (a_{t+1} + \Phi) \right] \right\},$$

with the convention that $\mu_t \geq 0$. This problem yields the first order conditions with respect to c_t and a_{t+1} :

$$\begin{aligned} u'(c_t) &= \lambda_t \\ \frac{\lambda_t}{1+r_t} &= \beta E_t \lambda_{t+1} + \mu_t, \end{aligned}$$

where λ_t and μ_t are Lagrange multipliers on the budget constraint and the borrowing constraint, respectively.

Combining these two conditions, we obtain the consumption Euler equation

$$u'(c_t) = \beta (1+r_t) E_t [u'(c_{t+1})] + \underbrace{\tilde{\mu}_t}_{\equiv \mu_t(1+r_t)},$$

and using the complementary slackness condition

$$u'(c_t) \geq \beta (1+r_t) E_t [u'(c_{t+1})],$$

which holds with equality if the borrowing constraint is not binding.

2 The Permanent Income Hypothesis (PIH)

In order to derive a better intuition of the forces at work in the model, let us first specialize the model a little to what is commonly called the “permanent income hypothesis”. To

simplify the notation for the next sections, we assume away fluctuations in the aggregate endowment $Y_t(s^t)$, either deterministic or stochastic, hence $r_t = r$. We assume that the only source of uncertainty comes from idiosyncratic risk in individual endowments y_t . For now, we will be a bit loose with our treatment of uncertainty, in that we will not explicitly discuss probability spaces that form the formal basis of our representation of uncertainty.

The strict version of the PIH is a special case of the bond economy with three key assumptions:

1. households have quadratic utility

$$u(c_t) = b_1 c_t - \frac{1}{2} b_2 c_t^2,$$

Note that $u' > 0$ requires $c_t < b_1/b_2$ and $u'' < 0$ requires $b_2 > 0$.

2. the interest rate on the one-period bond equals the inverse of the discount rate, or $\beta(1+r) = 1$.
3. there is no borrowing constraint $\mu_t = 0, \forall t$.

Consumption as a random walk— In this case, the consumption Euler equation is:

$$u'(c_t) = \beta(1+r) E_t[u'(c_{t+1})],$$

and using the quadratic utility assumption:

$$b_1 - b_2 c_t = \beta(1+r) E_t(b_1 - b_2 c_{t+1}) \Rightarrow E_t c_{t+1} = c_t. \quad (7)$$

from which we recover the well known result that consumption is a martingale.⁸

It is useful to note that from the law of iterated expectations and the martingale property:

$$E_t c_{t+2} = E_t [E_{t+1} c_{t+2}] = E_t c_{t+1} = c_t$$

and, more in general:

$$E_t c_{t+j} = c_t, \text{ for any } j \geq 0. \quad (8)$$

Recall the budget constraint:

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t \Rightarrow a_t = c_t - y_t + \frac{a_{t+1}}{1+r}.$$

⁸A martingale is a stochastic process (i.e., a sequence of random variables) $\{x_t\}$ which satisfies, at every t , $E_t x_{t+1} = x_t$.

Iterating forward one period on the budget constraint, we obtain

$$c_t = y_t + a_t - \frac{1}{1+r}a_{t+1} = y_t + a_t - \frac{1}{1+r}E_t \left[c_{t+1} - y_{t+1} + \frac{a_{t+2}}{1+r} \right]$$

and rearranging

$$c_t + \frac{1}{1+r}E_t c_{t+1} = a_t + y_t + \frac{1}{1+r}E_t y_{t+1} - \left(\frac{1}{1+r} \right)^2 E_t a_{t+2}$$

If we keep iterating J times, we arrive at

$$\sum_{j=0}^J \left(\frac{1}{1+r} \right)^j E_t c_{t+j} = a_t + \sum_{j=0}^J \left(\frac{1}{1+r} \right)^j E_t y_{t+j} - \left(\frac{1}{1+r} \right)^{J+1} E_t a_{t+J+1}$$

Taking the limit as $J \rightarrow \infty$ and using the No-Ponzi scheme condition, we arrive at:

$$\begin{aligned} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j E_t c_{t+j} &= a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j E_t y_{t+j} \\ c_t &= \frac{r}{1+r} \left[a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j E_t y_{t+j} \right] = \frac{r}{1+r} (a_t + H_t) \end{aligned} \quad (9)$$

where the LHS of the second row uses property established in (8), and in the last equality we denoted human wealth, i.e. the expected discounted value of future earnings, with H_t . Recall that financial wealth is a_t . This implies that consumption is the annuity value of lifetime (human and financial) wealth $W_t \equiv (a_t + H_t)$, also referred to as permanent income.⁹ This is a very important result: consumption depends on the entire discounted value of future income, not just current income. Also, this equation has some very specific implications on the joint dynamics of consumption and expectations, as we are going to see. Therefore, we have the following result:

Result 2.1: If preferences are quadratic, and $\beta(1+r) = 1$, then consumption follows a martingale process and equals permanent income, i.e. the annuity value of human and financial wealth.

Certainty equivalence— Notice that, if one solves the non-stochastic version of the PIH problem stated earlier, from the FOC (7) one obtains $c_{t+1} = c_t$ and, by iterating

⁹The annuity value is defined as that portion that, when consumed every period, keeps the asset value constant. For example, the annuity value w_t of wealth W_t is such that

$$W_{t+1} = (1+r)(W_t - w_t) = W_t \Rightarrow w_t = \frac{r}{1+r}W_t.$$

forward on the budget constraint,

$$c_t = \frac{r}{1+r} \left[a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j y_{t+j} \right].$$

Compared to equation (9), the above equation suggests that the consumption satisfies *certainty equivalence* in the sense that to obtain the solution of the stochastic problem, one can 1) solve the deterministic problem and 2) substitute conditional expectations of the forcing variables (y_{t+j}) in place of the variables themselves. Put differently, *the variance and higher moments of the income process do not matter for the determination of consumption*. This property descends directly from the linear-quadratic objective function.

Consumption dynamics– From (9), the change in consumption at time t equals

$$\Delta c_t = c_t - c_{t-1} = c_t - E_{t-1}c_t = \frac{r}{1+r} [W_t - E_{t-1}W_t],$$

where we have used the random walk property. Now, use the definition of total wealth W_t to define the innovation (i.e. the unexpected change) in permanent income, at time t as

$$\begin{aligned} W_t - E_{t-1}W_t &= a_t - E_{t-1}a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j [E_t y_{t+j} - E_{t-1}(E_t y_{t+j})], \\ &= \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j (E_t y_{t+j} - E_{t-1}y_{t+j}), \end{aligned} \quad (10)$$

where we have used the law of iterated expectations $E_{t-1}(E_t y_{t+j}) = E_{t-1}y_{t+j}$, and the fact that $a_t = E_{t-1}a_t$, since there is no uncertainty at time t about the evolution of wealth next period.¹⁰ Putting together (10) and the expression above for the change in consumption we arrive at

$$\Delta c_t = \frac{r}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j (E_t y_{t+j} - E_{t-1}y_{t+j}). \quad (11)$$

This equation states another useful result:

¹⁰Recall that the budget constraint reads $a_{t+1} = (1+r)(y_t + a_t - c_t)$, so:

$$E_t a_{t+1} = (1+r) E_t (y_t + a_t - c_t) = (1+r)(y_t + a_t - c_t) = a_{t+1}$$

Result 2.2: Under the PIH, the change in consumption between $t - 1$ and t is proportional to the revision in expected earnings due to the new information (“news”) accruing in that same time interval.

2.1 Example with a Specific Income Shock

At this point, to make further progress, we need to make some assumptions on the statistical properties of the labor income process. We choose a specification that is very common in labor economics. We assume that labor income is the sum of two orthogonal components, a permanent component y_t^p which follows a martingale, and a transitory component u_t that is independently distributed over time:

$$\begin{aligned} y_t &= y_t^p + u_t, \\ y_t^p &= y_{t-1}^p + v_t. \end{aligned} \tag{12}$$

Note that v_t is the innovation to the permanent component, independently distributed over time. Assume that $E(v_t) = E(u_t) = 0$ and that the two shocks are orthogonal, $u_t \perp v_\tau$ for all pairs (t, τ) .

Using $y_t^p = y_t - u_t$ from the first equation into the second equation, we obtain the representation

$$\begin{aligned} y_t - u_t &= y_{t-1} - u_{t-1} + v_t \\ y_t &= y_{t-1} + u_t - u_{t-1} + v_t \end{aligned} \tag{13}$$

With this structure of shocks in hand, we can express the change in consumption from (11) only as a function of the permanent innovation v_t and the transitory innovation u_t . Focusing on the RHS of (11) and ignoring for now the constant $r/(1+r)$, we have

$$\begin{aligned} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j (E_t y_{t+j} - E_{t-1} y_{t+j}) &= E_t y_t - E_{t-1} y_t + \frac{1}{1+r} (E_t y_{t+1} - E_{t-1} y_{t+1}) \\ &+ \left(\frac{1}{1+r} \right)^2 (E_t y_{t+2} - E_{t-1} y_{t+2}) + \dots \end{aligned} \tag{14}$$

Using (13) into the first term of the RHS of equation (14)

$$E_t y_t - E_{t-1} y_t = y_t - (y_{t-1} - u_{t-1}) = (y_{t-1} + u_t - u_{t-1} + v_t) - (y_{t-1} - u_{t-1}) = u_t + v_t$$

In other words, the unexpected change in income y_t (compared to the one-step-ahead forecast $E_{t-1}y_t$) is the sum of the permanent and the transitory innovations at time t .

Using (13) into the second term of the RHS of (14):

$$\begin{aligned} E_t y_{t+1} - E_{t-1} y_{t+1} &= E_t y_{t+1} - E_{t-1} y_{t+1} \\ &= E_t (y_t + u_{t+1} - u_t + v_{t+1}) - E_{t-1} (y_t + u_{t+1} - u_t + v_{t+1}) \\ &= E_t y_t - E_{t-1} y_t - u_t = u_t + v_t - u_t = v_t, \end{aligned}$$

since all the terms indexed by $t + 1$ drop out because $E_t(x_{t+1}) = E_{t-1}(x_{t+1}) = 0$. At this point, it is easy to see that

$$E_t y_{t+j} - E_{t-1} y_{t+j} = v_t \quad \text{for any } j \geq 1$$

In other words, the forecast revision between $t - 1$ and t about income beyond time t equals the permanent innovation at time t .

Going back to expression (11), the innovation to permanent income is

$$\begin{aligned} \Delta c_t &= \frac{r}{1+r} \left[u_t + v_t + \frac{1}{1+r} v_t + \left(\frac{1}{1+r} \right)^2 v_t + \dots \right] \\ &= \frac{r}{1+r} \left[u_t + v_t \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j \right] \\ &= \frac{r}{1+r} u_t + v_t. \end{aligned}$$

Hence, households adjust their consumption responding to the annuitized change in income. This means that they will respond only weakly to purely transitory shocks (u_t), whereas they will respond one for one to permanent shocks (v_t). Indeed, the former shocks have only a small effect on permanent income, while the latter change permanent income one for one, by definition.

Identification of the shocks through panel data— Suppose that one has panel data on consumption and income for a cohort of households, $i = 1, \dots, N$ followed over periods $t = 0, \dots, T$. Suppose that the variances of (u_t^i, v_t^i) change over time. Let var_t denote the cross-sectional variance (i.e., the variance across individuals of this cohort) at

time t . Then, note that for $t = 1, \dots, T$ we can write:

$$\begin{aligned} \text{var}(\Delta c_t^i) &= \left(\frac{r}{1+r}\right)^2 \text{var}(u_t^i) + \text{var}(v_t^i) \simeq \text{var}(v_t^i), \\ \text{var}(\Delta y_t^i) &= \text{var}(v_t^i) + \text{var}(u_t^i) + \text{var}(u_{t-1}^i), \\ \text{cov}(\Delta c_t^i, \Delta y_t^i) &= \left(\frac{r}{1+r}\right) \text{var}(u_t^i) + \text{var}(v_t^i) \end{aligned}$$

where the approximate equality in the first row holds for r “small”. Therefore, it is easy to see that with data on consumption and income one can separately identify the variances of the underlying structural income shocks.

For example, if over a certain period of time we observe the variance of income rising, but the variance of consumption approximately flat, we should conclude that the rise in income uncertainty was mostly transitory.

2.2 When are borrowing constraints binding?

So far, we have ignored the presence of borrowing constraints. We imposed a no-Ponzi scheme condition, but we never checked whether it’s actually binding. Is it a good abstraction? The answer is that, as we show below, it depends on the income process.

Wealth dynamics with borrowing constraints— First, recall the budget constraint

$$a_{t+1} = (1+r)(y_t + a_t - c_t).$$

Rearranging, we obtain an expression for the change in wealth

$$\Delta a_{t+1} = (1+r)y_t + ra_t - (1+r)c_t. \quad (15)$$

The optimal consumption choice from (9) is

$$c_t = \frac{r}{(1+r)} \left[a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r}\right)^j E_t y_{t+j} \right]. \quad (16)$$

Now, suppose that the income process follows a random walk, $y_t = y_{t-1} + \varepsilon_t$, with ε_t iid and $E(\varepsilon_t) = 0$. Then, from from (16) we obtain

$$c_t = \frac{r}{1+r} a_t + y_t$$

since $E_t y_{t+j} = y_t$. Substituting into equation (15) this yields $a_{t+1} = a_t$. Therefore, the initial wealth endowment perpetuates itself (i.e., it is constant) so if the individual starts

above the borrowing constraint, it will never be binding. The reason for this result is that wealth changes only if the individual is consuming just a part of its income (and saving the remaining part) in order to smooth consumption. With permanent shocks, all the income shock is consumed in every period.

However, if the income process is *iid*, ($y_t = \varepsilon_t$) from (16) we have

$$c_t = \frac{r}{1+r}(a_t + y_t)$$

since $E_t y_{t+j} = 0$. Substituting into equation (15) yields $\Delta a_{t+1} = \varepsilon_t$. This means that wealth follows a random walk and, as a result, any constraint on asset holdings will be binding with probability one sooner or later.

To conclude, whether ignoring borrowing constraint is troublesome or not depends on the specific income process. However, in general this result highlights the fact that borrowing constraints cannot be ignored.

Appendices

A Borrowing constraints, No-Ponzi and Transversality Conditions

It is common to impose a number of constraints on the feasible consumption plans when solving the income fluctuation problems. In particular, we want to prevent agents from raising arbitrarily large amounts of resources by issuing arbitrarily large amounts of debt. The objective of this short note is to clarify a number of common confusions between borrowing constraints, no-Ponzi conditions, intertemporal budget constraints and transversality conditions. I will consider only the incomplete market case of the bond economy seen in class, in which we are solving the following problem:

$$\begin{aligned} \max_{c_t(s^t), a_{t+1}(s^t)} \quad & \mathbb{E}_0 \left[\sum \beta^t u(c_t) \right] \\ \text{s.t.} \quad & c_t(s^t) + \frac{a_{t+1}(s^t)}{R_t(s^t)} = y_t(s^t) + a_t(s^t), \forall (t, s^t) \end{aligned}$$

A.1 Definitions

Although some economists would deem it a crime to consider together notions as different as borrowing constraints and transversality conditions, I am going to do so here because there seems to be much confusion/overlap in the literature. Let me first remind you why these notions are very different things.

A.1.1 Transversality conditions

Transversality conditions are sufficient conditions that guarantee the optimality of an allocation. We know for example that an allocation satisfying the Euler equation and the TVC in the neoclassical growth model is optimal. TVCs are thus part of the optimality conditions, which are based on solid mathematical foundations. They are not constraints that we impose on the problem, even though they might sometimes look like no-ponzi conditions. The exact TVCs of a given problem have to be carefully derived in every case. TVCs are not ad-hoc restrictions that one imposes to rule out undesired solutions, as it is unfortunately the case in various papers. Good references for the derivation of the TVC

are SLP (1989) or Sims (2006).¹¹ Following SLP, the transversality condition of the bond economy should be written

$$\lim_{j \rightarrow \infty} \beta^j \mathbb{E}_t [\lambda_{t+j} a_{t+j+1}] = 0$$

where λ_t should be understood as the Lagrange multiplier on the budget constraint. According to Sims (2006) the above condition is not absolutely right and a more appropriate condition would be

$$\lim_{j \rightarrow \infty} \beta^j \mathbb{E}_t [\lambda_{t+j} a_{t+j+1}] \leq 0. \quad (17)$$

A.1.2 Borrowing constraints

These are ad-hoc constraints that we impose to capture frictions that are not explicitly modeled in a reduced form. The type of frictions that these constraints are supposed to capture are asymmetric information, moral hazard, adverse selection, defaults, etc : all sorts of frictions that can limit trade in assets. In general, borrowing constraints take the following form:

$$a_t(s^t) \geq -\Phi,$$

where Φ is some positive number. In many cases, we set $\Phi = 0$ (no borrowing constraint) or we use what is commonly called the *natural debt limit*:

$$a_t(s^t) \geq -\sum_{j=0}^{\infty} \frac{y_{min}}{R^j} = -\frac{R}{R-1} y_{min}.$$

This constraint is obtained by iterating on the budget constraint of the agent in the bond economy and by considering the worst case scenario in which the agent receives always the worst realization of income y_{min} and sets future consumption to 0 (see below).

A.1.3 No-Ponzi conditions

No-Ponzi conditions are hybrids between borrowing constraints and transversality conditions. In the complete market setup, they are sometimes imposed to obtain the equivalence between time-0 trading and the sequential budget constraint (more precisely, to have the equivalence between the intertemporal budget constraint and the sequential budget constraints). In principle, these conditions are imposed to specifically rule out Ponzi

¹¹This is a short note by Chris Sims that you can find at <http://sims.princeton.edu/yftp/Macro2010/rlg.pdf> and in which you will find a quite general approach to TVCs.

schemes. Indeed, without further restrictions, the solution of the consumption-saving problem stated above would be to issue arbitrarily large amounts of debt (let a_t diverge to $-\infty$) to finance consumption. Of course, this is not a very reasonable solution. In the literature, No-Ponzi conditions are used to rule out such outcomes in two ways:

1. As ad-hoc constraints: often in partial equilibrium setups or in environments in which the actual frictions that should prevent such outcomes are not explicitly modeled. In that case, No-Ponzi conditions end up looking very much like borrowing constraints and there is no uniquely reasonable way to write them.
2. Derived from first principles: there are deep economic reasons why Ponzi schemes should not be observed in equilibrium. One explanation is simply that no lender would ever be willing to make these loans: Ponzi schemes should probably violate the transversality conditions of the lenders. We will see an example of that. Another way to proceed would be to model financial market frictions explicitly and, for instance, find how much defaults lenders would be willing to tolerate. This would tell us which path should be ruled out in equilibrium.

A.2 No-Ponzi conditions as ad-hoc borrowing constraints

To understand better the meaning of the most widely used no-Ponzi conditions, start with the sequential budget constraint

$$c_t(s^t) + \frac{a_{t+1}(s^t)}{R_t(s^t)} = y_t(s^t) + a_t(s^{t-1})$$

and iterate to obtain the following intertemporal budget constraint

$$\sum_{\tau=0}^{\infty} \frac{c_{t+\tau}(s^{t+\tau})}{R_t^{t+\tau}} + \lim_{\tau \rightarrow \infty} \frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}} = \sum_{\tau=0}^{\infty} \frac{y_{t+\tau}(s^{t+\tau})}{R_t^{t+\tau}} + a_t(s^{t-1}), \quad (18)$$

where $R_t^{t+\tau}(s^{t+\tau}) = \prod_{s=0}^{\tau-1} R_{t+s}(s^{t+s})$. Notice here that this equation is valid pathwise: for any possible realization of the shocks $(s_0, \dots, s_t, \dots)$.

It is common to find in the literature the following no-Ponzi conditions. Either:

1. By making the assumption that $\lim_{\tau \rightarrow \infty} \frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}} \geq 0$. This effectively implies that

debt can be repaid on any possible path

$$\underbrace{-a_t(s^{t-1})}_{\text{borrowing}} \leq \sum_{\tau=0}^{\infty} \frac{y_{t+\tau}(s^{t+\tau}) - c_{t+\tau}(s^{t+\tau})}{R_t^{t+\tau}}. \quad (19)$$

This is arguably a quite restrictive condition since it implies that the agent must be able to pay back its debt even in the worst case scenario. No default is ever observed on the equilibrium path. Since this condition has to hold pathwise, it should hold in particular in the worst case scenario:

$$-a_t(s^{t-1}) \leq \sum_{\tau=0}^{\infty} \frac{y_{\min} - c_{t+\tau}(s^{t+\tau})}{R_t^{t+\tau}} \leq \sum_{\tau=0}^{\infty} \frac{y_{\min}}{R_t^{t+\tau}}$$

which we see, together with the non-negativity of consumption, can be shown to be equivalent to the *natural borrowing limit* (in a slightly general form though, since we allow R_t to vary).

2. To relax the previous condition slightly, a number of authors impose instead the condition $\mathbb{E}_t \left[\lim_{\tau \rightarrow \infty} \frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}} \right] \geq 0$.¹² Taking the expectations at time t in (18), this implies the following

$$-a_t(s^{t-1}) \leq \mathbb{E}_t \left[\sum_{\tau=0}^{\infty} \frac{y_{t+\tau}(s^{t+\tau}) - c_{t+\tau}(s^{t+\tau})}{R_t^{t+\tau}} \right]. \quad (20)$$

This is a weaker condition since it only implies that the agent can repay its debt in expectation. In principle, this does not rule out paths with explosive debt. So what should we do with these paths? In some cases, one could argue that these paths could be tolerated. In reality, people might effectively start following such paths, but they would default at some point. This condition implicitly assumes that creditors accept defaults on the equilibrium path and that the loan is effectively not *risk-free*.

To understand the difference between the two constraints, let us consider the following example. Assume that y_t follows an iid process from $t \geq 0$ onwards such that $y_t \in$

¹²I will ignore here the technical details that ensure that $\mathbb{E}_t \left[\lim_{\tau \rightarrow \infty} \frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}} \right] \geq 0$ is equivalent to $\lim_{\tau \rightarrow \infty} \mathbb{E}_t \left[\frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}} \right] \geq 0$. We can use the argument here that expectation is a continuous operator on the space of Lebesgue integrable functions L^1 . We would then need to make sure that the interest rate and income process we assume guarantee that the process $\frac{a_{t+\tau+1}(s^{t+\tau+1})}{R_t^{t+\tau+1}}$ is integrable and that its limit exists.

$[y_{min}, y_{max}]$ with $E[y_t] = \bar{y}$. Then, the pathwise no-Ponzi condition can be written as

$$\forall (t, s^t), \quad -a_t(s^{t-1}) \leq \frac{R}{R-1} y_{min} \text{ (worst case scenario: } y_{t+\tau} = y_{min}, \forall \tau \geq 0)$$

while the expectational no-Ponzi condition can be written as

$$\forall (t, s^t), \quad -a_t(s^{t-1}) \leq y_{min} + \frac{\bar{y}}{R-1} \text{ (worst case scenario: } y_t = y_{min}, \text{ today only).}$$

We see here that the expectational condition is indeed weaker: $y_{min} + \frac{\bar{y}}{R-1} \geq \frac{R}{R-1} y_{min}$. Pick some wealth level $\bar{a} \in (-y_{min} - \frac{\bar{y}}{R-1}, -\frac{R}{R-1} y_{min})$ and consider the allocation $a_t(s^{t-1}) = \bar{a}, \forall t$, letting $c_t(s^t)$ be defined recursively by the sequential budget constraint, $c_t(s^t) = y_t(s^t) + \bar{a}(1 - \frac{1}{R})$. This consumption plan will be feasible for values of the interest rate close to 1, but sustainable only under the expectational no-Ponzi condition. This is quite unfortunate: this is a very reasonable consumption plan, in which c_t and a_t are well behaved (not a Ponzi scheme) and, yet, this allocation is ruled out by the pathwise no-Ponzi condition.

This shows us that we have some flexibility in setting the no-Ponzi condition, depending on whether we find the pathwise no-Ponzi condition too restrictive or not. The pathwise condition does remove all the possible explosive paths, but removes too many “good” allocations too. On the other hand, the expectational no-Ponzi condition, being less restrictive, does not rule out all the Ponzi schemes, but it might be more realistic and better able to fit the data. The extent at which we will run into troubles will eventually depend on the assumptions we make on the income process, interest rates, etc.

A.3 No-Ponzi conditions and Transversality

The above discussion is not entirely satisfactory. Are we ruling out potentially important consumption paths? How should we discipline these choices? One way to do that is to go back to first principles. We show in this section that some form of no-Ponzi condition can be obtained from the transversality condition of the lenders.

Let us assume that there are two types in the economy $i = 1, 2$. Agents face the idiosyncratic income process $\{y_t^i(s^t)\}$. We write the consumption-saving problem of both

types as usual

$$\begin{aligned} & \max_{c_t^i(s^t), a_{t+1}^i(s^t)} \mathbb{E}_0 \left[\sum \beta^t u(c_t^i) \right] \\ & \text{s.t. } c_t^i(s^t) + \frac{a_{t+1}^i(s^t)}{R_t(s^t)} = y_t^i(s^t) + a_t^i(s^t), \forall (t, s^t). \end{aligned}$$

Assuming that the aggregate supply of assets is 0 in equilibrium, we require the following market clearing condition:

$$a_t^1(s^t) + a_t^2(s^t) = 0.$$

Deriving the optimality conditions of the above maximization, we know that the following transversality condition

$$\lim_{j \rightarrow \infty} \beta^j \mathbb{E}_t [\lambda_{t+j}^i a_{t+j+1}^i] \leq 0$$

is sufficient to guarantee that an allocation satisfying the Euler equation is optimal. What does agent 2's TVC imply for the borrowing of agent 1? Using the market clearing constraint in agent 2's TVC, we obtain:

$$\lim_{j \rightarrow \infty} \beta^j \mathbb{E}_t [\lambda_{t+j}^2 a_{t+j+1}^1] \geq 0. \quad (21)$$

This looks very much like our expectational no-Ponzi condition: this is a constraint that prevents agent 1 from borrowing in the limit. Notice that Ponzi schemes might still be observed in equilibrium since the condition only has to hold in expectation. This tells us that lenders will be willing to tolerate such divergent path as long as they have a low probability to arise. In some way, this example provides a justification for the use of the expectational no-Ponzi condition.

Notice, however, that the above condition differs from the expectational no-Ponzi condition in one important way: the presence of the Lagrange multiplier λ . What this condition tells us exactly is that we should discount future assets using the proper “pricing kernel” and that discounting the asset with the interest rate R may not always be appropriate. One could argue, though, that condition (20) approximates (21) well enough, which might explain why the expectational no-Ponzi condition is so widely used in the literature.

Notice, finally, that this discussion provides a microfoundation for the no-Ponzi condition found in the literature in the deterministic case. When fluctuations in income are

deterministic, we can combine the above TVC condition with the FOC in a_t : $\lambda_t^i = \beta R_t \lambda_{t+1}^i$, and obtain the condition

$$\lim_{j \rightarrow \infty} \frac{a_{t+j+1}^1}{R_t^{t+j}} \geq 0$$

which is precisely the condition used in the literature.

1 Precautionary Savings: Prudence and Borrowing Constraints

In this section we study conditions under which savings react to changes in income uncertainty. Recall that in the PIH, when you abstract from borrowing constraints, certainty equivalence implies that “mean preserving spreads” of the income distribution do not impact on saving. But quadratic utility is very special, what happens with more general utility functions?

1.1 Prudence

1.1.1 A two-period model

Consider the simple two-period consumption-saving problem studied by Leland (1968):

$$\begin{aligned} \max_{\{c_0, c_1, a_1\}} \quad & u(c_0) + \beta u(c_1) \\ \text{s.t.} \quad & \\ c_0 + \frac{a_1}{1+r} = & y_0 \\ c_1 = & a_1 + \bar{y}_1 \end{aligned}$$

where y_0 is given, and income next period $\bar{y}_1$ is also exogenous and deterministic. If we retain the assumption $\beta(1+r) = 1$ to simplify the algebra, the Euler equation gives

$$u' \left(y_0 - \frac{a_1}{1+r} \right) = u'(a_1 + \bar{y}_1),$$

which is one equation in one unknown, a_1 . The LHS is increasing in a_1 since $u'' < 0$, and the RHS is decreasing for the same reason, hence a_1^* is uniquely determined.

Note that current consumption c_0 is determined by the period-zero budget constraint

$$c_0^* = y_0 - \frac{a_1^*}{1+r},$$

hence a rise in savings leads to a fall in current consumption.

Mean-preserving spread– What happens to optimal consumption at $t = 0$ if the uncertainty over income next period rises, i.e. as future income becomes more risky? Consider a mean-preserving spread of $\bar{y}_1$. Define

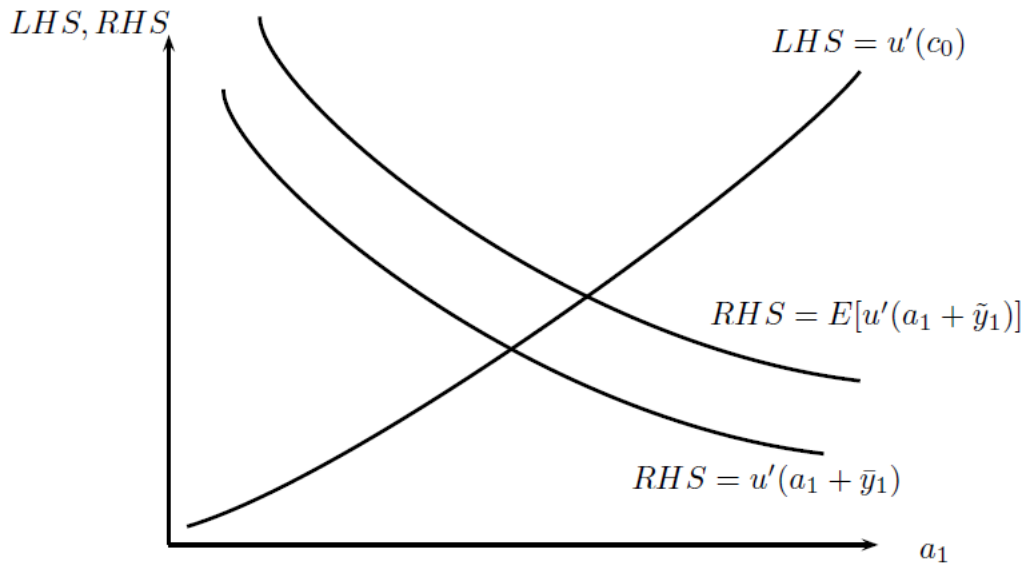
$$\tilde{y}_1 = \bar{y}_1 + \varepsilon_1,$$

where ε_1 is the stochastic component and $\bar{y}_1$ is the mean. Assume that $E(\varepsilon_1) = 0$ and $var(\varepsilon_t) = \sigma_\varepsilon^2$. The Euler equation becomes

$$u' \left(y_0 - \frac{a_1}{1+r} \right) = E[u'(a_1 + \bar{y}_1 + \varepsilon_1)],$$

which shows that *if u' is convex* (a property that quadratic utility does not satisfy), then by Jensen's inequality, a mean-preserving spread of ε_1 will increase the value of the RHS which shifts upward, inducing a rise in a_1^* and a fall in c_0^* . This is an application of the famous result by Rothschild and Stiglitz (1970).

Figure 1: Effect of a mean preserving spread



Prudence– The convexity of the marginal utility (or $u''' > 0$) is called “prudence” and is a property of preferences, like risk aversion: risk-aversion refers to the curvature of the utility function, whereas prudence refers to the curvature of the marginal utility function.¹

Result 2.3: If the marginal utility is convex ($u''' > 0$), then the individual is “prudent” and a rise in future income uncertainty leads to a rise in current savings and a decline in current consumption.

Prudence is a motive for additional savings in order to take precaution against possible negative realizations of the income shock next period. In this sense, savings induced by

¹Precisely, Kimball (1990) defines the index of absolute prudence as the ratio $-u'''(c)/u''(c)$, so in a similar vein to the Arrow-Pratt index of absolute risk-aversion $-u''(c)/u'(c)$.

prudence are called “precautionary savings” or “self-insurance”. In this simple, two-period partial equilibrium model one can define precautionary wealth due to income uncertainty σ_ε as the difference between the optimal asset choice under uncertainty $a_1^*(\sigma_\varepsilon)$ and the optimal asset choice under certainty over next period income, i.e. $a_1^*(0)$.

It can be easily seen that any utility function with decreasing absolute risk aversion, i.e. in the DARA class (which includes CRRA), displays positive third derivative. Let $\alpha(c)$ be the coefficient of absolute risk aversion. Then:

$$\alpha(c) = \frac{-u''(c)}{u'(c)} \Rightarrow \alpha'(c) = \frac{-u'''(c)u'(c) + [u''(c)]^2}{[u'(c)]^2}.$$

Since with DARA $\alpha'(c) < 0$, then we have that

$$-u'''(c)u'(c) + [u''(c)]^2 < 0 \Rightarrow u'''(c) > \frac{[u''(c)]^2}{u'(c)} > 0.$$

Intuitively, a rise in uncertainty reduces the certainty-equivalent consumption next period and with DARA effectively increases the degree of risk-aversion of the agent, inducing him to save more.

1.1.2 Multi-period case

Let's generalize the two-period model to a multiperiod model. In the multi-period case, the problem of the household is

$$\begin{aligned} & \max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ & s.t. \\ & c_t + \frac{a_{t+1}}{1+r} = a_t + y_t \end{aligned}$$

From the FOC's with respect to c_t and a_{t+1} , we obtain:

$$u'(c_t) = \beta(1+r)E_t[u'(c_{t+1})]. \quad (1)$$

Applying the same logic as in the two-period model, it is easy to conclude that precautionary saving arises as long as $u''' > 0$. Note that here we still assume there are no borrowing constraints besides a no-Ponzi scheme condition.

1.2 Borrowing Constraints

To isolate the role of borrowing constraints for precautionary saving, we abstract from prudence altogether and focus on the quadratic utility case. To account for the possibility that the borrowing constraint is binding, the Euler equation needs to be modified. Suppose households face a *no-borrowing constraint* $a_{t+1} \geq 0$. Then, consumption is

$$c_t = \begin{cases} E_t c_{t+1} & \text{if } a_{t+1} > 0 \\ y_t + a_t & \text{if } a_{t+1} = 0 \end{cases}$$

where the first line is just the FOC of the agent when the constraint is not binding, while the second line follows directly from the budget constraint $a_{t+1} = (1+r)(y_t + a_t - c_t)$ when the constraint is binding ($a_{t+1} = 0$). The constrained household would like to borrow to finance consumption, but it is not allowed, so it consumes all its resources.

The above pair of conditions can be written in compound form as

$$c_t = \min \{y_t + a_t, E_t c_{t+1}\} = \min \{y_t + a_t, E_t [\min \{y_{t+1} + a_{t+1}, E_{t+1} c_{t+2}\}]\}.$$

Now, suppose that the uncertainty about income y_{t+1} increases. Very low realizations of income y_{t+1} become more likely, which makes the borrowing constraint more likely to bind in the future and reduces the value of $E_t [\min \{y_{t+1} + a_{t+1}, E_{t+1} c_{t+2}\}]$. This, in turn, reduces the value of $E_t c_{t+1}$.² Thus, if the borrowing constraint is not already binding at time t but it may be binding in the future, then agents consume less today.

Intuitively, when agents face borrowing constraints, they fear getting several consecutive bad income realizations which would push them towards the constraint and force them to consume their income *without the ability of smoothing consumption*. To prevent this situation, they save for self-insurance (precautionary motive). Thus, we have an important result: prudence is not strictly necessary for precautionary saving behavior, or:

Result 2.4: Even in absence of prudence (e.g. with quadratic preferences), in presence of borrowing constraints a rise in future income uncertainty leads to a rise in current savings for precautionary reasons and to a decline in current consumption.

Even though we showed this result for quadratic utility, it is a general results that holds for any concave utility.

²See the Appendix for a more detailed discussion.

1.2.1 A Natural Debt Limit

This is a good place to discuss how to model debt limits. We started by imposing an exogenous borrowing constraint like $a_{t+1} \geq -\Phi$, where Φ is a parameter (in our previous case $\Phi = 0$). However, one may wonder if there is a “natural” borrowing limit that the household faces.

Suppose the income process $\{y_t\}_{t=0}^{\infty}$ is deterministic. By imposing non-negativity of consumption throughout the life of the household, i.e. $c_t \geq 0$ for all t , we have

$$c_t = a_t + y_t - \frac{a_{t+1}}{1+r} \geq 0 \quad \Rightarrow \quad a_t \geq -y_t + \frac{a_{t+1}}{1+r} \forall t.$$

Iterating forward under the restriction of non-negative consumption for all t we have

$$\begin{aligned} a_t &\geq -y_t + \frac{a_{t+1}}{1+r} \geq -y_t + \frac{1}{1+r} \left[-y_{t+1} + \frac{a_{t+2}}{1+r} \right] \geq \dots \\ a_t &\geq -\sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j y_{t+j}. \end{aligned}$$

In other words, by imposing this constraint, the household is not allowed to accumulate more debt than what she will ever be able to repay by consuming just zero every period.

If the income process is stochastic, then how can we be sure that whatever the household borrows she will repay *almost surely* (i.e. with probability 1)? Then, we need to substitute y_t at each t with the lowest possible realization of the income shock, call it $y_{\min}$, and we have the *natural debt limit*

$$a_t \geq -\left(\frac{1+r}{r} \right) y_{\min}. \tag{2}$$

This is the loosest possible debt limit. No exogenous borrowing constraint can ever be looser than the natural debt limit. Note however that if $y_{\min} = 0$, then the natural debt limit is zero!

Inada conditions and natural borrowing limit– Keep in mind an important property: if the utility function satisfies the Inada condition $\lim_{c \rightarrow 0} u'(c) = +\infty$, then the consumer will never want to borrow up to the natural debt limit.³ Suppose she has borrowed up to $a_t = -\left(\frac{1+r}{r} \right) y_{\min}$ and suppose the income realization y_t is precisely $y_{\min}$

³For example, CRRA utility $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ with $\gamma \geq 1$ satisfies the Inada condition.

which has positive probability. From the budget constraint at date t :

$$\begin{aligned}
c_t &= a_t + y_{\min} - \frac{a_{t+1}}{1+r} = -\left(\frac{1+r}{r}\right)y_{\min} + y_{\min} - \frac{a_{t+1}}{1+r} \\
&= -\frac{1}{r}y_{\min} - \frac{a_{t+1}}{1+r} \leq -\frac{1}{r}y_{\min} - \frac{1}{1+r} \underbrace{\left[-\left(\frac{1+r}{r}\right)y_{\min}\right]}_{\text{max that can be borrowed}} \\
&= 0
\end{aligned}$$

which shows that, with positive probability the consumer will have to consume zero. However that would lead to an infinitely negative utility, and so the consumer will never to reach that state.

The preferences alone will insure that the *natural* borrowing limit will never bind. In other words, in solving for the optimal consumption you can safely assume interior solutions for the Euler equation. This is not true for ad-hoc debt limits!

1.3 Saving motives

This is a good time to make a short remark about “saving motives”. The saving motive of the pure PIH where utility is quadratic (hence uncertainty has no role) and $\beta(1+r) = 1$ is called *smoothing motive*. The individual wants to smooth consumption as he navigates through income shocks. As explained above, the saving motive associated to future income uncertainty is called *precautionary or self-insurance motive*. If we also assume that $\beta(1+r) \neq 1$, this pushes the individual to either postpone or bring forward consumption because of patience and/or returns to savings. This saving motive is called *intertemporal motive*. We add that in a life-cycle model where the individual faces a retirement period, during the working stage of the life-cycle the individual would have a *life-cycle motive* for saving associated to the desire of smoothing consumption between working life and retirement. In presence of altruism towards their offsprings, we have an additional saving motive aimed at leaving behind some assets as *bequest*.

2 The Income Fluctuation Problem

We are now ready to formally examine the general problem. In what follows, we will adopt the timing convention that interest income is paid in the beginning of the period,

before the consumption-saving decision is made.⁴ (See the Appendix for more details on the timing convention).

2.1 Formulations of The Income Fluctuation Problem

Below we state three equivalent formulations of the income fluctuation problem.

The sequential formulation– The consumption-saving problem of an agent is

$$\begin{aligned} \max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \quad & E_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & \\ c_t + a_{t+1} = & Ra_t + y_t \\ a_{t+1} \geq & 0 \end{aligned}$$

where we combine general preferences (i.e., we just impose $u' > 0, u'' < 0$) with a no-borrowing constraint. The value for the interest rate is *exogenously given* and constant, $R \equiv (1 + r)$.⁵

The first order condition for optimality is

$$u'(c_t) = \beta RE_t[u'(c_{t+1})] + \lambda_t, \quad (3)$$

where $\lambda_t > 0$ is the multiplier on the no-borrowing constraint. Condition (3) implies the Euler equation

$$u'(c_t) \geq \beta RE_t[u'(c_{t+1})]. \quad (4)$$

The recursive formulation– Written recursively, the consumption-saving problem of an agent is

$$\begin{aligned} V_t(a_t, y_t) = \max_{\{c_t, a_{t+1}\}} \quad & u(c_t) + \beta E[V_{t+1}(a_{t+1}, y_{t+1})] \\ \text{s.t.} \quad & \\ c_t + a_{t+1} = & Ra_t + y_t \\ a_{t+1} \geq & 0 \end{aligned}$$

⁴All subsequent results are robust to the timing convention we used in the previous sections, where interest was paid at the end of the period, after the consumption-saving decision was made. It is a matter of convenience which one to choose. The timing convention that interest income is paid in the beginning of the period is slightly more widely used in practice.

⁵Later in the course, we will endogenize the interest rate

The Lagrangian is:

$$\mathcal{L} = \max_{\{c_t, a_{t+1}\}} u(c_t) + \beta E[V_{t+1}(a_{t+1}, y_{t+1})] - \lambda_t(c_t + a_{t+1} - y_t - Ra_t) + \mu_t a_{t+1}$$

From the FOC's with respect to c_t and a_{t+1} , we obtain:

$$u'(c_t) = \beta E \left[\frac{\partial V_{t+1}(a_{t+1}, y_{t+1})}{\partial a_{t+1}} \right] + \mu_t. \quad (5)$$

The envelope condition is:

$$\begin{aligned} \frac{\partial V_t(a_t, y_t)}{\partial a_t} &= u'(c_t) \left(R - \frac{\partial a_{t+1}}{\partial a_t} \right) + \beta E \left[\frac{\partial V_{t+1}(a_{t+1}, y_{t+1})}{\partial a_{t+1}} \frac{\partial a_{t+1}}{\partial a_t} \right] \\ &= u'(c_t) \end{aligned}$$

The envelope condition implies $\frac{\partial V_t(a_t, y_t)}{\partial a_t} = Ru'(c_t)$. Updating forward and plugging into (5) gives the usual Euler equation

$$u'(c_t) \geq \beta RE[u'(c_{t+1})].$$

The cash-on-hand formulation— Note that when the income shocks $\{y_t\}$ are *iid*, or when income is deterministic, we can define a unique state variable which is a sufficient statistics for the household choice, “cash-on-hand” $x_t \equiv Ra_t + y_t$ since (a_t, y_t) always enter additively and current levels of y_t do not provide any information about the future realizations of income shocks.

In this case, the household budget constraint becomes:

$$c_t + a_{t+1} = x_t$$

and the new state variable, cash-on-hand, evolves according to the following law of motion:

$$x_{t+1} = Ra_{t+1} + y_{t+1} = R(x_t - c_t) + y_{t+1},$$

The utility maximization problem can then be formulated as

$$\begin{aligned} V_t(x_t) &= \max_{\{c_t, a_{t+1}\}} u(c_t) + \beta E[V_{t+1}(x_{t+1})] \\ &\quad s.t. \\ c_t + a_{t+1} &= x_t \\ x_{t+1} &= Ra_{t+1} + y_{t+1} \\ a_{t+1} &\geq 0 \end{aligned}$$

The Langrangian is

$$\mathcal{L} = \max_{a_{t+1}} u(x_t - a_{t+1}) + \beta EV_{t+1}(Ra_{t+1} + y_{t+1}) + \mu_t a_{t+1}$$

From the FOC's with respect to a_{t+1} , we obtain

$$u'(c_t) = \beta RE[V'_{t+1}(x_{t+1})] + \mu_t.$$

The envelope condition implies $\frac{\partial V_t(x_t)}{\partial x_t} = u'(c_t)$.

An alternative and even more compact way of formulating the problem with cash-on-hand as state variable is to let s_t denote the fraction of cash-on-hand saved. Then $c_t = (1 - s_t)x_t$ and $a_{t+1} = s_t x_t$. We can then cast the problem in term of choosing s_t as follows

$$V_t(x_t) = \max_{\{s_t\}} u((1 - s_t)x_t) + \beta E[V_{t+1}(Rs_t x_t + y_{t+1})]$$

$s.t.$

$$s_t \geq 0 \quad \text{this guarantees that the borrowing constraint is satisfied}$$

$$s_t \leq 1 \quad \text{this guarantees that consumption is non-negative}$$

2.2 Wealth and Consumption Dynamics

In what follows, we want to further understand the dynamics of individual wealth and consumption in the income fluctuation problem. In particular, we want to understand what are the conditions under which the endogenous state space for assets $[0, \bar{a}]$ is compact, i.e. there exists an upper bound for assets $\bar{a}$ which is finite. This is a crucial requirement for the existence of an equilibrium in economies populated by many agents who choose optimal consumption and savings by solving income fluctuation problems. If each agent has a tendency to accumulate assets such that their wealth sequence a_{t+1} diverges in the long-run, then in such an economy there will be an infinite supply of assets which prevents existence of an equilibrium.

The convergence properties of the consumption and wealth sequences will depend on the value of the term βR . We always examine three separate cases: βR above, equal to or below one.

We start from a problem where income fluctuations are deterministic, i.e., perfectly foreseen. Next we move to stochastic income fluctuations. We will focus throughout on the case in which at any point in time net present discounted value of income from that point on is finite i.e.

$$\sum_{j=0}^{\infty} \frac{1}{R^j} y_{t+j} < \infty.$$

A special case which obviously satisfies this assumption is a constant income y .

2.2.1 Deterministic Income Fluctuations

Case $\beta R > 1$: Without uncertainty, since u is strictly concave, the Euler equation (4) implies

$$u'(c_t) \geq \beta R u'(c_{t+1}) > u'(c_{t+1}) \Rightarrow c_{t+1} > c_t,$$

thus consumption grows indefinitely.⁶ Since borrowing is limited at zero, assets must grow to finance consumption, so also assets diverge. The individual is “too patient” or/and the rate of return on savings is too high. Both forces push her to accumulate too much wealth.

To see this mathematically, note first that the budget constraint can be written as

$$\sum_{j=0}^{\infty} \frac{1}{R^j} c_{t+j} \leq R a_t + \sum_{j=0}^{\infty} \frac{1}{R^j} y_{t+j}. \quad (6)$$

Since $c_t \rightarrow \infty$, then $\sum_{j=0}^{\infty} \frac{1}{R^j} c_{t+j} \rightarrow \infty$, but $\sum_{j=0}^{\infty} \frac{1}{R^j} y_{t+j} < \infty$, and so it must be that $a_t \rightarrow \infty$.

Case $\beta R = 1$: From the Euler Equation, $u'(c_t) \geq u'(c_{t+1})$. Ideally, households want perfectly smooth consumption ($c_{t+1} = c_t$) when the liquidity constraint is not binding. If it is binding, then $c_{t+1} > c_t$. Overall, consumption is a *nondecreasing* sequence. One can prove that the existence of the borrowing constraint affects the problem only until a given time τ and vanishes thereafter. Until $t = \tau$, consumption will grow and then it remains constant thereafter. From (6) it follows that the asset sequence is also bounded.

How is τ determined? Define

$$h_t = \frac{r}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j y_{t+j}$$

⁶Suppose the limit of consumption is finite, i.e. $c_t \rightarrow c^*$. Then, as $t \rightarrow \infty$, the Euler equation is $u'(c^*) \geq \beta R u'(c^*) \Leftrightarrow \beta R \leq 1$, which is a contradiction.

as the annuity value of the of human wealth (i.e., the discounted present value of future income). Then τ satisfies

$$\tau = \arg \sup_t h_t.$$

See LS (16.3.1) for a formal proof of this result.

So, consumption and assets converge to a finite value. To gain some intuition, consider a sequence for y_t that increases until time t^* and then is decreasing from there on. It is easy to see that $\tau < t^*$. Households would like to borrow initially against future higher income, but they cannot so the constraint will bind for a while and they will keep increasing consumption until τ and thereafter they will follow a constant consumption path where they will save at the beginning to finance future consumption when income is low.

Clearly, there may be some paths for $\{y_t\}$ such that h_t is always increasing, in which case the individual will always be constrained (or, $\tau \rightarrow \infty$). However, assuming that the sequence $\{y_t\}$ is bounded above (reasonable, since y_t is individual income) is enough to guarantee that h_t has a maximum for finite τ .

Case $\beta R < 1$: It is fairly intuitive that in this case the consumption and asset sequences will converge as consumers value early consumption more than late consumption. To see this more precisely, consider the simple case where the endowment sequence is constant at y . Let's consider the formulation with cash-on-hand as a state variable, i.e., $x \equiv Ra + y$. Then:

$$\begin{aligned} V(x) &= \max_{c, a'} \{u(c) + \beta V(x')\} \\ &\quad s.t. \\ c + a' &= x \\ x' &= Ra' + y \\ a' &\geq 0 \end{aligned}$$

If at time t the liquidity constraint is binding, then next period $a_{t+1} \equiv a' = 0$, and $c_{t+1} \equiv c' = y$. Since y is constant, this situation perpetuates and c is constant forever: the individual is always constrained. In this case the consumption sequence is bounded and, trivially, so are assets.

The interesting case is when the liquidity constraint is not binding and wealth is

positive. In that case, the optimal choice of consumption is dictated by the FOC⁷

$$u_c(c) = \beta R V_x(x').$$

Using the envelope condition $u_c(c) = V_x(x)$, the FOC for consumption can be equivalently rewritten as

$$V_x(x) = \beta R V_x(x') < V_x(x') \implies x' < x,$$

which shows that as long as the borrowing constraint is not binding, then cash-on-hand decreases over time, i.e. if $a'(x) > 0$, then $x' < x$.⁸ To see what that implies for consumption, differentiate the envelope condition wrt to x to obtain

$$u_{cc}(c) \frac{dc}{dx} = V_{xx}(x) \implies \frac{dc}{dx} = \frac{V_{xx}(x)}{u_{cc}(c)} > 0. \quad (7)$$

Therefore, consumption is increasing in cash-in-hand, under concavity of V , so if we start from a positive level of assets a_0 , cash-on-hand will fall over time as assets are optimally decumulated and consumption will decline because of (7). Therefore, the consumption sequence is bounded from above, and from (6) so are assets.

Finally, we can also show that when x reaches y from above, then $a'(y) = 0$ and $c(y) = y$, i.e. in the limit as assets get depleted and reach zero (and cash-on-hand equals y), the consumption sequence converges to the constant endowment stream.⁹ We prove it *by contradiction*. Suppose that at the point where $x = y$, $a'(y) > 0$. Then, the FOC holds with equality and

$$\begin{aligned} u_c(c) &= \beta R V_x(x') \\ V_x(y) &= \beta R V_x(Ra' + y) < V_x(Ra' + y) < V_x(y), \end{aligned}$$

where the second line uses the envelope condition, the fact that $\beta R < 1$ and the strict concavity of the value function. The second line contains the contradiction. Thus, when $\beta R < 1$ consumption converges to a finite value.

It is not difficult to generalize this result, and show that in this case the consumption sequence will converge even if the y_t sequence is time-varying.

⁷We use $f_x(x)$ instead of $f'(x)$ to denote the derivative of $f(x)$ wrt x , in order to avoid confusion with the timing convention of denoting by x' the value of x_{t+1} .

⁸Here we are assuming the concavity of V but we know that under certain conditions the Bellman operator preserves strict concavity of u .

⁹You should try to prove that the strictly decreasing cash-on-hand sequence will reach y .

Taking stock- How can we take stock of these results? In the deterministic case, the desire to save is increasing in patience (β) and the interest rate (R). When βR is too large, households' desire to accumulate is "too strong" and both assets and consumption diverge to infinity. To prevent divergence, we need $\beta R \leq 1$.

2.2.2 Stochastic Income Fluctuations

We now turn to the stochastic case. Here there is an additional motive for saving, the *precautionary motive*, due to 1) prudence (for some utility functions) and 2) the interaction between risk-aversion (i.e., aversion to consumption fluctuations) and the borrowing constraint. Given the extra saving motive, we should expect that the conditions under which wealth and consumption are bounded will be more stringent than in the deterministic case. It turns out that we will need $\beta R < 1$.

A useful supermartingale- Multiply both sides of

$$u'(c_t) \geq \beta R E_t[u'(c_{t+1})]$$

by $(\beta R)^t$ and define $M_t \equiv (\beta R)^t u'(c_t) > 0$. Then equation (4) can be written as

$$M_t \geq E_t M_{t+1}$$

which asserts that M_t follows a *supermartingale*.

There is one essential theorem that is extremely useful for our purpose: the supermartingale convergence theorem (Doob, 1995).¹⁰ Since M_t is non-negative, by the supermartingale convergence theorem (Doob, 1995), this stochastic process converges almost surely to a non-negative finite limit $\bar{M}$, i.e.,

$$\lim_{t \rightarrow \infty} M_t = \bar{M} < \infty. \quad (8)$$

Case $\beta R > 1$: According to the convergence theorem above, $(\beta R)^t u'(c_t)$ has a finite limit. Since $(\beta R)^t \rightarrow \infty$, then marginal utility $u'(c_t)$ can only converge to $u'(\bar{c}) = 0$ or, given the Inada condition, $c_t \rightarrow \infty$. Since debt is limited (at zero, in our benchmark, but could be any finite limit), this large consumption cannot be financed by debt but must be financed by saving. And hence divergence of consumption means $a_t \rightarrow \infty$, which in

¹⁰Let $\{x_t\}$ be a non-negative supermartingale. Then there is a random variable x with $E|x| < \infty$ such that $\lim_{t \rightarrow \infty} x_t = x$ almost surely.

turn implies that there is no upper bound in the asset space. This is the same result we found for the certainty case.

Case $\beta R = 1$: We can give a simple proof of this result for general income process if we assume that $u''' > 0$. From the Euler Equation

$$u'(c_t) \geq E_t[u'(c_{t+1})].$$

From convexity of the marginal utility, by Jensen's inequality

$$u'(c_t) \geq E_t[u'(c_{t+1})] > u'(E_t(c_{t+1})),$$

and by concavity of u , we have that $E_t(c_{t+1}) > c_t$, so consumption will always tend to ratchet upward over time which prevents the consumption sequence from converging almost surely to a finite number. The reason for this is exactly the same reason that generates precautionary saving, which means that risk acts as an increase in the subjective discount factor β .

In the case $\beta R = 1$, the most general result is the one provided by Chamberlain and Wilson (2000) who show, using a bounded utility function, that if the process for income is sufficiently volatile consumption and assets will diverge.¹¹ Finally, note that the convergence result with $\beta R = 1$ of the deterministic case does not hold in the stochastic case.

Case $\beta R < 1$: This case is ambiguous. Under additional restriction on utility and on the income process we can prove that the assets are bounded but one cannot prove it in general. Consider the case of iid income shocks and, as before, let x be cash-on-hand. From the Euler Equation:

$$u_c(c(x)) = \beta R E[u_c(c(x'))],$$

where $x' = Ra'(x) + y'$ is cash-on-hand next period given that today's cash in hand is x . Let $\bar{x}' = Ra'(x) + \bar{y}$ be the cash-on-hand associated to the maximum realization of income next period, given that today's cash in hand is x . Then the Euler equation can be written as

$$u_c(c(x)) = \beta R \frac{E[u_c(c(x'))]}{u_c(c(\bar{x}'))} u_c(c(\bar{x}')). \quad (9)$$

¹¹See LS (16.5) for a discussion of this result.

Suppose that the limit

$$\lim_{x \rightarrow \infty} \frac{E[u_c(c(x'))]}{u_c(c(\bar{x}'))} = 1. \quad (10)$$

Then, for x large enough, since $\beta R < 1$, the Euler equation (9) yields

$$u_c(c(x)) = \beta R u_c(c(\bar{x}')) < u_c(c(\bar{x}')).$$

Concavity of u and the fact that $c_x(x) > 0$ (proved in the deterministic case earlier) implies that

$$c(\bar{x}') < c(x) \Rightarrow \bar{x}'(x) < x,$$

and we would be done, because we have demonstrated that cash-on-hand does not increase forever: when x is large enough $x' < x$ for sure.

Therefore, we only need to establish conditions under which the limit in (10) holds. Consider $u_c(c(x'))$ and compute a first-order Taylor approximation around $x' = \bar{x}'$:

$$u_c(c(x')) \simeq u_c(c(\bar{x}')) + u_{cc}(c(\bar{x}')) c_x(\bar{x}') (x' - \bar{x}').$$

Taking expectations of both sides

$$\begin{aligned} E[u_c(c(x'))] &\simeq u_c(c(\bar{x}')) - u_{cc}(c(\bar{x}')) c_x(\bar{x}') E[\bar{x}' - x'] \\ &= u_c(c(\bar{x}')) - u_{cc}(c(\bar{x}')) c_x(\bar{x}') E[\bar{y} - y'] \end{aligned} \quad (11)$$

where in the first line we use the fact that $\bar{x}'$ is deterministic since it is implied by the specific income realization $\bar{y}$. And in the second line we use the fact that $x' \equiv Ra' + y'$ which implies $\bar{x}' - x' \equiv \bar{y} - y'$.

Dividing equation (11) by $u_c(c(\bar{x}'))$ we obtain:

$$\frac{E[u_c(c(x'))]}{u_c(c(\bar{x}'))} \simeq 1 + \alpha(c(\bar{x}')) c_x(\bar{x}') [\bar{y} - E(y')],$$

where $\alpha(c)$ is the coefficient of absolute risk aversion at consumption level $c(\bar{x}')$. Since both $[\bar{y} - E(y')]$ and $c_x(\bar{x}')$ are positive and finite,¹² a sufficient condition for the limit in (10) to hold is

$$\lim_{x \rightarrow \infty} \alpha(c(x)) = 0. \quad (12)$$

If, for example, *absolute risk aversion is monotonically decreasing with asset holdings* then condition (12) holds. The faster α decreases, the smaller the upper bound on the asset

¹²See Carroll and Kimball (1996) for results on the concavity of the consumption function.

space. The intuition is clear: DARA means that the agent is less worried about income uncertainty as she gets rich because she becomes less risk averse, so she will consume more and accumulate less. This force offsets, and eventually overcomes, precautionary accumulation as wealth increases.

Note that CRRA utility has DARA, so it satisfies condition (12). Remember that DARA is a sufficient condition and that this result holds for iid shocks. Huggett (1993) generalizes this result to a 2-state Markov chain for the income process (but only for the CRRA utility case).

We conclude by summarizing our findings in:

Result 2.5: In presence of borrowing constraints and uncertain income, the condition $\beta R < 1$ is necessary for the optimal consumption sequence and for the asset space to be bounded. Moreover, if income shocks are iid and absolute risk-aversion is decreasing (DARA utility), then $\beta R < 1$, is also a sufficient condition for the asset space to be bounded. More general, even with Markov shocks, as long as absolute risk aversion decreases fast enough with c , the state space will remain bounded.

2.2.3 Notes

Leland (1968) and Sandmo (1970) showed in a two-period setting that a positive third derivative is needed to obtain precautionary savings. Sibley (1975) extended the result to a multi-period setting. Kimball (1990) further generalized these results. Schechtman and Escudero (1977) characterized the income fluctuation problem with *iid* shocks. Huggett (1993) generalizes their proofs to the case of a 2-state Markov chain for the income process and CRRA utility. Chamberlain and Wilson (2000) extend it to any arbitrary endowment sequence that is sufficiently stochastic.

References

- [1] Carroll, Christopher and Kimball, Miles (1996); On the Concavity of the Consumption Function, *Econometrica*
- [2] Chamberlain, Gary and Charles Wilson (2000); Optimal Intertemporal Consumption under Uncertainty, *Review of Economic Dynamics*

- [3] Huggett, Mark (1993); The risk-free rate in heterogeneous-agent incomplete-insurance economies, *Journal of Economic Dynamics and Control*
- [4] Kimball, Miles (1990); Precautionary Saving in the Small and in the Large, *Econometrica*
- [5] Leland, H. (1968); Savings and Uncertainty: The Precautionary Demand for Saving, *Quarterly Journal of Economics*
- [6] Sandmo Agnar (1970); The Effect of Uncertainty on Saving Decisions, *Review of Economic Studies*.
- [7] Schechtman J. and V. Escudero (1977); Some Results on An Income Fluctuation Problem, *Journal of Economic Theory*
- [8] Sibley David (1975); Permanent and Transitory Income Effects in a Model of Optimal Consumption with Wage Income Uncertainty, *Journal of Economic Theory*

Appendices

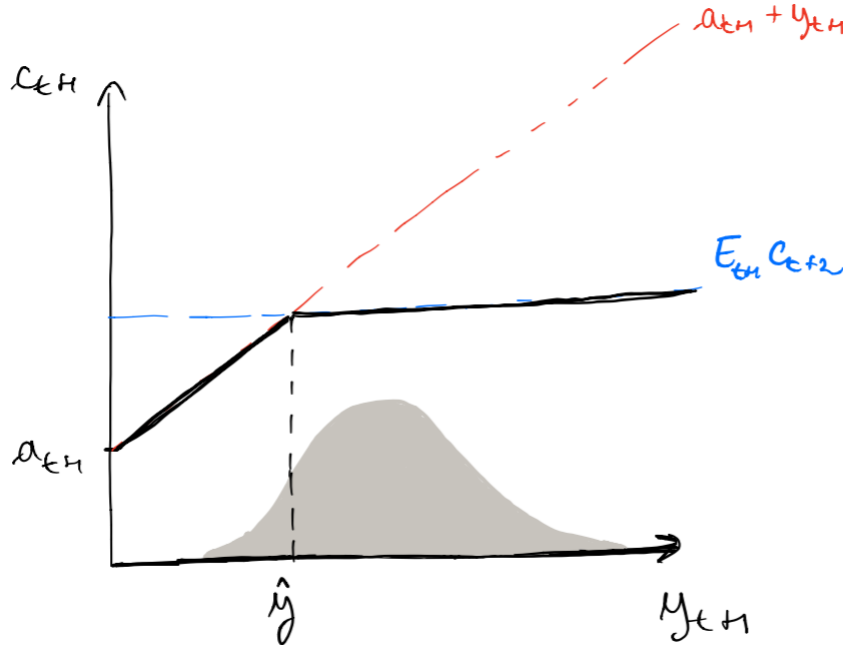
A Precautionary savings and borrowing constraints

Recall that consumption is:

$$c_t = \min \{y_t + a_t, E_t c_{t+1}\} = \min \{y_t + a_t, E_t [\min \{y_{t+1} + a_{t+1}, E_{t+1} c_{t+2}\}]\}.$$

We want to study the effect of a mean-preserving spread of y_{t+1} on c_t . To that end, we start backwards by first studying the choice of c_{t+1} . The black line in Figure 2 depicts the consumption choice at $t + 1$ as a function of y_{t+1} for an agent that enters the period with assets a_{t+1} . If the realization of y_{t+1} is below a threshold $\hat{y}$ (endogenously determined), then the agent is constrained and sets $c_{t+1} = a_{t+1} + y_{t+1}$. If instead the realization of y_{t+1} is above the threshold $\hat{y}$, the agent is on the Euler equation and chooses $c_{t+1} = E_{t+1} c_{t+2}$. The shaded area superimposed in the figure is the distribution of y_{t+1} .

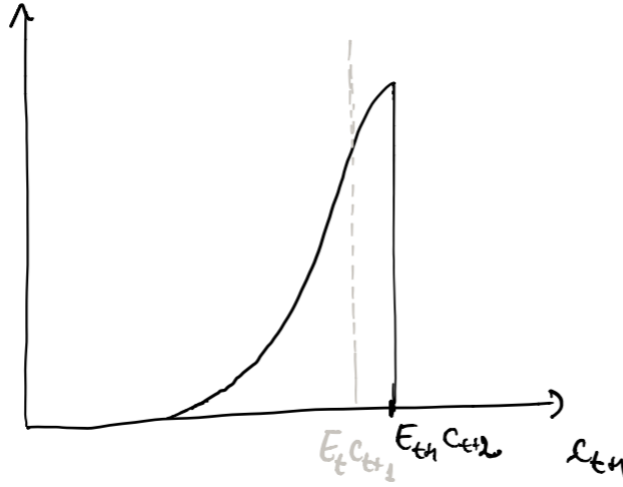
Figure 2: Consumption plan



To visualize the implications of the distribution of income for $E_t c_{t+1}$, Figure 3 depicts the implied distribution of c_{t+1} . At $t + 1$ consumption can be at most $E_{t+1} c_{t+2}$ and the probability of that happening is equal to the probability that $y + 1 > \hat{y}$. Otherwise,

consumption is $a_{t+1} + y_{t+1}$, and is lower the lower the realization of y_{t+1} . Note that the probability of these lower consumption realizations is a reflection of the probability of lower realizations of y_{t+1} . The gray vertical line denotes the value of $E_t c_{t+1}$ calculated using this distribution. Note that this is equal to c_t for a household who is not constrained in period t .

Figure 3: Distribution of c_{t+1}



Consider now a mean preserving spread of y_{t+1} , depicted in orange in Figure 4. Note that Figure 4 is just Figure 2 with another distribution superimposed. In this case, the probability that y_{t+1} is below the threshold $\hat{y}$, and therefore the probability that the agent is constrained, is higher.

This is reflected in Figure 5, which plots the new distribution of c_{t+1} . As lower values of c_{t+1} are more likely, $E_t c_{t+1}$ is lower and, therefore, so is c_t for a household that is not constrained in period t .

Figure 4: Consumption plan, mean preserving spread

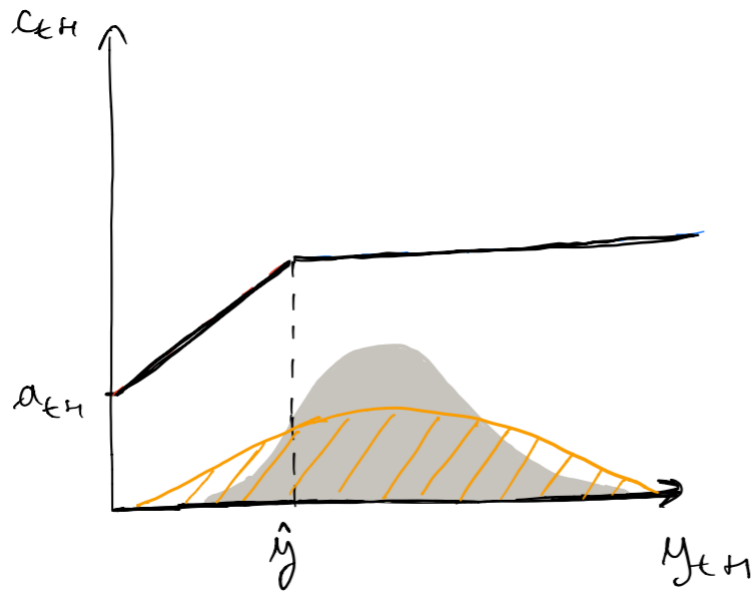
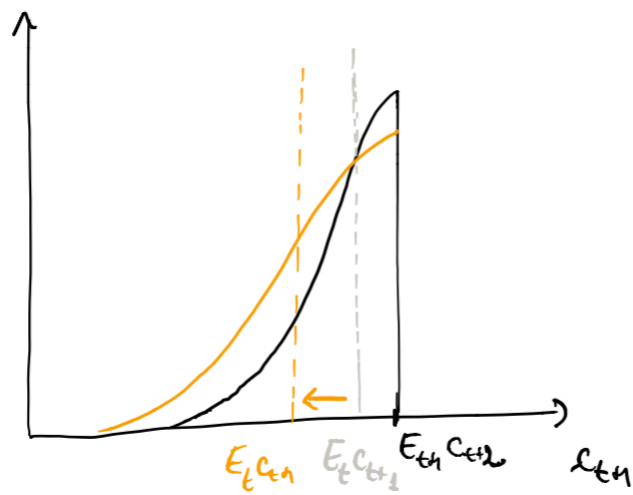


Figure 5: Distribution of c_{t+1} , mean preserving spread



B Note on timing

Our baseline model is a bond economy. Recall that a bond pays the principal (in our notation a_t) and a coupon (in our notation the interest r).

We began with the following budget constraint:

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t.$$

This formulation implicitly assumes that the coupon of the bonds purchased at time t (i.e. a_{t+1}) is paid in the current period t and the principal is paid in the next period $t+1$. Alternatively, you can think of a bond in this arrangement as having a price $q < 1$ at the time of purchase in period t and paying a non-state contingent return of 1 at $t+1$.

The budget constraint we will use for the rest of the class is:

$$c_t + a_{t+1} = y_t + (1+r)a_t.$$

This formulation implicitly assumes that both the coupon and the principal of the bonds purchased at time t (i.e. a_{t+1}) are paid in the next period $t+1$. Alternatively, you can think of a bond in this arrangement as having a price of 1 when purchased in period t and paying a non-state contingent return r in period $t+1$.

Numerical Methods

As we progress in our study of heterogeneous agents economies, it will become increasingly difficult to characterize equilibria and solutions analytically. In these notes, we cover a number of useful numerical methods that people use to compute equilibria in such economies.

1 Discretization of a continuous stochastic process

In many of the problems we studied in the previous chapter, we postulated that agents face a continuous stochastic income process. A typical example would be assuming that income is given by

$$y_t = \exp(\hat{y}_t),$$

where $\hat{y}_t$ follows a first-order autoregressive process of the class

$$\hat{y}_t = \rho \hat{y}_{t-1} + \varepsilon_t, \tag{1}$$

where $|\rho| < 1$ and ε_t is a white noise process with variance σ_ε^2 . This $AR(1)$ process is covariance-stationary with mean zero and variance $\sigma_{\hat{y}}^2 = \sigma_\varepsilon^2 / (1 - \rho^2)$.

In what follows, we will learn how to approximate discretely a continuous process through a finite-state Markov chain that will mimic closely the underlying process. To discretize the continuous process in (1) we need two ingredients: (i) the points on the finite state space and (ii) the transition probabilities.

1.1 Tauchen's method

Assume that ε_t is drawn from a distribution G . Let $\Pr\{\varepsilon_t \leq \bar{\varepsilon}\} = G(\bar{\varepsilon}) = F(\bar{\varepsilon}/\sigma_\varepsilon)$, where F is the “standardized” version of $G(\varepsilon_t)$ with unit variance. Let $\tilde{y}$ be the discrete-valued process that approximates $\hat{y}$ and let $\{\bar{y}_1, \bar{y}_2, \dots, \bar{y}_N\}$ be the finite set of possible realizations of $\tilde{y}$.

Tauchen (1986) suggests to select values $\bar{y}_j$ so that $\bar{y}_N$ is a multiple m (e.g., $m = 3$) of the unconditional standard deviation, i.e.

$$\bar{y}_N = m \left(\frac{\sigma_\varepsilon^2}{1 - \rho^2} \right)^{\frac{1}{2}},$$

and let $\bar{y}_1 = -\bar{y}_N$ (assuming G is symmetric), and $\{\bar{y}_2, \bar{y}_3, \dots, \bar{y}_{N-1}\}$ be located in a equispaced manner over the interval $[\bar{y}_1, \bar{y}_N]$. For example, if $\bar{y}_N = 3 \left(\frac{\sigma_\varepsilon^2}{1-\rho^2} \right)^{\frac{1}{2}}$, we will choose $\bar{y}_1 = -3 \left(\frac{\sigma_\varepsilon^2}{1-\rho^2} \right)^{\frac{1}{2}}$, and set grid points $\bar{y}_k$, $1 < k < N$ according to

$$\bar{y}_k = \bar{y}_1 + (k-1) \frac{\bar{y}_N - \bar{y}_1}{N-1},$$

where $\frac{\bar{y}_N - \bar{y}_1}{N-1}$ is the distance between successive points in the state space. Denote this distance with d .

Once we set the grid, we need to compute transition probabilities. Let

$$\begin{aligned} \pi_{jk} &= \Pr \{ \tilde{y}_t = \bar{y}_k | \tilde{y}_{t-1} = \bar{y}_j \} = \Pr \{ \bar{y}_k - d/2 < \rho \bar{y}_j + \varepsilon_t \leq \bar{y}_k + d/2 \} = \\ &\Pr \{ \bar{y}_k - d/2 - \rho \bar{y}_j < \varepsilon_t \leq \bar{y}_k + d/2 - \rho \bar{y}_j \} \end{aligned}$$

be the generic transition probability. Then, if $1 < k < N-1$, for each j choose

$$\pi_{jk} = F \left(\frac{\bar{y}_k + d/2 - \rho \bar{y}_j}{\sigma_\varepsilon} \right) - F \left(\frac{\bar{y}_k - d/2 - \rho \bar{y}_j}{\sigma_\varepsilon} \right),$$

while for the boundaries of the interval $k=1$ and $k=N$ choose:

$$\begin{aligned} \pi_{j1} &= F \left(\frac{\bar{y}_1 + d/2 - \rho \bar{y}_j}{\sigma_\varepsilon} \right), \\ \pi_{jN} &= 1 - F \left(\frac{\bar{y}_N - d/2 - \rho \bar{y}_j}{\sigma_\varepsilon} \right). \end{aligned}$$

Clearly, as $d \rightarrow 0$ (and therefore $N \rightarrow \infty$), the approximation becomes better and better until it converges to the true continuous process y_t . It is useful to notice that other integration rules (e.g., Gaussian quadrature) could lead to a more efficient placement of the points on the the interval $[\bar{y}_1, \bar{y}_N]$.

Accuracy of approximation: To assess the adequacy of the approximation, one would like to check if the first and second moments of the approximate process are close to that of the original one. To do so, one can simulate a long sample for $\{\tilde{y}_t\}$ (for example a sample of 10,000 draws with a burn-in period of 1,000 draws to avoid initial conditions to affect the outcome), compute some key statistics and compare them with those in the original process.

Another method is to note that the approximate process $\tilde{y}$ admits a representation

$$\tilde{y}_t = \tilde{\rho} \tilde{y}_{t-1} + \tilde{\varepsilon}_t,$$

and compute the first and second moments algebraically using that $\tilde{\rho} = \text{cov}(\tilde{y}_{t-1}, \tilde{y}_t) / \text{var}(\tilde{y}_t)$ and $\tilde{\sigma}_\varepsilon = (1 - \tilde{\rho}^2) \text{var}(\tilde{y}_t)$. To do so, it is useful to compute the stationary distribution of $\tilde{y}_t$, which can be done as follows. Let Θ_N be the transition matrix of dimension $(N \times N)$ with generic element π_{jk} . Then, the stationary distribution $\pi^* = \{\pi_1^*, \pi_2^*, \dots, \pi_N^*\}$ solves

$$\pi^* = \pi^* \Theta_N \rightarrow \pi^* (I - \Theta_N) = 0$$

which is a linear system of N equations. Using this method, one can directly compare ρ to $\tilde{\rho}$ and σ_ε to $\tilde{\sigma}_\varepsilon$ and gauge the quality of the approximation.

The easiest way to compute π^* is to iterate on

$$\pi_{t+1} = \pi_t \Theta_N$$

This is done by starting at some initial distribution π_0 and iterating until the π s don't change much.

Multivariate processes: Tauchen describes how to approximate also a multivariate process. This strategy that can be used to approximate higher-order autoregressive processes as well. For example, consider the $AR(2)$ process

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \varepsilon_t.$$

Define a column vector $Z_t = [y_t \ y_{t-1}]'$. Then one can write the $AR(2)$ process above in multivariate form as

$$Z_t = \begin{bmatrix} \rho_1 & \rho_2 \\ 1 & 0 \end{bmatrix} Z_{t-1} + [\varepsilon_t \ 0]'$$

1.2 Rouwenhorst method

This is the best method to discretize a continuous stochastic process, in particular those with very high persistence, typical in macro (Kopecky and Suen, 2010). It is accurate and fast. Consider again the $AR(1)$ process above, but this time we do not need to make any distributional assumption. We want to approximate it through a discrete-space Markov chain over a symmetric and evenly-spaced state space $\{\bar{y}_1, \bar{y}_2, \dots, \bar{y}_N\}$ with $-\bar{y}_1 = \bar{y}_N = \psi$. Construct the Markov chain Θ_N recursively as a function of three parameters only (p, q, ψ) as follows:

- **Step 1** For $N = 2$, define Θ_2 as

$$\Theta_2 = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$$

Step 2 For $N \geq 3$, define recursively Θ_N as the $N \times N$ matrix

$$\Theta_N = p \begin{bmatrix} \Theta_{N-1} & \mathbf{0} \\ \mathbf{0}' & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & \Theta_{N-1} \\ 0 & \mathbf{0}' \end{bmatrix} + (1-q) \begin{bmatrix} \mathbf{0}' & 0 \\ \Theta_{N-1} & \mathbf{0} \end{bmatrix} + q \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & \Theta_{N-1} \end{bmatrix},$$

where $\mathbf{0}$ is a $N-1$ column vector and $\mathbf{0}'$ is a $N-1$ row vector.

For example, for $N = 3$

$$\begin{aligned} \Theta_3 = & p \begin{bmatrix} p & 1-p & 0 \\ 1-q & q & 0 \\ 0 & 0 & 0 \end{bmatrix} + (1-p) \begin{bmatrix} 0 & p & 1-p \\ 0 & 1-q & q \\ 0 & 0 & 0 \end{bmatrix} \\ & + (1-q) \begin{bmatrix} 0 & 0 & 0 \\ p & 1-p & 0 \\ 1-q & q & 0 \end{bmatrix} + q \begin{bmatrix} 0 & 0 & 0 \\ 0 & p & 1-p \\ 0 & 1-q & q \end{bmatrix} \end{aligned}$$

Note that the elements in the first and last row sum up to 1, while the elements in the second row sum up to 2. This motivates Step 3 below.

Step 3 Divide all but the top and bottom rows by 2 so that the elements in each row sum up to 1.

One can show that this Markov chain converges to the invariant binomial distribution

$$P(i-1; N-1, s) = \binom{N-1}{i-1} s^{i-1} (1-s)^{N-i}$$

where

$$s = \frac{1-p}{2-(p+q)}.$$

Recall that a *binomial distribution* gives the discrete probability distribution of obtaining exactly $i-1$ successes out of $N-1$ Bernoulli trials (where the result of each Bernoulli trial is true with probability s and false with probability $1-s$).

We can compute analytically all the moments of this distribution. For example

$$\begin{aligned} E(\tilde{y}_t) &= \frac{(q-p)\psi}{2-(p+q)} \\ E(\tilde{y}_t^2) &= \psi^2 \left[1 - 4s(1-s) + \frac{4s(1-s)}{N-1} \right] \\ \text{Corr}(\tilde{y}_t, \tilde{y}_{t+1}) &= p+q-1 \end{aligned}$$

See Appendix C of Kopecky and Suen (2009) for the derivation of these and other moments.¹

For any continuous stochastic process we can also compute analytically a number of unconditional and conditional moments and we can use a method of moments estimator to choose the three parameters of the Markov chain (p, q, ψ) that minimize the distance between true and approximated selected moments.

For example, consider the $AR(1)$ process $\{\hat{y}_t\}$ above and note that

$$\begin{aligned} E(\hat{y}_t) &= 0 \rightarrow \frac{(q-p)\psi}{2-(p+q)} = 0 \rightarrow q = p \\ Corr(\hat{y}_t, \hat{y}_{t+1}) &= \rho \rightarrow p + q - 1 = \rho \rightarrow p = q = \frac{1+\rho}{2} \\ var(\hat{y}_t) &= \sigma_{\hat{y}}^2 \rightarrow \psi^2 \left[1 - 4s(1-s) + \frac{4s(1-s)}{N-1} \right] = \sigma_{\hat{y}}^2 \rightarrow \psi = \sigma_{\hat{y}} \sqrt{N-1}. \end{aligned}$$

Therefore, in this example, one can replicate exactly unconditional mean, variance, and first-order auto correlation of the original continuous process. Finally, since we know that the binomial converges to a Normal distribution for $N \rightarrow \infty$, this method is particularly apt at approximating log-normal (in levels) income processes, but it works with other distributions as well.

2 Global Solution Methods for the Income Fluctuation Problem

Up to now, the solution methods in macroeconomics that most of you have seen are probably value function iteration and policy function iteration. There are in fact many more ways to solve standard macroeconomic problems and it is good to have, at least, a vague idea of what they are.

First, it is common to distinguish between *local* and *global* methods. Local methods typically approximate some important object, like value functions, policy functions or equilibrium conditions, around a particular point using simpler functions like polynomials. In the *linear-quadratic approximation* case (LQ), one would approximate the objective function with a second order polynomial around some steady-state (see Kydland and Prescott, 1982). In the stochastic growth model, for example, a natural point would be to

¹Kopecky and Suen (2009) is the working paper version of Kopecky and Suen (RED, 2010).

use the deterministic steady-state level of capital k^* and the mean of the shock (usually normalized to 1). Fluctuations due to aggregate productivity shocks move the system in a small neighborhood of k^* , so we may hope to have a reasonably good approximation. Another widely used method is *linearization* or *log-linearization*, in which one would simply linearize the first-order conditions of the problem around its steady-state and solve a system of (possibly forward looking) linear equations. This is what most people use to solve New Keynesian DSGE models. *Perturbation methods* exploit this idea and push it further by looking at higher-order Taylor expansions of the policy functions and equilibrium conditions.

These solution methods may, however, lead to a number of issues, in particular when solving the consumption-saving problem with uncertainty. First, choosing a particular point on the asset grid where to approximate individual behavior is much less obvious. Second, quantitatively, individual income uncertainty is much larger than aggregate uncertainty, which means that the system moves over a large interval in the state space. Third, in LQ approximations the marginal utility is linear and the role of uncertainty as a motive for saving is artificially reduced (in fact, it completely disappears in absence of borrowing constraints).

For all these reasons, we need to resort to *global* solution methods for the consumption-saving problem. The method still relies heavily on value or policy function iteration, but we will use a global approximation of the value/policy function. One can choose among several alternative ways to approximate the policy function or the value function:

- *discretization* means that the state is assumed to take only values on a pre-specified grid
- *interpolation* is a procedure that finds a function that goes exactly through the grid points and defines the function outside the grid by interpolating locally between two adjacent grid points
- *L^p -approximation* is any procedure that finds a function close to the original one in the sense of a L^p norm, so it's a more global interpolation method.² A typical

²A distance between two functions evaluated over the N points $\{x_k\}_{k=1}^N$ in the L^p norm is

$$\|f(x) - \hat{f}(x)\|_p = \left(\sum_{k=1}^N |f(x_k) - \hat{f}(x_k)|^p \right)^{1/p}.$$

example would be to approximate the value function with Chebyshev or Hermite polynomials. This method is based on the Weierstrass approximation theorem, that states that every continuous function defined on a closed interval can be uniformly approximated as closely as desired by a polynomial function.

Solving the income fluctuation problem

Consider the problem in its recursive formulation, with states (a, y) . In general, with two state variables, we must approximate/interpolate a function in two dimensions, which is a hard problem.³ This is where the discretization of the income process comes in handy. We maintain that asset holdings a are a continuous variable and discretize the income process. In particular, let

$$\begin{aligned} y_t &= \exp(\hat{y}_t) \\ \hat{y}_t &= \rho \hat{y}_{t-1} + \varepsilon_t, \end{aligned}$$

where $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. Discretizing the income process yields:

- a grid for income in logs, $\tilde{y}_t$. Taking the exponential of each element in this grid gives a grid for income in level, y_t . Let Y denote the grid for the income level.
- a probability transition matrix Θ , with generic element $\pi(\tilde{y}'|\tilde{y}) = \pi(y'|y)$.

Then, we can write the income fluctuation problem in recursive form as:

$$\begin{aligned} V(a, y) &= \max_{\{c, a'\}} u(c) + \beta \sum_{y' \in Y} \pi(y'|y) V(a', y') \\ \text{s.t.} \\ c + a' &\leq Ra + y \\ a' &\geq -\phi \end{aligned}$$

The Euler equation, once we substitute the budget constraint, reads

$$u_c(Ra + y - a') - \beta R \sum_{y' \in Y} \pi(y'|y) u_c(Ra' + y' - a'') \geq 0.$$

³If the income process is iid, remember that you can reduce the dimensionality of the state space to one variable, cash in hand $x = Ra + y$. In general, whenever you can reduce the dimensionality of the state-space you should do so.

Equality holds if $a' > -\phi$. The objective is to find a decision rule for asset holding $a'(a, y)$, i.e., an invariant function of the states (a, y) that satisfies the Euler Equation. The strategy will be to guess a decision rule for a'' , and solve for a' . At the solution, the two policy functions must be identical.

We present here in detail the three variants of the policy-function iteration method: discretization, interpolation, and approximation.

2.1 Discretization

1. Construct a grid on the asset space $\{a_1, a_2, \dots, a_M\}$, with $a_1 = -\phi$. The best way to construct the grid varies problem by problem. In general, one must put more points where the policy function is expected to display more curvature (i.e., where it is far from linear). In our case, this happens for values of a near the debt constraint $-\phi$. You can set asset grid points a_k , $1 < k < M$ according to

$$a_k = a_1 + (a_M - a_1) \left(\frac{k-1}{M-1} \right)^\nu, \quad \nu \geq 1.$$

If $\nu = 1$, the the grid is evenly spaced. If $\nu > 1$ then the grid will have more points near $-\phi$.

2. Guess an initial decision rule for a'' on the grid points (i.e., a $M \times N$ matrix):

$$\{\hat{a}_0(a_1, y_1), \hat{a}_0(a_1, y_2), \dots, \hat{a}_0(a_1, y_N); \hat{a}_0(a_2, y_1), \dots, \hat{a}_0(a_2, y_N); \dots, \hat{a}_0(a_M, y_N)\}$$

by making sure that each decision rule maps into a point on the asset grid. The subscript denotes the iteration number, and “0” denotes the first iteration. A reasonable guess for $a''_0(a_i, y_j)$ would be

$$\hat{a}_0(a_i, y_j) = a_i.$$

3. Determine if the liquidity constraint is binding. For each point (a_i, y_j) on the grid, check whether

$$u_c(Ra_i + y_j - a_1) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra_1 + y' - \hat{a}_0(a_1, y')) > 0.$$

If this inequality holds, then it means that the borrowing constraint binds, set $a'_0(a_i, y_j) = a_1$ and repeat this check for the next grid point. If the equation instead

holds with the $<$ inequality, it means that there is an interior solution (it is optimal to save, or to borrow less than ϕ , for the household) and we proceed to the next step.

4. Determine $a'_0(a_i, y_j)$. Find the pair of adjacent grid points (a_k, a_{k+1}) such that

$$\begin{aligned}\delta(a_k) &\equiv u_c(Ra_i + y_j - a_k) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra_k + y' - \hat{a}_0(a_k, y')) < 0 \\ \delta(a_{k+1}) &\equiv u_c(Ra_i + y_j - a_{k+1}) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra_{k+1} + y' - \hat{a}_0(a_{k+1}, y')) > 0,\end{aligned}$$

This means that $a'_0(a_i, y_j) \in (a_k, a_{k+1})$.⁴ Since we only work with points on the grid, set

$$a'_0(a_i, y_j) = \arg \min_{l \in \{k, k+1\}} |\delta(a_l)|$$

5. Check convergence by comparing the guess $\hat{a}_0(a_i, y_j)$ to the solution $a'_0(a_i, y_j)$. If, for each pair (a_i, y_j) on the grid, $a'_0(a_i, y_j) = \hat{a}_0(a_i, y_j)$ the convergence is achieved, so the algorithm ends. Otherwise, go back to point 3 with the new guess $\hat{a}_1(a_i, y_j) = a'_0(a_i, y_j)$ and repeat until convergence.

This is a very simple and fast method because it avoids the calculation of the policy function outside the grid points, but at the same time it can be imprecise, unless M is a very large number, but in this case the method becomes computationally costly.

2.2 Policy Function Iteration with Linear Interpolation

1. Construct a grid on the asset space $\{a_1, a_2, \dots, a_M\}$ with $a_1 = -\phi$.
2. Guess an initial vector of decision rules for a'' on the grid points, call it $\hat{a}_0(a_i, y_j)$.
3. For each point (a_i, y_j) on the grid, check whether the borrowing constraint binds, as before. I.e., for each point (a_i, y_j) on the grid, check whether

$$u_c(Ra_i + y_j - a_1) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra_1 + y' - \hat{a}_0(a_1, y')) > 0.$$

If this inequality holds, then it means that the borrowing constraint binds, set $a'_0(a_i, y_j) = a_1$ and repeat this check for the next grid point. If the equation instead

⁴In other words, a_k is smaller than $a^*(a_i, y_j)$ because $u'(c)$ is too small and a_{k+1} is larger than $a^*(a_i, y_j)$ because $u'(c)$ is too large, relative to the optimal choice.

holds with the $<$ inequality, it means that we have an interior solution (it is optimal to save, or to borrow less than ϕ , for the household) and we proceed to the next step.

4. For each point (a_i, y_j) on the grid, use a *nonlinear equation solver* (e.g. bisection) to look for the solution a^* of the nonlinear equation

$$u_c(Ra_i + y_j - a^*) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra^* + y' - \hat{a}_0(a^*, y')) = 0, \quad (2)$$

and notice that the equation solver will try to evaluate many times the function $\hat{a}_0$ outside the grid points for assets $\{a_1, a_2, \dots, a_M\}$. Hence, the need for the linear interpolation.

- (a) Assume that, for every value y' , the function $\hat{a}_0(a, y')$ is *piecewise linear* between the asset grid points. So, every time the equation solver above requires evaluating the function on a^* which lies between grid points, do as follows. First, find the pair of adjacent grid points $\{a_k, a_{k+1}\}$ such that $a_k < a^* < a_{k+1}$, and then compute

$$\hat{a}_0(a^*, y') = \hat{a}_0(a_k, y') + (a^* - a_k) \left(\frac{\hat{a}_0(a_{k+1}, y') - \hat{a}_0(a_k, y')}{a_{k+1} - a_k} \right)$$

- (b) If the solution of the nonlinear equation in (2) is a^* , then set $a'_0(a_i, y_j) = a^*$ and iterate on the next new grid point.
5. Check convergence by comparing $a'_0(a_i, y_j) - \hat{a}_0(a_i, y_j)$ through some pre-specified norm. For example, declare convergence if

$$\max_{i,j} \{|a'_0(a_i, y_j) - \hat{a}_0(a_i, y_j)|\} < \varepsilon$$

for some small number ε which determines the degree of tolerance in the solution algorithm. If convergence is achieved, stop. Otherwise, go back to point 3 with the new guess $\hat{a}_1(a_i, y_j) = a'_0(a_i, y_j)$ and repeat until convergence.

Piecewise linear interpolation is fast, and obviously preserves positivity, monotonicity and concavity. However, the optimal policy obtained is not differentiable everywhere.

2.3 Policy function Iteration with Chebyshev Approximation (optional)

There are several families of polynomials with good properties for function approximation, e.g., Legendre, Chebyshev, Laguerre, Hermite. In what follows, we use Chebyshev since in most of the common applications, they have the best properties (see below and Judd, chapter 6).

Chebyshev Polynomials Suppose we want to approximate a function $f(x)$ over the interval $[-1, 1]$, through a polynomial function

$$\hat{f}(x) = \sum_{p=0}^N \kappa_p T_p(x) \quad (3)$$

where N is the order of the polynomial approximation, $T_p(x)$ is called the basis functions, i.e. it is a polynomial of order p , and κ_p are the coefficients that weight the various polynomials.

The Chebyshev polynomials are a family of basis functions that is very useful for this type of approximations. They are given by the simple formula

$$T_p(x) = \cos(p \arccos(x)),$$

which can be simply constructed sequentially (please verify) as

$$\begin{aligned} T_0(x) &= 1, \\ T_1(x) &= x, \\ T_{p+1}(x) &= 2xT_p(x) - T_{p-1}(x), \quad \text{for } p > 1. \end{aligned}$$

The polynomial $T_p(x)$ with $p > 0$ has p zeros located at the points $x_k = -\cos\left(\frac{2k-1}{2p}\pi\right)$, with $k = 1, 2, \dots, p$ and it has $(p+1)$ extrema. All the maxima equal 1 and all the minima equal -1 , so the range of the Chebyshev polynomials is $[-1, 1]$. It is precisely this property that makes the Chebyshev polynomials so useful in the approximation of functions. Note that the Chebyshev polynomials satisfy the orthogonality condition among them

$$\sum_{k=1}^M T_q(x_k) T_p(x_k) = 0, \quad \text{for } p \neq q,$$

which makes them basis functions.

One can prove that, if the κ_p coefficients are defined as

$$\kappa_p = \frac{\sum_{k=1}^M f(x_k) T_p(x_k)}{\sum_{k=1}^M T_p(x_k)^2}, \quad p = 0, \dots, N$$

then the approximation formula (3) is exact for those x_k equal to every zero of $T_N(x)$. Note that the expression for κ_p is that of an OLS estimator that solves

$$\min_{\{\kappa_p\}_{p=0}^N} \left\{ \sum_{k=1}^M \left[f(x_k) - \sum_{p=0}^N \kappa_p T_p(x_k) \right]^2 \right\}$$

i.e. the vector of κ_p minimizes the sum of squared residuals between the true function and the approximated function.

As we increase the order of the approximating Chebyshev polynomial toward $N = \infty$, we get closer to the true function. However, even for relatively small N , this procedure yields very good approximations. In particular, notice that since the T_p functions are all bounded by one in absolute value, if the coefficients κ_p decline fast with p (and they do), then the largest omitted term in the error has order $(N + 1)$. Moreover, $T_{N+1}(x)$ is an oscillatory function with $(N + 1)$ extrema distributed smoothly over the interval. This smooth spreading out of the error is a very important property of optimal approximations!⁵

The Algorithm

1. Compute the M Chebyshev interpolation nodes on the normalized interval $[-1, 1]$ which are given by the simple formula

$$x_k = -\cos\left(\frac{2k-1}{2M}\pi\right), \quad k = 1, \dots, M,$$

i.e. they are the points where the Chebyshev polynomials are exactly equal to zero.

- (a) Fix the bounds of the asset space $\{-\phi, a_{\max}\}$. Transform the Chebyshev nodes over $[-1, 1]$ into a grid over the $[-\phi, a_{\max}]$ interval, by setting

$$a_k = -\phi + (x_k + 1) \left(\frac{a_{\max} + \phi}{2} \right), \quad k = 1, \dots, M$$

⁵The best polynomial approximation is given by the *minimax* polynomial (see Judd, chapter 6) which has the property that the sign of the error term should alternate between points (the so-called Equioscillation Theorem).

In particular note that for $k = M$ and M large, $x_k \simeq 1$ and $a_M \simeq a_{\max}$; for $k = 1$ and M large, $x_k \simeq -1$ and $a_k \simeq -\phi$.⁶

2. Guess an initial vector of decision rules for a'' on the grid point, call it $\hat{a}_0(a_i, y_j)$.
3. For each point (a_i, y_j) on the grid, check whether the borrowing constraint binds. If not, continue.
4. For each point (a_i, y_j) on the grid, use a nonlinear equation solver to look for the solution a^* of the nonlinear equation

$$u_c(Ra_i + y_j - a^*) - \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(Ra^* + y' - \hat{a}_0(a^*, y')) = 0,$$

and notice that the equation solver will try to evaluate many times $\hat{a}_0$ off the grid. Here, we use the Chebyshev approximation.

- (a) Choose an order $N < M$ for the Chebyshev polynomials. Compute the Chebyshev coefficients

$$\kappa_p^0 = \frac{\sum_{k=1}^M \hat{a}_0(a_k, y_j) T_p(x_k)}{\sum_{k=1}^M T_p(x_k)^2}, p = 0, \dots, N \quad (4)$$

- (b) Use the coefficients κ_p^0 for the evaluation outside the grid as follows:

$$\hat{a}_0(a^*, y') = \sum_{p=0}^N \kappa_p^0 T_p \left(2 \frac{a^* + \phi}{a_{\max} + \phi} - 1 \right).$$

- (c) If the solution of the nonlinear equation in (2) is a^* , then set $a'_0(a_i, y_j) = a^*$ and iterate on a new grid point from step 4.
5. To check convergence, in this case, it is better to compare iteration after iteration the Chebyshev coefficients in order to get a sense of how globally distant are the policy functions obtained as a solution in successive iterations. For example, at iteration n , one could use criterion

$$\min_p |\kappa_p^n - \kappa_p^{n-1}| < \varepsilon.$$

⁶To understand this transformation, note that the reverse transformation is

$$x_k = 2 \frac{a_k + \phi}{a_{\max} + \phi} - 1.$$

As explained, the Chebyshev approximation can be extremely good. It is smooth and differential everywhere, but it may not preserve concavity.

2.4 The Endogenous Grid Method

This is a newly developed method that, when it can be applied (not always), is much faster than the traditional method just described because it does not require the use of a nonlinear equation solver. The essential idea of the method is to construct a grid on a' , next period's asset holdings, rather than on a , as is done in the standard algorithm. The method also requires the policy function for consumption to be at least weakly monotonic in current asset holdings. Consider the above income fluctuation problem and write the Euler equation as follows:

$$u_c(c(a, y)) \geq \beta R \sum_{y' \in Y} \pi(y'|y) u_c(c(a', y')).$$

Equality holds if $a' > -\phi$. We now solve for the consumption policy instead that for the next-period assets policy, i.e. we look for a decision rule for consumption $c(a, y)$, which is an invariant function of the states that satisfies the Euler Equation and that does not violate the borrowing constraint. We are going to start from a guess $\hat{c}_0(a, y)$ and will iterate on the Euler Equation until the decision rule for consumption that we solve for is essentially identical to the one in the previous iteration. The algorithm involves the following steps:

1. Construct a grid on (a, y) where $a \in A = \{a_1, \dots, \bar{a}\}$ with $a_1 = -\phi$ and $y \in Y = \{y_1, \dots, y_N\}$. Notice here that we are defining our grid over assets tomorrow.
2. Guess a policy function $\hat{c}_0(a_i, y_j)$. A good initial guess is to set $\hat{c}_0(a_i, y_j) = ra_i + y_j$ which is the solution under quadratic utility if income follows a random walk.
3. Here's the key difference. Instead of iterating over pairs of $\{a_i, y_j\}$, we iterate over pairs $\{a'_i, y_j\}$. For any pair $\{a'_i, y_j\}$ on the mesh $A \times Y$ construct the RHS of the Euler equation [call it $B(a'_i, y_j)$]

$$B(a'_i, y_j) \equiv \beta R \sum_{y' \in Y} \pi(y'|y_j) u_c(\hat{c}_0(a'_i, y'))$$

where the RHS of this equation uses the guess $\hat{c}_0$.

4. Use the Euler equation to solve for the value $\tilde{c}(a'_i, y_j)$ that satisfies

$$u_c(\tilde{c}(a'_i, y_j)) = B(a'_i, y_j)$$

and note that it can be done analytically, i.e. for $u_c(c) = c^{-\gamma}$ we have $\tilde{c}(a'_i, y_j) = [B(a'_i, y_j)]^{-\frac{1}{\gamma}}$. This is the step where the algorithm becomes much more efficient than the traditional one. First, we do not require a nonlinear equation solver, which takes a lot of computing time and could introduce numerical instabilities. Second, we only compute the expectation in step 3 once. In the traditional algorithm the expectation is computed within the nonlinear equation solver multiple times.

5. From the budget constraint, solve for $a^*(a'_i, y_j)$ such that

$$\tilde{c}(a'_i, y_j) + a'_i = Ra_i^* + y_j$$

which implicitly gives the function $c_0(a_i^*, y_j) = \tilde{c}(a'_i, y_j)$. $a^*(a'_i, y_j)$ is the value of assets today that would lead the consumer to have a'_i assets tomorrow if his income shock was y_j today. This is the endogenous grid and it changes on each iteration. Note that this function in general is not defined on the grid points of A .

6. Now we need to update our guess defined on the original grid. To that end, let a_1^* be the value of asset holdings that induces the borrowing constraint to bind, i.e. the value for a_1^* that solves

$$\tilde{c}(a'_1, y_j) + a'_1 = Ra_1^* + y_j.$$

To get the new guess $\hat{c}_1(a_i, y_j)$ on grid points $a_i > a_1^*$ we can use simple interpolation methods using values for $\{c_0(a_k^*, y_j), c_0(a_{k+1}^*, y_j)\}$ on the two most adjacent values $\{a_k^*, a_{k+1}^*\}$ that include the given grid point a_i . To define the new guess of the consumption policy function on grid values $a_i < a_1^*$, then we use the budget constraint

$$\hat{c}_1(a_i, y_j) = Ra_i + y_j - a'_1$$

since we cannot use the Euler equation as the borrowing constraint is binding for sure next period. The reason is that we found it was binding at a_1^* , therefore a fortiori it will be binding for $a_i < a_1^*$.

7. Check convergence by comparing $\hat{c}_{n+1}(a_i, y_j)$ to $\hat{c}_n(a_i, y_j)$. For example declare convergence at iteration $n + 1$ when

$$\max_{i,j} \{|\hat{c}_{n+1}(a_i, y_j) - \hat{c}_n(a_i, y_j)|\} < \varepsilon$$

for some small ε which determines the degree of tolerance in the solution algorithm.

3 Simulation

Once we solved for the policy function $a'(a, y)$ and obtained the consumption function $c(a, y)$ residually from the budget constraint (or viceversa if using the endogenous grid method), we can simulate paths for the variables of interest and compute statistics like the saving rate, the variance of consumption, the correlation between consumption change and income change, etc... This also allows us to do comparative statics exercises, i.e., asking questions such as what happens to average savings when uncertainty rises? Or, how do savings respond to changes in the interest rate? And so on.

One key step in the simulation is drawing paths of the income process. Suppose, for simplicity that we have a 2-state Markov chain with transition probabilities $\{\pi_{LL}, \pi_{LH}, \pi_{HL}, \pi_{HH}\}$ over two realizations (y_L, y_H) . Then, we implement the following algorithm:

1. Draw a long sequence of T realizations from a uniform distribution on $[0, 1]$, i.e. $\{\alpha_1, \alpha_2, \dots, \alpha_T\}$. Note that $\pi_{sL} + \pi_{sH} = 1$, with $s \in \{L, H\}$.
2. Start from $y_0 = y_L$. If $\alpha_1 < \pi_{LL}$, then set $y_1 = y_L$. Otherwise, set $y_1 = y_H$. Suppose $\alpha_1 > \pi_{LL}$ so that $y_1 = y_H$. Next, if $\alpha_2 < \pi_{HL}$, then set $y_2 = y_L$. Otherwise, set $y_2 = y_H$. And continue until you have mapped the α -sequence into a y -sequence.

To simulate series for consumption and asset holdings, start from an initial condition (a_0, y_0) and use the policy functions $c(a, y)$ and $a'(a, y)$ to compute:

$$\begin{aligned} a_0 &= a_0, & c_0 &= y_0 + Ra_0 - a'(a_0, y_0) \\ a_1 &= a'_0 = a'(a_0, y_0), & c_1 &= y_1 + Ra_1 - a'(a_1, y_1) \\ a_2 &= a'_1 = a'(a_1, y_1), & c_2 &= y_2 + Ra_2 - a'(a_2, y_2) \\ &\dots \end{aligned}$$

One important trick to keep in mind when doing comparative statics (e.g., to study the effect of a rise in r on the saving rate) is that the statistics of interest (e.g., average savings) must be computed based on simulations run *with the same seed of the random number generator for the α sequence*, i.e., one has to use the same paths for the shocks in order to minimize the simulation variance.

It is useful to discard a certain number of initial observations (the so called burn-in period) to avoid dependence of the result from the initial condition (a_0, y_0) and allow draws to occur from the stationary distribution. Later on, we will learn how to compute the stationary distribution directly, without resorting to simulation.

4 Checking Accuracy of the Numerical Solution

Suppose that we are using the piecewise linear interpolation. How can we determine when the solution is accurate enough that no more points in the grid are needed? Or, if we are using the Chebyshev approximation method, how do we determine that increasing the order of the polynomial would not lead to any significant improvement in accuracy?

4.1 Euler Equation Error Analysis

The numerically approximated Euler equation is

$$u_c(c_t) - \beta RE_t[u_c(c_{t+1})] \simeq 0.$$

The above equation will be very close to zero when evaluated on gridpoints, but it may be quite far from zero outside the grid.

One can define the relative approximation error ε_t as that value such that the equation holds exactly at t

$$u_c(c_t(1 - \varepsilon_t)) = \beta RE_t[u_c(c_{t+1})]$$

Let g denote the inverse function of the marginal utility, then we have

$$\varepsilon_t = 1 - \frac{g(\beta RE_t[u_c(c_{t+1})])}{c_t}.$$

For example an error of 0.01 means that the agent is making a mistake equivalent to \$1 for every \$100 consumed when choosing consumption and saving in period t . Santos

(2000) proves that, the implied cost in terms of household welfare is the square of the Euler equation error, i.e., the error in the value function is of the order ε^2 .

By running a long (but finite) simulation, one can compute a string of Euler equation errors and report either the maximum or the average of the errors in absolute value. This procedure may not explore the entire state space, but it will explore the region of the state space where the agent is more likely to find herself.

Alternatively, one can explore the entire state space by choosing a grid for assets different from the one used in the computation and calculate for each point a_i on this new grid, and for every y_j the Euler equation error $\varepsilon(a_i, y_j)$. Let $a^*(a, y)$ be the policy function obtained in the numerical solution. Then, compute:

$$u_c((Ra_i + y_j - a^*(a_i, y_j))(1 - \varepsilon(a_i, y_j))) - \beta RE_y[u_c(Ra^*(a_i, y_j) + y' - a^*(a^*(a_i, y_j), y'))] = 0.$$

Usually, absolute errors are reported in base 10 logarithm, so a value of -2 means an error of \$1 for every \$100, a value of -3 means an error of \$1 for every \$1,000, and so on. Acceptable average errors should be around values of -4 and lower. See Aruoba, Fernandez-Villaverde and Rubio-Ramirez (2006) for a comparison of various global solution methods of the growth model based on Euler Equation error analysis.

4.2 den Haan-Marcet Test

Den Haan and Marcet (1994) devise a simple test based on Hansen J test of overidentifying restrictions. We present here the test applied to the consumption-saving problem. The consumption Euler equation

$$u_c(c_t) = \beta RE_t[u_c(c_{t+1})]$$

implies that the residual

$$\varepsilon_{t+1} = u_c(c_t) - \beta Ru_c(c_{t+1})$$

should not be correlated with any variable dated t and earlier, since the expectation at time t is conditional on everything observable up to then. Therefore, we should have, for every t

$$E_t[\varepsilon_{t+1} \otimes h(z_t)] = 0, \tag{5}$$

where the symbol $\otimes$ denotes element-by-element product. The term $h(z_t)$ is a $(r \times 1)$ vector of functions of z_t , which can include all the variables in the information set of the

agent at time t like $\{y_j, c_j, a_j\}_{j=0}^t$. Clearly, this is true only for the *exact* solution. A badly approximated solution will not satisfy this property. This is the key idea of the test proposed by den Haan and Marcet.

One can obtain an estimate of the LHS of (5) through a simulation of length S of the model and the construction of

$$B_S = \frac{\sum_{t=1}^S \hat{\varepsilon}_{t+1} \otimes h(\hat{z}_t)}{S},$$

where $\hat{\varepsilon}_{t+1}$ and $\hat{z}_t$ are the simulated counterparts. It can be shown that, under mild conditions, $\sqrt{S}B_S \xrightarrow{d} N(0, V)$, and one can construct the appropriate quadratic form for the test statistics

$$SB'_S \hat{V}_S^{-1} B_S \xrightarrow{d} \chi_r^2$$

where $\hat{V}_S^{-1}$ is the inverse of some consistent estimate of V .

Some remarks are in order. First, the test does not require any knowledge of the true solution, which is a big advantage. Second, given a certain level of approximation in the solution, we can always find a number S large enough so that the approximation fails the accuracy test. This is not a problem when comparing solution methods, since one can fix the same S for both, or one can look for the smallest S such that the method fails the test and compare these thresholds. However, when we want to judge the accuracy of our unique solution, how large should S be? It's not clear: Den Haan and Marcet pick S to be 20 times larger than the typical sample period available.

1 Economies with Idiosyncratic Risk and Incomplete Markets: Stationary Equilibrium

We now return to our study of economies with incomplete markets and idiosyncratic risk, commonly known as Bewley economies.¹ We have studied the income fluctuation problem in partial equilibrium. We now close the economy by looking at the equilibrium on the goods and asset markets, while still focusing on the case without aggregate uncertainty.

We are interested in building a class of models whose equilibria feature a nontrivial endogenous distribution of income and wealth across agents in order to analyze questions such as:

1. What is the fraction of aggregate savings due to the precautionary motive?
2. How much of the observed wealth inequality can one explain through uninsurable earnings variation across agents?
3. What are the redistributive implications of various policies? How are inequality and welfare affected by such policies? It is key to have an equilibrium model to answer policy questions, because changes in policy affect equilibrium prices.
4. Can we generate a reasonable equity premium (i.e. excess return of stocks over a risk-free bond), once we introduce a risky asset?
5. How large are the welfare losses of a rise in labor market risk (e.g., unemployment)?

The model is constructed around three building blocks: 1) the “income-fluctuation problem”, 2) the aggregate neoclassical production function, and 3) the equilibrium of the asset market. For now, we focus on the *stationary equilibrium*, i.e. an economy without aggregate shocks.

Income fluctuation problem— This is the problem we studied in the previous chapter. Individuals are subject to exogenous income shocks. These shocks are not fully insurable because of the lack of a complete set of Arrow-Debreu contingent claims. There

¹Due to Bewley (1977), Bewley is the namesake of Bewley models, the class of incomplete markets general equilibrium models in which agents face idiosyncratic income shocks and achieve partial insurance via, for example, a risk-free bond or capital. Aiyagari (1994), Huggett (1993), and Krusell and Smith (1998) are famous examples of Bewley models.

is only a risk-free asset (i.e. with non-state contingent rate of return) in which the individual can save/borrow, and that the individual faces a borrowing (liquidity) constraint. A continuum of such agents subject to different shocks will give rise to a wealth distribution. Integrating wealth holdings across all agents will give rise to an *aggregate supply of capital*.

Aggregate production function: profit maximization of the competitive representative firm operating a CRS technology will give rise to an *aggregate demand for capital*.

Equilibrium in the asset market: When we let demand and supply interact in an asset market, an *equilibrium interest rate* will arise endogenously.

1.1 The Economy

Demographics: the economy is populated with a continuum of measure one of infinitely lived, ex-ante identical agents.

Preferences: the individual has time-separable preferences over streams of consumption

$$U(c_0, c_1, c_2, \dots) = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where the period utility function $u(c_t)$ satisfies $u' > 0, u'' < 0$ and the discount factor $\beta \in (0, 1)$. The expectation is over future sequences of shocks, conditional to the realization at time 0. The individual supplies labor inelastically.

Endowment: each individual has a stochastic endowment of efficiency units of labor $\varepsilon_t \in E \equiv \{\varepsilon^1, \varepsilon^2, \dots, \varepsilon^{N-1}, \varepsilon^N\}$. The shocks follow a Markov process with transition probabilities $\pi(\varepsilon' \mid \varepsilon) = \Pr(\varepsilon_{t+1} = \varepsilon' \mid \varepsilon_t = \varepsilon)$. Shocks are *iid* across individuals. We assume a law of large numbers to hold, so that $\pi(\varepsilon' \mid \varepsilon)$ is also the fraction of agents in the population subject to this particular transition.² We assume that the Markov transition is well-behaved, so there is a unique invariant distribution $\Pi^*(\varepsilon)$. As a result, the aggregate endowment of efficiency units

$$H_t = \sum_{i=1}^N \varepsilon_i \Pi^*(\varepsilon_i), \text{ for all } t$$

²There are some tricky issues with laws of large numbers in this setting. Please, refer to Judd (1985) and Uhlig (1996) for a discussion.

is constant over time, i.e. there is no aggregate uncertainty. Note in particular, that H_t is exogenously determined.

Budget constraint: For individual i at time t , the budget constraint reads

$$c_t + a_{t+1} = (1 + r_t) a_t + w_t \varepsilon_t,$$

where c_t is current consumption, a_{t+1} is next period wealth, $(1 + r_t)$ is the gross interest rate and w_t is the wage rate at period t . Wealth is held in the form of a one-period risk-free bond whose price is one and whose return, next period, will be $(1 + r_{t+1})$, independently of the individual state (i.e., r_{t+1} does not depend on the realization of ε_{t+1}). In this sense, the asset a is non state-contingent.

Borrowing constraint: At every t , agents face the borrowing limit

$$a_{t+1} \geq -\phi$$

where ϕ is exogenously specified. Alternatively, we could assume the agents face the “natural” borrowing constraint.

Technology: The representative competitive firm produces with CRS production function $Y_t = F(K_t, H_t)$ with decreasing marginal returns in both inputs and standard Inada conditions. Physical capital depreciates geometrically at rate $\delta \in (0, 1)$. The firm’s problem is

$$\max_{K_t, H_t} F(K_t, H_t) - (r_t + \delta) K_t - w_t H_t$$

Market structure: Final goods market (consumption and investment goods), labor market, and capital market are all competitive.

Aggregate resource constraint: The aggregate feasibility condition in this economy reads:

$$F(K_t, H_t) = C_t + I_t = C_t + K_{t+1} - (1 - \delta) K_t,$$

where capital letters denote aggregate variables.³

³This can be obtained by integrating the budget constraint across all households and using that $F(K_t, H_t) = (r_t + \delta) K_t + w_t H_t$.

1.2 Stationary Equilibrium

We are now ready to define the stationary equilibrium of this economy through the concept of *Recursive Competitive Equilibrium* (RCE). Most of the requirement of this RCE definition will be standard (agents optimize, markets clear). Moreover, in the stationary equilibrium of this economy we require the distribution of agents across states to be invariant.⁴ This probability measure will permanently reproduce itself. It is in this sense that the economy is in a rest-point, i.e. a steady state.

Some Preliminaries

Let us introduce some useful notation. The individual is characterized by the pair (a, ε) – the individual states. What characterizes the aggregate state is the distribution of agents over individuals states. Let $\lambda(a, \varepsilon)$ denote this distribution. We would like this object to be a *probability measure*.

First we need to define an appropriate measurable space on which the measure λ is defined. Define the set $A \equiv [-\phi, \bar{a}]$ of possible asset holdings and the countable set E of possible realizations of the endowment of efficiency units, and let the state space $S = A \times E$ be the Cartesian product of A and E .⁵ Define by P_E the power set of E (i.e. the set of all subsets of E) and by B_A the Borel σ -algebra on A . Finally, let $\Sigma_s = P_E \otimes B_A := \sigma(P_E \times B_A)$ be the smallest σ -algebra that contains all measurable sets in $S = A \times E$. Then, the space (S, Σ_s) is a measurable space. Let $\mathcal{S} = \mathcal{A} \times \mathcal{E}$ be the typical subset of Σ_s . For any element $\mathcal{S} \in \Sigma_s$, $\lambda(\mathcal{S})$ is the measure of agents in the set $\mathcal{S}$.

The definition of stationary RCE will impose that λ is invariant. It is therefore important to characterize the way individuals transit across states over time. That is, how do we obtain next period's distribution, given this period's distribution? We need a transition function. Define $Q((a, \varepsilon), \mathcal{A} \times \mathcal{E})$ as the probability that an individual with current state (a, ε) transits to the set $\mathcal{A} \times \mathcal{E}$ next period

$$Q((a, \varepsilon), \mathcal{A} \times \mathcal{E}) = I_{\{a'(a, \varepsilon) \in \mathcal{A}\}} \sum_{\varepsilon' \in \mathcal{E}} \pi(\varepsilon' | \varepsilon) \quad (1)$$

⁴However, individuals move up and down in the earnings and wealth distribution, so “social mobility” can be meaningfully defined. Recall that with complete markets, there is no social mobility: initial rankings persist forever.

⁵Assume for now that the upper bound $\bar{a}$ exists.

where $I_{\{\cdot\}}$ is the indicator function, and $a'(a, \varepsilon)$ is the optimal saving policy.⁶ Then the distribution evolves according to

$$\lambda_{t+1}(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} Q((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda_t(da, d\varepsilon). \quad (2)$$

Let us now re-state the problem of the individual in recursive form, i.e. through dynamic programming

$$\begin{aligned} V(a, \varepsilon; \lambda) &= \max_{c, a'} \left\{ u(c) + \beta \sum_{\varepsilon' \in E} V(a', \varepsilon'; \lambda') \pi(\varepsilon' | \varepsilon) \right\} \\ &\quad s.t. \\ c + a' &= (1 + r(\lambda))a + w(\lambda)\varepsilon \\ a' &\geq -\phi \\ \lambda' &= H(\lambda) \end{aligned} \quad (3)$$

where $H(\cdot)$ characterizes the law of motion of the distribution of agents and where, for clarity, we have made explicit the dependence of prices from the distribution of agents. In a stationary environment this dependence will be *redundant* and can be omitted since the probability measure λ will be constant. We are now ready to proceed to the definition of equilibrium.

Definition of Stationary RCE

A **stationary recursive competitive equilibrium** is a value function $V : S \rightarrow \mathbb{R}$; policy functions for the household $a' : S \rightarrow \mathbb{R}$, and $c : S \rightarrow \mathbb{R}_+$; firm's choices H and K ; prices r and w ; and, a stationary measure λ^* such that:

- given prices r and w , the policy functions a' and c solve the household's problem (3) and V is the associated value function,
- given r and w , the firm chooses optimally its capital K and its labor H , i.e. $r + \delta = F_K(K, H)$ and $w = F_H(K, H)$,
- markets clear:

⁶That Q is a transition function requires that (i) $Q(s, \cdot)$ is a probability measure on (S, Σ_S) for all $s \in S$, and (ii) $Q(\cdot, \mathcal{S})$ is a Σ_S -measurable function for all $\mathcal{S} \in \Sigma_S$. This last condition means that the inverse image of any open set $\mathcal{S}$ is open. Continuity of $a'(a, \varepsilon)$ guarantees that. Note that $Q((a, \varepsilon), A \times E) = 1$.

- labor market: $H = \int_{A \times E} \varepsilon \lambda^*(da, d\varepsilon)$
- asset market: $K = \int_{A \times E} a'(a, \varepsilon) \lambda^*(da, d\varepsilon)$
- goods market: $\int_{A \times E} c(a, \varepsilon) \lambda^*(da, d\varepsilon) + \delta K = F(K, H)$ ⁷
- for all $(\mathcal{A} \times \mathcal{E}) \in \Sigma_s$, the invariant probability measure λ^* satisfies

$$\lambda^*(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} Q((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda^*(da, d\varepsilon),$$

where Q is the transition function defined in (1).

1.3 Existence and Uniqueness of the Stationary Equilibrium

Characterizing the conditions under which an equilibrium exists and is unique boils down, like in every general equilibrium model, to show that the excess demand function (of the price) in each market is continuous, strictly monotone and intersects “zero”. Equilibrium in the labor market is trivial: aggregate labor supply is exogenous and labor demand is strictly decreasing in wages. By Walras law, if we prove that the equilibrium in the asset market exists and is unique, we are done.

Demand for capital– Consider first the aggregate demand of capital. From the optimal choice of the firm, we have

$$F_K(K, H) = r + \delta,$$

Note that for $r = -\delta$, then $K \rightarrow +\infty$, while for $r \rightarrow +\infty$, $K \rightarrow 0$. Whether the demand for capital $K(r)$ is continuous and monotonic in the interest rate r depends on the properties of the production function. For, example, if $F(K, H) = K^\alpha H^{1-\alpha}$, then

$$K(r) = \left(\frac{\alpha}{\delta + r} \right)^{\frac{1}{1-\alpha}} H$$

is a continuous, strictly decreasing function of the interest rate r .⁸

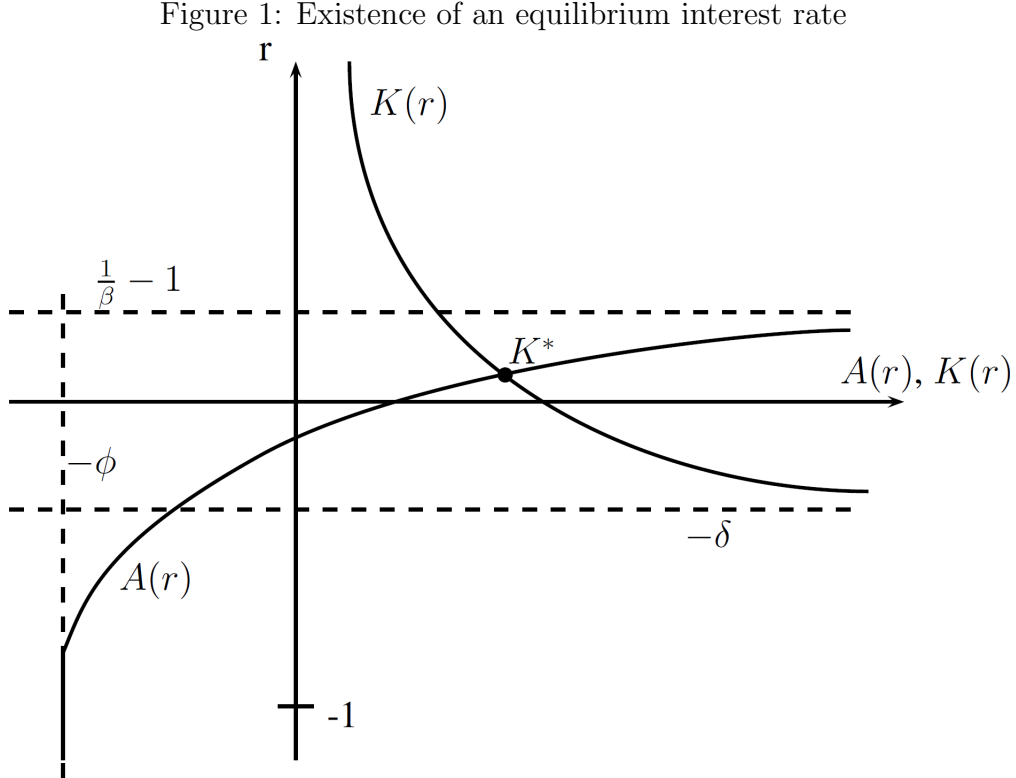
Supply of capital– If we could show that the aggregate supply function

$$A(r) = \int_{A \times E} a'(a, \varepsilon; r) \lambda_r^*(da, d\varepsilon)$$

⁷This condition is redundant by Walras law.

⁸In Huggett (1993) assets are in zero net supply, so $K(r) = 0$ trivially has continuity and monotonicity properties.

is continuous in r and crosses the aggregate demand function, then we would prove existence. Suppose first $(1+r)\beta = 1$, i.e. $r = \frac{1}{\beta} - 1$, then we know that the aggregate supply of assets goes to infinity, i.e. $A(\frac{1}{\beta} - 1) \rightarrow \infty$, for a sufficiently volatile income process (Chamberlain and Wilson, 2000).⁹ For $r = -1$ the individual would like to borrow until the limit, as repayment is costless and every unit saved vanishes, so $A(-1) \rightarrow -\phi$.¹⁰ Thus, if $A(r)$ is continuous, we can represent the demand and supply schedules of capital in Figure 1.



This figure plots the equilibrium in the asset market graphically. By continuity of $K(r)$ and $A(r)$, the Intermediate Value Theorem tells us that the two lines will cross at least once and we have an existence result for the equilibrium interest rate r^* . Furthermore, if we could prove the monotonicity of $A(r)$, we would conclude further that the equilibrium interest rate is unique. Now, can we prove these properties?

⁹Strictly speaking, in this case there is no stationary distribution, hence $A(\frac{1}{\beta} - 1)$ is not well defined. Aiyagari argues, invoking a continuity argument, that $\lim_{r \rightarrow \frac{1}{\beta} - 1} A(r) = \infty$.

¹⁰For values of the interest rate $r < 0$, the agent may still want to hold some wealth for precautionary reasons. It depends on the exact parametrization.

Continuity and Monotonicity of $A(r)$

Continuity – Recall the definition of $A(r)$

$$A(r) = \int_{A \times E} a'(a, \varepsilon; r) \lambda_r^*(da, d\varepsilon).$$

Standard results in dynamic programming ensure that, if u is continuous, $u' > 0$ and $u'' < 0$, the solution to the household problem is unique and that the policy function $a'(a, \varepsilon; r)$ is continuous in r (by the Theorem of the Maximum). Now what about λ_r^* ? Does such a stationary distribution even exist? Is it unique? We must first ensure that the stationary distribution λ_r^* is well defined, i.e. we have to establish that the operator T^* defined by

$$T^*(\lambda_r)(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} Q_r((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda_r(da, d\varepsilon)$$

has a unique fixed point, $\lambda_r^* = T^*(\lambda_r^*)$.

Proposition 1 (SLP Theorem 12.10) *If (i) the state space S is compact and (ii) the transition function Q_r satisfies the Feller property, then there exists a probability measure λ_r^* invariant under Q_r .*

We verify these properties one at a time:

- *Compactness*: when $\beta(1+r) < 1$ and preferences display decreasing absolute risk aversion (or asymptotically bounded relative risk aversion) we showed that an upper bound $\bar{a}$ on the asset space exists—recall our discussion in the previous chapter—so the state space is a compact subset of $\mathbb{R}^2$.¹¹
- *Feller property of Q_r* : The Feller property requires that the operator

$$T(f)(s) = \int_S f(s') Q_r(s, ds')$$

maps continuous and bounded functions f into continuous and bounded functions. Intuitively, $T(f)(s)$ tells us the expected value of the function in the next period, conditional on the current state being s . The condition ensures that small changes in the initial state s lead to small changes in the expected value of f under the transition function Q_r .

¹¹Recall that this has been proven only for certain income processes.

To verify this, we can apply Theorem 9.14 in SLP which states that if E is countable and $P(E)$ is the discrete σ -algebra on E , A is compact and B_A is the Borel σ -algebra on A , and a' is continuous, then Q_r has the Feller property.

In practical terms, if a transition function of a stochastic process satisfies the Feller property, it means that the process does not exhibit discontinuous behavior as a function of the state.

We can now turn to the uniqueness of λ_r^* , which we establish in the following proposition:

Proposition 2 (SLP Theorem 12.12) *If (i) Q_r is monotone, (ii) Q_r satisfies the Feller property and (iii) Q_r satisfies the monotone mixing condition (MMC), then there exists a unique invariant probability measure λ_r^* under Q_r and $T^{*n}\lambda_0 \rightarrow \lambda_r^*$, for all λ_0 .*

As before, we briefly discuss these properties:

- *Monotonicity of Q_r* : Monotonicity of Q_r requires that the associated operator T maps increasing functions into increasing functions. This property is rather straightforward given that: (i) the Markov chain is monotone¹² and (ii) $a'(a, \varepsilon; r)$ is increasing in both arguments. Intuitively, higher pairs (a, ε) make Q_r put more probability mass on higher pairs (a', ε') next period.
- *MMC*: We need this condition to ensure that there are no “traps”, i.e., cases in which initial conditions persist for an infinite number of periods, leading to multiple ergodic distributions. To prevent this, we must ensure that there is enough social mobility or “mixing”. In general, this condition is going to be satisfied if starting from, say, a mass 1 of households either at the top or at the bottom of the state space, it takes only a finite number of steps to move a positive fraction of this mass away from the top or the bottom. Loosely speaking, suppose the household starts from $(\bar{a}, \varepsilon_{\max})$ and receives a long stream of the worse realization of the the shock

¹²A Markov chain ε is monotone iff, for any increasing function $f(\varepsilon')$ the conditional expectation

$$\mathbb{E}[f(\varepsilon') | \varepsilon] = \sum_{\varepsilon'} f(\varepsilon') \pi(\varepsilon' | \varepsilon)$$

is increasing in ε . Note that this restriction on the Markov chain corresponds to positive autocorrelation in the income process.

$\varepsilon_{\min}$. Then, she will keep decumulating wealth until she reaches some neighborhood of the lower bound. The reason for decumulation is that the household knows that this income realization is well below average, her permanent income is higher and consumption is dictated by permanent income. Suppose now that the household starts with $(-\phi, \varepsilon_{\min})$ and receives a long stream of the best shock $\varepsilon_{\max}$. Then, she will accumulate wealth until she reaches some neighborhood of the upper bound. The reason for accumulation is similar: the household realizes that this good realization is “transitory” and her expected income is below the current income, so she saves a fraction of these lucky draws.

After establishing existence and uniqueness of λ_r^* , we may now return to our initial goal: is λ_r^* continuous in r ? This result is provided in the following proposition:

Proposition 3 (SLP Theorem 12.13) *Under the conditions of the previous propositions if for any sequence $r_n \rightarrow r$, $Q_{r_n}(s, \cdot)$ converges weakly to $Q_r(s, \cdot)$, then the invariant probability distributions $\lambda_{r_n}^*$ converge weakly to λ_r^* .*

Note that continuity of $a'(a, \varepsilon; r)$ is sufficient to guarantee this property.

Weak convergence of probability measures is obtained if for any continuous and bounded real valued function f on S , we have

$$\int f(s) \lambda_{r_n}^*(ds) \rightarrow \int f(s) \lambda_r^*(ds), \text{ as } n \rightarrow \infty.$$

We can now turn to the continuity of $A(r)$. Weak convergence of $\lambda_{r_n}^*$ together with the uniform convergence of $a'(a, \varepsilon; r_n)$ implies that, for any sequence $r_n \rightarrow r$,

$$A(r_n) = \int a'(a, \varepsilon; r_n) \lambda_{r_n}^*(da, d\varepsilon) \rightarrow \int a'(a, \varepsilon; r) \lambda_r^*(da, d\varepsilon) = A(r).$$

In other words, the aggregate supply of capital is continuous and we have proved the existence of an equilibrium interest rate r^* for Bewley models.

Monotonicity – There are no results on the monotonicity of the aggregate supply of capital with respect to r , so uniqueness is never guaranteed. Intuitively, a higher r has both income and substitution effects on savings: the relative dominance between the two could switch at a certain level of assets, so $a'(a, \varepsilon; r)$ may not be monotone in r for every a . Even if we make sure that preferences are such that one of the two effects always

dominates, it is very hard to assess what a change in r does to the distribution of assets. In practice, one can use the computer to plot aggregate supply as a function of the interest rate on a fine grid for a reasonably large range of values of r to check its slope.

1.4 An Algorithm for the Computation of the Equilibrium

How do we compute, in practice, this equilibrium? The algorithm that can be used is a fixed point algorithm over the interest rate.

1. Fix an initial guess for the interest rate $r^0 \in \left(-\delta, \frac{1}{\beta} - 1\right)$, where these bounds follow from our previous discussion. The interest rate r^0 is our first candidate for the equilibrium (the superscript denotes the iteration number).
2. Given the interest rate r^0 , obtain the wage rate $w(r^0)$ using the CRS property of the production function (recall that H is given exogenously with inelastic labor supply). Specifically, the firm's optimal choices of capital and labor are:

$$F_K(K, H) = f_K\left(\frac{K}{H}\right) = r + \delta \quad (5)$$

$$F_N(K, H) = f_H\left(\frac{K}{H}\right) = w \quad (6)$$

Let $k = \frac{K}{H}$. Invert the first order condition (5) to obtain $k(r^0)$ and plug into the first order condition (6) to obtain the wage rate $w(r^0)$.

Example: with a Cobb-Douglas production function $k = \left(\frac{\alpha}{r^0 + \delta}\right)^{\frac{1}{1-\alpha}}$ and $w = (1 - \alpha)k^\alpha$.

3. Given prices $(r^0, w(r^0))$, you can now solve the dynamic programming problem of the agent (3) to obtain $a'(a, \varepsilon; r^0)$ and $c(a, \varepsilon; r^0)$. We described several solution methods.
4. Given the policy function $a'(a, \varepsilon; r^0)$ and the Markov transition over productivity shocks $\pi(\varepsilon' | \varepsilon)$, we can construct the transition function $Q(r^0)$ and, by successive iterations over (2), we obtain the fixed point distribution $\lambda(r^0)$, conditional on the candidate interest rate r^0 .

- (a) The easiest method (though not the fastest) to implement this step, in practice, is by simulation of a large number of households N (say 10,000) and track

them over time, like survey data do. Initialize each individual in the sample with a pair (a_0, ε_0) and, using the decision rule $a'(a, \varepsilon)$ and a random number generator that replicates the Markov chain $\pi(\varepsilon' | \varepsilon)$, update their pair of individual states at every period t .

- (b) For every t , compute a set of cross-sectional moments J_t^N which summarize the distribution of assets (e.g., mean, variance, various percentiles). Stop when J_t^N and J_{t-1}^N are close enough. At that point, the cross-sectional distribution has converged. We know that for any given r , a unique invariant distribution will be reached for sure.
- 5. Compute the aggregate demand of capital $K(r^0)$ from the optimal choice of the firm who takes as given r^0 , i.e.

$$K(r^0) = f_k^{-1}(r^0 + \delta) H$$

- 6. Compute the integral

$$A(r^0) = \int_{A \times E} a'(a, \varepsilon; r^0) \lambda(r^0) (da, d\varepsilon)$$

which gives the aggregate supply of assets. Clearly, this can be easily done by exploiting the model-generated data from the invariant distribution obtained in step 4.

- 7. Compare $K(r^0)$ with $A(r^0)$ to verify the asset market clearing condition. If $A(r^0) > (<) K(r^0)$, then the next guess of the interest rate should be lower (higher), i.e. $r^1 < (>) r^0$. To obtain the new candidate r^1 a good choice is, for example,

$$r^1 = \omega r^0 + (1 - \omega) [F_K(A(r^0), H) - \delta],$$

where $\omega \in [0, 1]$ is the weight on the previous guess. For example, you can begin by setting $\omega = \frac{1}{2}$ (bisection) and increase it if the algorithm is not stable (dampening). Note that r^0 and $F_K(A(r^0), H) - \delta$ are, by construction, on opposite sides of the steady-state interest rate r^* .

- 8. Update your guess to r^1 and go back to step 1. Keep iterating until one reaches convergence of the interest rate, i.e. until

$$|r^{n+1} - r^n| < \varepsilon,$$

for ε small. Typically, you need less than 10 iterations for convergence.

9. All the equilibrium statistics of interest, like aggregate savings, inequality measures, etc. can be computed from stationary distribution (or simulated data) calculated in step 4.

1.5 Calibration of the model

To solve the model numerically, one needs first to choose values for the parameters. Here's some guidance on how to pick values. Suppose you set the model's period to one year.

Technology— With Cobb-Douglas production function, pick the capital share α to be equal to $1/3$. Set the depreciation rate δ to 6%.

Preferences— Typically, we work with CRRA utility. Let γ be the coefficient of relative risk aversion. Acceptable values range between 1 and 5, with values at the low end of the range (say $\gamma = 1$ or $\gamma = 2$) being the most commonly chosen. As for the discount rate, you could set choose it so that the K/Y ratio replicates the one for the U.S. economy which is around 3.¹³ This means that the parameter is not calibrated externally, but internally.

Labor income process— We want to calibrate the labor endowment shocks to replicate the typical dynamics of individual earnings in the U.S. economy. The best source of data for this is the *Panel Study of Income Dynamics*. A decent approximation to U.S. individual earnings dynamics is an AR(1) process like

$$\ln \varepsilon_t = \rho \ln \varepsilon_{t-1} + \epsilon_t, \quad \text{with } \epsilon_t \sim N(0, \sigma_\epsilon)$$

where the autocorrelation coefficient is $\rho = 0.95$ and the standard deviation of the shocks is $\sigma_\epsilon = 0.20$. More sophisticated estimates include a transitory component to capture measurement error, as well as less persistent shocks, and a fixed individual component to capture the effect of education, ability, etc.

Borrowing constraint— If the natural borrowing constraint is not a good choice for the problem at hand, one could calibrate the borrowing constraint in order to match, say,

¹³To be precise, the size of the K/Y ratio depends on whether we include residential capital. If we do, then the K/Y ratio is closer to 4.

the fraction of agents with negative net worth which is around 15% in the U.S. economy. The difficulty is that this strategy requires, again, an internal calibration.¹⁴

1.6 Notes

This class of model with a continuum of agents facing individual income shocks and trading a risk-free asset (money, in the original model) was introduced by Bewley (1986). Laitner (1992) studies one version of this economy with altruism. These papers are referenced by Huggett and Aiyagari. Bewley's economy is discussed in L-S textbook. Huggett (1993) analyzes the equilibrium in an endowment economy where agents trade an asset in zero net supply. Aiyagari (1994) generalizes the model to a production economy with an aggregate production function (the model described here). Huggett (1996) presents an OLG version of this model. The chapter by Rios-Rull in the book edited by Cooley (1995) contains a careful description of how to compute equilibria in this class of economies. So does Section 7 in Herr and Maussner.

¹⁴Net worth is total assets (cash, bonds, stocks, cars, retirement accounts, housing, etc.) minus total liabilities (credit card debt, mortgages, etc.). This is a one-asset model, so when we confront it to the data, in the data we collapse all types of wealth into a unique asset, net worth.

1 Some Applications of Bewley Models

1.1 How much saving for self-insurance (precautionary saving) in the US?

Aiyagari (1994) uses the economy we discussed to study the effect of idiosyncratic income risk on the aggregate saving rate. To calculate the saving rate, recall the resource constraint in this economy is

$$C_t + I_t = F(K_t, H_t).$$

The saving rate is defined as $s_t = \frac{I_t}{F(K_t, H_t)} = \frac{K_{t+1} - (1-\delta)K_t}{F(K_t, H_t)}$. In the stationary equilibrium this is $s = \frac{\delta K}{F(K, H)}$.

Aiyagari interprets the model period as 1 year and chooses $\beta = 0.96$ and $\phi = 0$. The production function is Cobb-Douglas with capital income share $\alpha = 0.36$, depreciation rate $\delta = 0.08$. The utility function is CRRA $U(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $\gamma = \{1, 3, 5\}$. The earnings process is a Markov chain with 7 states with a transition probability chosen according to the Tauchen method to mimic the moments of

$$\log(\varepsilon_t) = \rho \log(\varepsilon_{t-1}) + \sigma \sqrt{1 - \rho^2} \epsilon_t, \epsilon_t \sim N(0, 1)$$

Aiyagari considers a several values for $\sigma = \{0.2, 0.4\}$ and $\rho = \{0, 0.3, 0.6, 0.9\}$.

Aiyagari contrasts the equilibrium interest rate and the saving rate in the model with idiosyncratic income risk and incomplete markets with those that prevail in a model with complete markets. Recall that when markets are complete $r = \frac{1}{\beta} - 1$ in the steady state.¹ For $\beta = 0.96$, the complete markets interest rate is $r^{CM} \approx 4.17\%$. Therefore, in the complete markets model, the steady state saving rate is $s^{CM} = \frac{\delta K}{K^\alpha H^{1-\alpha}} = \frac{\delta}{\left(\frac{K}{H}\right)^{\alpha-1}} = \frac{\delta \alpha}{r + \delta}$, where the last step uses the firm's first order condition with respect to capital. Under the given parameter values, $s^{CM} = 23.67\%$.

The tables below show how the equilibrium interest rate and the saving rate vary with parameters in the incomplete market's model.

Aiyagari finds that with log utility and *iid* shocks ($\rho = 0$, $\gamma = 1$) the precautionary saving rate is approximately zero. The reason is that agents with low risk aversion

¹As we saw in the first lectures, the Arrow-Debreu version of the neoclassical growth model yields $\frac{p_t}{p_{t+1}} = \frac{1}{\beta} \frac{u'(c_t)}{u'(c_{t+1})}$ and $\frac{p_t}{p_{t+1}} = F_K(K_{t+1}, H_{t+1}) + 1 - \delta$. The sequential version of the same model yields $1 + r_{t+1} = F_K(K_{t+1}, H_{t+1}) + 1 - \delta$. Combining the two, we have that in the steady state $\frac{1}{\beta} = 1 + r$.

Table 1: Interest rate and saving rate in Aiyagari (1994)

ρ	σ	γ	r	s
0	0.2	1	4.17	23.67
0.9	0.2	1	3.93	23.73
0.9	0.4	1	3.30	25.47
0	0.2	5	4.09	23.83
0.9	0.2	5	2.53	27.36
0.9	0.4	5	-0.35	37.63

who face income shocks with low persistence are not too concerned about the missing insurance markets and accumulate little capital for self-insurance. With *CRRA* utility, risk-aversion parameter equal to 5, and autocorrelation of the income shocks of 0.9, the size of precautionary savings is larger. These households are much more concerned about self-insurance since with high persistence income can remain low for a long time, plus for them consumption fluctuations are very costly. In general, the departures from complete markets are largest when shocks are more volatile/persistent and when households have a strong precautionary motive.²

Note the importance of equilibrium considerations here: in the income fluctuation problem, one can always set r close enough to $1/\beta - 1$ in order to generate any desired amount of savings (even $+\infty$). The equilibrium model imposes more discipline because, given demand for capital, there is only one value of the interest rate consistent with the consumption/saving decisions of the households.³

1.1.1 Comparative Statics on Precautionary Saving

The Figure depicting the equilibrium is helpful to perform some comparative statics on the equilibrium. We are interested in the effects on the borrowing constraint ϕ , risk aversion, the persistence and variance of the idiosyncratic shocks.

²With *CRRA* utility the coefficients of relative risk aversion and relative prudence are tightly linked.

³In particular, the logic that the closer is $1+r$ to $1/\beta$, the higher are savings is reversed in equilibrium because $(1+r)\beta = 1$ is the complete-markets benchmark which has the lowest saving rate (no saving for self-insurance).

Borrowing limit– Suppose we increase ϕ , i.e. we slacken the liquidity constraint and increase the maximum amount that can be borrowed by the individual. Graphically, the asset supply curve $A(r)$ shifts upward, with $K(r)$ constant, which leads to a rise in the interest rate and a reduction of precautionary savings. The interest rate increases because more individuals have negative wealth, so the supply of capital falls at any given r . Note here an important point: agents can both save and borrow for self-insurance. The availability of a generous borrowing limit reduces the need for precautionary saving.

Risk aversion– The $K(r)$ curve is unaffected by changes in preferences. If we raise risk aversion, the $A(r)$ curve shifts downward: individuals are more concerned about consumption smoothing, so they accumulate higher buffer-stock savings: for any given r , $A(r)$ is larger. This leads to a lower equilibrium interest rate.

Changes in the income process– Suppose we increase the variance of the uninsurable income shock. $K(r)$ is unchanged, but the supply of capital $A(r)$ would go up (curve shifts down), as individuals accumulate more savings to cope with the higher uninsurable uncertainty of their income. Increasing the persistence of the shock has a similar effect, but in general it is quantitatively bigger.

1.2 Understanding Wealth Inequality

The Bewley models contain a theory of consumption and wealth inequality. In the model, agents are ex-ante equal and become different as time goes by due to variation in the realization of their income shocks. It is a theory of inequality based largely on luck. In response to shocks, agents choose optimally how much to consume and how much to save. Hence, different paths of shocks induce different levels of consumption and wealth across agents. Note that, in the absence of endogenous labor supply, the model has nothing to say on earnings (wages times hours worked) inequality.

For a full description of facts on earnings and wealth inequality, one should refer to a paper by Kuhn and Rios-Rull (2015). The following table, reproduced from the paper above, provides the key statistics for the US (from the *Survey of Consumer Finances*, 2013).⁴ The key fact to observe is that wealth is much more unequally distributed than earnings. Both distributions are positively (right) skewed (mean > median), but the wealth

⁴Updated statistics for more recent years are available at <https://sites.google.com/site/kuhnecon/home/us-inequality?authuser=0>.

distribution much more so that the earnings distribution: the top 1% of the wealth distribution owns 35% of the US wealth.

U.S. SCF (2013). Values of earnings and wealth are in thousands of 2013 dollars								
	Mean	$\frac{Mean}{Median}$	Gini	CV	Top 95-99	Top 99-100	Bottom 20%	Share top 1%
Earnings	63.9	1.96	0.67	3.69	293.4	1,203	-0.3	18.8
Wealth	528.2	6.49	0.85	6.81	3,614.7	18,734	-17.8	35.5

After the model is calibrated and simulated, we can use it as a measurement tool for following question: how far can we go in explaining wealth inequality when idiosyncratic earnings shocks are the only source of heterogeneity among households? The typical answer is that the standard model generates too much asset holdings at the bottom and too little at the top: the Gini generated by the model economy is around 0.6 –much smaller than the data value 0.85 (Cagetti and De Nardi, 2006).⁵

This standard model needs to be modified to 1) introduce an extra incentive for the rich to accumulate capital, 2) reduce the incentives for the poor to save for self-insurance purposes.

Inequality at the bottom– Modelling carefully the welfare state goes a long way in generating the right amount of asset holdings at the bottom. Once we introduce public insurance schemes (e.g., social security, housing benefits, child benefits, unemployment insurance), the incentives for private self-insurance are much reduced because some of these benefits are means-tested. See, for example, Hubbard, Skinner and Zeldes (1995).

Inequality at the top– To improve the quantitative explanation of inequality at the top, several alternatives have been proposed. 1) Quadrini (2000) explores the role of entrepreneurship. Implicitly, entrepreneurs have a higher return on their investment, hence a stronger incentive to accumulate. Empirically, a large fraction of wealth at the top is held by entrepreneurs. 2) Krusell and Smith (1997) study heterogeneity in discount factors. A Markov process regulates transitions between two levels of patience $\{\beta_L, \beta_H\}$. In the β_H state, households are more patient and save more. Small differences in β lead to a jump in the wealth Gini. 3) De Nardi (2004) studies the role of bequest. If (rich) households have a stronger bequest motive than poor households, this represents

⁵See the Appendix for a brief discussion of the Gini coefficient.

an additional reason to save. 4) Castaneda, Diaz-Jimenez and Rios-Rull (2003) add to an otherwise standard income process a very high realization of earnings (roughly 200 times larger than the mean) which occurs with a very low probability. They argue that income data are top-coded, so one does not observe these realizations in the data, even though they exist. 5) Benhabib, Bisin and Zhu (2016) and Benhabib, Bisin and Luo (2019) show that idiosyncratic returns to wealth are important for matching the tails of the wealth distribution. Models of entrepreneurship in the spirit of Quadrini (2000) are a way to endogenize such returns.

1.3 A Bewley Model with Entrepreneurship

We have argued that, empirically, a large fraction of wealth at the top is held by entrepreneurs. Here we want to write down a model where we have both entrepreneurs (individuals owning their own business) and workers (individuals working for someone else) and argue that it can replicate the upper tail in the wealth distribution. This model is a simplified version of Kitao (2008).

Demographics— Agents are infinitely lived. Every period agents choose an occupation. Entrepreneurs run their own business, and workers supply labor in the market. Entrepreneurs can manage one project which combines her managerial ability, capital and labor.

Endowments— Each agent is endowed with labor productivity $\varepsilon \in E$ and entrepreneurial ability $\theta \in \Theta$. Each one is regulated by a Markov chain, which we denote, respectively, by π_ε and π_θ .

Preferences— They are standard, time-separable preferences defined over streams of consumption, with discount factor β

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

Technology— There are two production sectors. The corporate sector is CRS with production $Y = F(K, H)$. Capital is rented by the corporate firms at the risk-free rate r and depreciates at rate δ . The non-corporate sector is composed by many entrepreneurs who run their own project according to the production function $y = f(k, n, \theta) = \theta k^{\nu_1} n^{\nu_2}$, with $\nu_1 + \nu_2 < 1$. This formulation mirrors the classic span of control model developed by

Lucas (1978), i.e. entrepreneurial ability determines the size of the firm. Also, note that $(1 - \nu_1 - \nu_2)$ is the share of output retained by the entrepreneurs as rents or profits from his managerial skills.

Financial markets— Agents cannot borrow to finance consumption. A competitive banking sector uses the deposits from households to rent capital to entrepreneurs and corporations. Because the banking sector is competitive, competition among banks drives bank profits to zero, so the lending rate at which entrepreneurs and corporations rent capital is the same as the rate that households receive for their deposits and is equal to r . Entrepreneurs are subject to the borrowing constraint $k \leq \lambda a$, $\lambda \geq 1$. Hence, they can invest in their project up to λa .⁶

Problems of the worker and entrepreneur— The problem of the worker is written as

$$\begin{aligned} V^w(a, \varepsilon, \theta) &= \max_{c, a', i} u(c) + \beta \{i E[V^w(a', \varepsilon', \theta')] + (1 - i) E[V^e(a', \varepsilon', \theta')]\} \\ &\quad s.t. \\ c + a' &= (1 + r)a + w\varepsilon \\ a' &\geq 0 \end{aligned}$$

where i governs the occupational choice: $i = 1$ if the worker chooses to remain a worker next period and $i = 0$ if he chooses to become an entrepreneur.

The problem of the entrepreneur is written as:

$$\begin{aligned} V^e(a, \varepsilon, \theta) &= \max_{c, a', i} u(c) + \beta \{i E[V^w(a', \varepsilon', \theta')] + (1 - i) E[V^e(a', \varepsilon', \theta')]\} \\ &\quad s.t. \\ c + a' &= (1 + r)a + \pi(a, \varepsilon, \theta) \\ a' &\geq 0 \end{aligned}$$

where $\pi(a, \varepsilon, \theta)$ are profits after payments of factors of production and loans and are

⁶Such a constraint is isomorphic to collateral constraint on debt and can be motivated as arising from a limited enforcement problem. Consider an entrepreneur with wealth a who rents k units of capital. The entrepreneur can steal a fraction $\frac{1}{\lambda}$ of rented capital. As a punishment, he would lose his wealth. In equilibrium, the financial intermediary will rent capital up to the point where individuals would have an incentive to steal the rented capital, implying a collateral constraint $\frac{k}{\lambda} \leq a$ or $k \leq \lambda a$.

determined as:

$$\begin{aligned} \pi(a, \varepsilon, \theta) &= \max_{k, n} f(k, n, \theta) - (r + \delta)k - w \max\{n - \varepsilon, 0\} \\ &\quad s.t. \\ k &\leq \lambda a. \end{aligned}$$

The borrowing limit is proportional to the entrepreneur's net worth and the parameter λ determining the tightness of the borrowing limit is common across agents.

An important implication of the way we wrote the problem of the entrepreneur is that an entrepreneur dedicates his time endowment to his own project, but cannot sell his efficiency units in the labor market, which amplifies the risk of entrepreneurship, in case θ turns out to be very low. A limit of this model is that entrepreneurs never make an operating loss within a period, as they can always choose $k = n = 0$ and earn the risk free rate on saving a .⁷

Who becomes an entrepreneur? Entrepreneurs in this model are characterized by a higher level of assets a and entrepreneurial ability θ than workers. Because entrepreneurial ability determines the size of the firm, conditional on wealth, more productive entrepreneurs (with higher values of θ) undertake larger projects and rely more heavily on outside financing. The fact that entrepreneurial investment is limited by the entrepreneur's wealth can lead to scenarios in which productive but wealth-poor entrepreneurs are not able to invest as much as they would want to, thus generating misallocation.

Why are entrepreneurs richer than workers? There are various channels that make entrepreneurs hold significantly more wealth than workers. First, entrepreneurs have incentives to accumulate wealth in order to grow out of the borrowing constraint. Second, as above, entrepreneurship is risky, so there is precautionary saving against entrepreneurial risk. Third, individuals with high entrepreneurial ability have access to a saving/investment technology with higher return than workers (who save at rate r), and therefore they accumulate wealth faster.

⁷If there was a fixed entry cost of becoming an entrepreneur and a lower operating cost that must be paid every period, entrepreneurs may choose to stay in business even though, for one or more periods, $\pi < 0$. This will happen if θ is mean reverting.

1.4 Role of Redistributive Taxation

In this model-economy where some of the earnings risk is uninsurable because of market incompleteness, there could be scope for public insurance, i.e. government intervention through taxation and redistribution from the rich-lucky to the poor-unlucky.

In a model with exogenous labor supply, suppose that the government (as in a Ramsey-style optimal taxation problem) chooses a labor income tax τ and a lump-sum subsidy t in order to maximize the ex-ante welfare of the households. Clearly, the optimal tax rate in the Aiyagari economy would be $\tau = 1$. The government would tax away all income and redistribute equally across all agents. This policy would achieve the first-best because taxation entails no distortions and no loss of efficiency and, at the same time, generates full insurance.

A more interesting economy is one with an endogenous margin of labor/leisure choice where there is a *trade-off between insurance and efficiency*. To evaluate this trade-off, we develop a variation of the benchmark Aiyagari economy with endogenous labor supply which is a simplified version of Floden and Linde (2001).

Model— First of all, period utility is given by $u(c, l)$ i.e., we introduce leisure $l \in (0, 1)$ in order to have a margin where distortions matter. This means that we will have an optimal policy for labor supply $h(a, \varepsilon)$. Notice also that the agents might be using their elastic labor supply to self-insure. Take an agent who is liquidity constrained and has a low realization of the productivity shock: to keep his consumption high, he could intensify his labor supply.

The new budget constraint reads

$$c + a' = (1 + r)a + (1 - \tau)w\varepsilon h + t,$$

where τ is a flat earnings tax, and t is the lump-sum transfer of the government. Note that the tax scheme is progressive because the least productive agents pay less taxes but get the same transfer. In other words, the average tax rate faced by a household $(\tau - t/w\varepsilon h)$ is increasing in labor income $w\varepsilon h$.

With leisure, the new equilibrium condition in the labor market becomes

$$H = \int_{A \times E} \varepsilon h(a, \varepsilon) \lambda^*(da, d\varepsilon).$$

The government budget constraint (balanced in equilibrium) reads

$$T = \tau wH,$$

where T denotes aggregate transfers (and in equilibrium it equals t).

The definition of the recursive competitive equilibrium for this economy is very similar to the benchmark case (the main differences are the existence of a decision rule for leisure and of the balanced budget condition of the government).

Computation— Suppose τ is a parameter of the problem. Then, we need to guess $\{r^0, H^0\}$. Given r^0 if the production function is CRS we get w^0 . From the government budget constraint, we recover t , which is equal to T given our assumption of measure 1 population, and we have all the inputs needed for solving the household problem. The rest of the computation algorithm is the same as before, with the caveat that we need to check two equilibrium conditions, asset market and labor market clearing.

Question— Floden and Linde ask the following question: what is the level of government redistribution that maximizes welfare? What welfare gains does such redistribution imply for individuals compared to the status quo?

In an economy with heterogeneous agents there is not a unique welfare function, it all depends on what weights are assigned to each type. Floden and Linde assume an equal-weight social welfare function, i.e. they solve

$$\max_{\tau} W(\tau) = \int_{A \times E} V^*(a, \varepsilon; \tau) \lambda_{\tau}^*(da, d\varepsilon),$$

where $V^*(a, \varepsilon; \tau)$ is the value function associated to the competitive equilibrium indexed by τ . It is a version of the Ramsey taxation problem with incomplete markets.

Intuitively, for low levels of redistribution, welfare is low because individuals have a large amount of undesired consumption fluctuations; for very high levels of taxes, consumption insurance is very high but at the same time heavy distortions on labor supply are imposed, which reduce output. So, there will be an interior level of τ , call it τ^* , that maximizes welfare.

Results— Floden and Linde calibrate two versions of the model: one to the US economy and one to Sweden. The major difference between the two countries is the wage process: shocks are much less variable and less persistent than the U.S., so wage fluctuations in Sweden are more insurable through precautionary savings. The key reason,

perhaps, is that unions and other wage compressing institutions reduce wage volatility already before taxes.

Table 2: Optimal Policy in Floden and Linde

	U.S.	Sweden
τ^*	46%	27%
τ (Calibrated status quo)	36%	57%
$t^* \times \text{GDP}$	15%	2%
$t \times \text{GDP}$ (Data)	8%	21%
Welfare gain	2%	8%

The results suggest that the the Swedish government does “too much” redistribution, as the current status quo features tax rates and transfers that are higher than optimal. Through the lens of the model, the US government appears to do “too little” redistribution, with status quo tax rates and transfers being lower than optimal. Switching to the optimal tax-transfer system entails welfare gains in both countries. Specifically, one would have to increase the life-time consumption of US (Swedish) households by 2% (8%) in order to make them indifferent between remaining in the status quo and being under the optimal allocation.

Keep in mind that the authors only considered a flat tax on labor income. Often governments use capital income tax for redistribution, which is much more distortionary, but also more progressive since the rich have a higher capital-income to labor-income ratio in this model. As far as transfers are concerned, it would be more efficient to condition the transfer on ε (i.e. agent with low ε would receive more), but Floden-Linde assume that ε is private information, hence unobservable to the government, so the government cannot condition on ε . A more sophisticated approach to optimal taxation (called the Mirleesian approach) would ask the question of what is the optimal tax scheme that maximizes welfare, given this private information constraint. In other words, the planner would not tie its hands to tax with a flat tax and redistribute lump sum.

1.5 Optimal Quantity of Debt

Motivated by concerns about the high level of debt in the US economy, Aiyagari and McGrattan (1998) study the quantity of government debt that maximizes welfare (same social welfare function as Floden and Linde) in the U.S. economy. The economy is like the one in Floden and Linde, except for the government sector.

Every period the government has two type of outlays: transfers T , and interest payments on the stock of existing public (one-period) debt B . These outlays need to be financed by distortionary taxes on labor income at rate τ . The government budget constraint reads

$$T + (1 + r) B = B' + \tau w H,$$

where in the stationary equilibrium $B' = B$. Here capital letters denote aggregate quantities.

Government debt is an additional risk-free asset and, by no arbitrage, it must carry the same rate of return as capital in equilibrium. Debt has a number of effects on the equilibrium:

1. debt is costly because financing interest payments on debt requires distortionary taxes.
2. public debt crowds-out productive capital via higher interest rates because some of the savings are shifted away from production to government debt. Note that the equilibrium condition in the asset market is now

$$K(r) + B = A(r) \Rightarrow K(r) = \tilde{A}(r) \equiv A(r) - B.$$

Note that, in the graphical representation of the stationary equilibrium, the aggregate demand is still $K(r)$. The aggregate supply shifts to the left by an amount B , so the interest rate rises unambiguously.

3. government debt has the advantage of providing households with additional means of smoothing consumption. The shift to the left of the aggregate supply curve is as if the effective borrowing constraint is looser.

After calibrating their model, Aiyagari and McGrattan conclude that the optimal quantity of debt is very close to the actual one for the U.S. economy, around 2/3 of GDP.

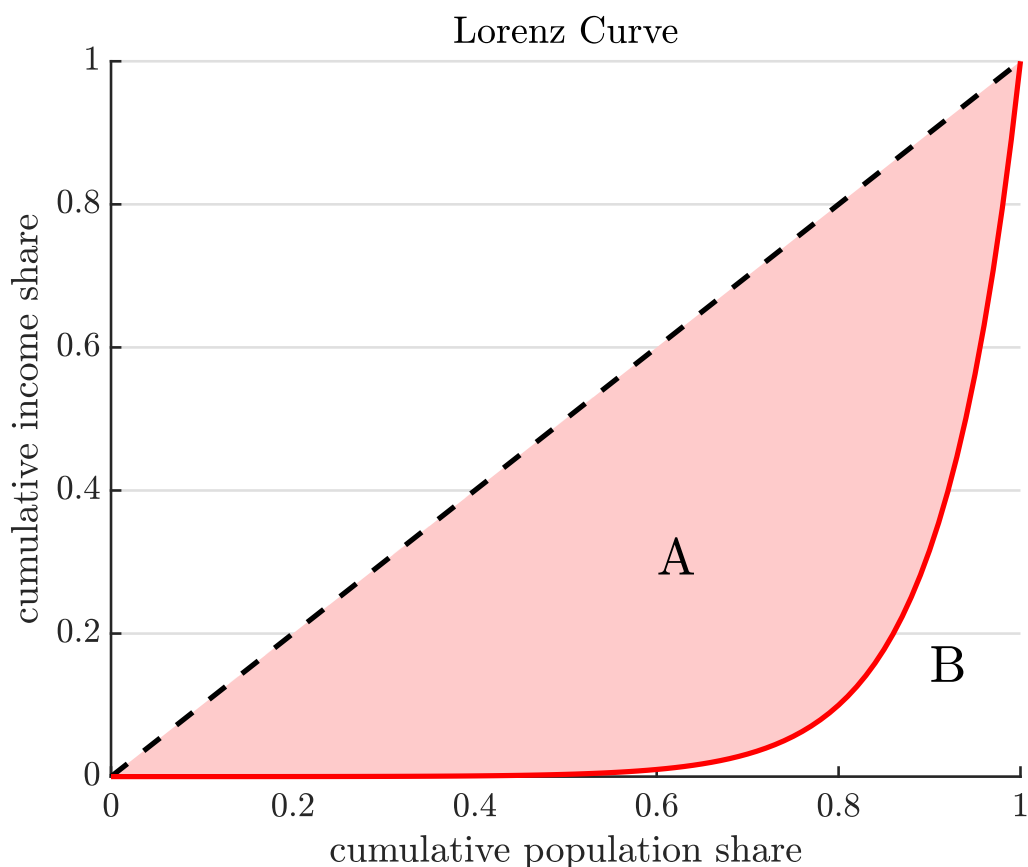
References

- [1] Aiyagari, Rao and Ellen Mc Grattan (1998); The Optimum Quantity of Debt, *Journal of Monetary Economics*.
- [2] Benhabib, Jess and Alberto Bisin and Shenghao Zhu (2016): The Wealth Distribution in Bewley Models with Investment Risk, *Journal of Economic Theory*
- [3] Benhabib, Jess and Alberto Bisin and Mi Luo (2019): Wealth distribution and social mobility in the US: A quantitative approach, *American Economic Review*
- [4] Cagetti, Marco and Mariacristina De Nardi (2006); Entrepreneurship, Frictions and Wealth, *Journal of Political Economy*
- [5] Castaneda, Ana, Javier Diaz-Jimenez and Jose-Victor Rios-Rull (2003); Accounting for earnings and wealth inequality, *Journal of Political Economy*
- [6] De Nardi, Mariacristina (2004); Wealth Inequality and Intergenerational Links, *Review of Economic Studies*
- [7] Floden Martin and Jesper Linde (2001); Idiosyncratic Risk in the U.S. and Sweden: Is there a Role for Government Insurance?, *Review of Economic Dynamics*
- [8] Hubbard, Skinner and Zeldes (1995); Precautionary saving and social insurance, *Journal of Political Economy*
- [9] Kitao, Sagiri (2008); Entrepreneurship, taxation and capital investment, *Review of Economic Dynamics*
- [10] Krusell, Per and Tony Smith (1997); Income and wealth heterogeity, portfolio choice, and equilibrium asset returns, *Macroeconomic Dynamics*

Appendices

The Lorenz curve is a graphical representation of the cumulative distribution of income, wealth, consumption. Suppose we want to depict the Lorenz curve of income, as in Figure 1. On the horizontal axis we list the cumulative percentage of the population arranged in increasing order of income, i.e. points on the x-axis refer to the poorest 10%, 20%, etc. of the population. On the vertical axis we measure the percentage of total income that accrues to the fraction of the population depicted on the x-axis. The 45° line represents an equal income distribution. The area between the line of perfect equality and the observed Lorenz curve captures the amount of inequality, i.e. how far the economy is from perfect equality.

Figure 1: Lorenz Curve



The Gini coefficient is the ratio of the area between the line of perfect equality and the observed Lorenz curve to the area between the line of perfect equality and the line

of perfect inequality. The higher the coefficient, the more unequal the distribution is. In the diagram, this is given by the ratio $A/(A+B)$, where A and B are the areas of regions as marked in the diagram. Since $A+B = 0.5$, the Gini coefficient can also be expressed as $2A$ or $1-2B$.

Non-Stationary Equilibria in Bewley Models

The problem we studied so far is of the form:

$$\begin{aligned}
 V(a, \varepsilon) &= \max_{c, a'} u(c) + \beta \sum_{\varepsilon' \in E} \pi(\varepsilon' | \varepsilon) V(a', \varepsilon') \\
 &\quad s.t. \\
 c + a' &= (1 + r) a + w\varepsilon \\
 a' &\geq -\phi
 \end{aligned} \tag{1}$$

We said that a stationary equilibrium is a value function $V : S \rightarrow \mathbb{R}$; policy functions for the household $a' : S \rightarrow \mathbb{R}$, and $c : S \rightarrow \mathbb{R}_+$; policies for the firm H and K ; prices r and w ; and, a stationary measure λ such that all agents optimize, markets clear, and the invariant distribution λ satisfies

$$\lambda(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} Q((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda(da, d\varepsilon)$$

where Q is the transition function:

$$Q((a, \varepsilon), \mathcal{A} \times \mathcal{E}) = I_{\{a'(a, \varepsilon) \in \mathcal{A}\}} \sum_{\varepsilon' \in \mathcal{E}} \pi(\varepsilon' | \varepsilon)$$

What made this problem tractable was:

1. our the assumption that ε is iid across households, so no aggregate uncertainty
2. our focus on a stationary equilibrium in which λ does not change over time, even though individual agents are subject to shocks that change their idiosyncratic a and ε from one period to another. Because the distribution is stationary, r and w are constant and the algorithm we use to solve for a stationary equilibrium is simply a search over the value of r that clears the asset market.

We would next like to extend the scope of this framework to study aggregate dynamics. For example, suppose that we want to study the welfare effect of an unexpected permanent rise in income taxes, or to analyze the consequences of a financial liberalization, i.e. a sudden relaxing of the borrowing constraints. Another interesting application is to introduce aggregate fluctuations (business cycles) in our incomplete markets economy and ask how different the results are from the business cycles results one obtain in

a representative agent framework. For example in the representative agent model one famous conclusion (due to Robert Lucas) is that eliminating business cycles has a small welfare effect. Is this conclusion still true in an incomplete markets economy where the representative agent construct no longer applies? We will study two different applications:

1. *Transition dynamics*: we consider one-time unexpected changes in the environment and study how the economy evolves in response to such changes. To fix ideas, we will think of one-time changes in the tax rate the government levies on labor earnings. Here, the path for prices is deterministic, and we do not need to keep track of the distribution as an additional state, as time is a sufficient statistic.
2. *Aggregate uncertainty*: we consider aggregate shocks (business cycles) and study how the economy evolves in response to such shocks. Here, the path for prices is stochastic, so we need to keep track of the distribution as an additional state.

1 Transitional Dynamics in Bewley Models

In this section we consider a one time permanent policy change and study the entire transition path across steady states. This is particularly useful if, for example, we want to study the welfare effect of a tax reform, e.g. a permanent rise in the labor income tax rate from τ^* to τ^{**} . Suppose that taxes are used to finance government transfers t which are lump-sum. It is important that the policy change is *unexpected*. If it was anticipated, agents would take actions in advance, as soon as the new information is revealed, to guarantee that their consumption path throughout the transition is as smooth as possible.

Steady-state comparison— The simplest way to do this is computing a stationary RCE for these two levels of the tax rate and compare aggregate variables and welfare between the two steady-states. However, this approach is not fully satisfactory. It can only be used to assess whether a household would prefer to live in the stationary equilibrium of an economy with tax rate equal to τ^* or in the stationary equilibrium of an economy with tax rate equal to τ^{**} . For example, it is useful to answer the question whether an individual would prefer to live, forever, in a country with a given tax rate τ^* or in another country with tax rate τ^{**} .

Transition— A more interesting and relevant policy question is: consider a household living in the stationary equilibrium of an economy with initial tax rate τ^* . What is the welfare change (gain or loss), for this household, associated to a rise in the tax rate from τ^* to τ^{**} ? To answer this question properly, one needs to compute the whole transition: the new policy will change the individual consumption/saving and labor supply decision, hence aggregate prices and will induce dynamics away from the current steady-state towards the new one (assuming the system has stable dynamics). See Domeij and Heathcote (2003) for an example that applies these techniques.

How do we attack this problem? Since the transition is characterized by a sequence of aggregate prices and quantities, we need to modify appropriately the definition of recursive competitive equilibrium.

1.1 Definition of Equilibrium with Transition

The economy at time $t = 0$ is in steady-state with stationary distribution $\lambda_0 = \lambda^*$ over assets and individual productivities and tax rate τ^* . At the end of period $t = 0$, the government announces that from $t = 1$ onward the tax policy will change to $\tau^{**} > \tau^*$ and that the additional revenues will augment the lump-sum transfer t_t . Hence, the relevant tax sequence needed to compute the equilibrium is

$$\tau_t = \begin{cases} \tau^*, & \text{for } t = 0 \\ \tau^{**}, & \text{for } t \geq 1. \end{cases}$$

Since policies (τ_t, t_t) are now a function of time, value functions, policy functions and aggregate prices (r_t, w_t) are time-varying.

First, let's define the household problem at time t in recursive form

$$\begin{aligned} V_t(a, \varepsilon) &= \max_{c_t, a_{t+1}} u(c_t(a, \varepsilon)) + \beta \sum_{\varepsilon_{t+1} \in E} \pi(\varepsilon_{t+1} | \varepsilon) V_{t+1}(a_{t+1}(a, \varepsilon), \varepsilon_{t+1}) \\ &\quad s.t. \\ c_t(a, \varepsilon) + a_{t+1}(a, \varepsilon) &= (1 + r_t)a + (1 - \tau_t)w_t\varepsilon + t_t \\ a_{t+1}(a, \varepsilon) &\geq -\phi \end{aligned}$$

It is important to understand that the transitional dynamics induced by the tax reform are *deterministic*. Since we know the exact future path for taxes, we know there will be a deterministic path for prices and for the distribution λ_t . Therefore, we do not need to

keep track of the distribution as an additional state, as time is a sufficient statistic. We will see that in presence of aggregate uncertainty, instead, the distribution λ becomes an aggregate state.

Let's denote the initial stationary distribution with λ^* . Given an initial distribution λ^* , and a sequence of tax rates $\{\tau_t\}_{t=0}^\infty$, a *recursive competitive equilibrium* is a sequence of value functions $\{V_t\}_{t=0}^\infty$ and policy functions for households $\{c_t, a_{t+1}\}_{t=0}^\infty$, firm choices $\{H_t, K_t\}_{t=0}^\infty$, prices $\{w_t, r_t\}_{t=0}^\infty$, government transfers $\{t_t\}_{t=0}^\infty$ and distributions $\{\lambda_t\}_{t=0}^\infty$ such that, for all t :

1. given prices $\{r_t, w_t\}$ and policies $\{\tau_t, t_t\}$, the policy functions $a_{t+1}(a, \varepsilon)$ and $c_t(a, \varepsilon)$ solve the household's problem (2) and $V_t(a, \varepsilon)$ is the associated value function
2. given prices $\{r_t, w_t\}$, the firm chooses optimally capital K_t and labor H_t , such that $r_t + \delta = F_K(K_t, H_t)$ and $w_t = F_H(K_t, H_t)$
3. markets clear:
 - labor market: $H_t = \int_{A \times E} \varepsilon \lambda_t(da, d\varepsilon) = H$
 - asset market: $K_{t+1} = \int_{A \times E} a_{t+1}(a, \varepsilon) \lambda_t(da, d\varepsilon)$
 - goods market: $\int_{A \times E} c_t(a, \varepsilon) \lambda_t(da, d\varepsilon) + K_{t+1} - (1 - \delta) K_t = F(K_t, H_t)$
4. the government budget constraint is balanced: $t_t = \tau_t w_t H$
5. for all $\mathcal{A} \times \mathcal{E} \in \Sigma_S$, the probability measure λ_{t+1} satisfies

$$\lambda_{t+1}(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} Q_t((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda_t(da, d\varepsilon),$$

where Q_t is the transition function defined as

$$Q_t((a, \varepsilon), \mathcal{A} \times \mathcal{E}) = I_{\{a_{t+1}(a, \varepsilon) \in \mathcal{A}\}} \sum_{\varepsilon_{t+1} \in \mathcal{E}} \pi(\varepsilon_{t+1} \mid \varepsilon) \quad (3)$$

and $\lambda_0 = \lambda^*$.

1.2 Numerical Computation of the Transition Path

We will assume that after T periods, with T arbitrarily large but finite, the economy will settle to the final steady-state. This assumption is helpful because it allows us to guess a finite sequence of aggregate capital stocks and use backward induction for the solution of the household problem.

To compute the equilibrium follows these steps:

1. Fix T (say $T = 200$).
2. Compute the initial steady state objects $\{V^*, c^*, a'^*, \lambda^*, K^*\}$ corresponding to $\tau = \tau^*$ and the final steady state objects $\{V^{**}, c^{**}, a'^{**}, \lambda^{**}, K^{**}\}$ corresponding to $\tau = \tau^{**}$.
3. Guess a sequence of aggregate capital stocks $\{\hat{K}_t\}_{t=1}^T$ of length T such that $\hat{K}_1 = K^*$ (capital at time 1 is predetermined at time $t = 0$ which is a steady-state) and $\hat{K}_T = K^{**}$. For example, you can make a guess based on the representative agent equivalent of your economy. Note that $H_t = H$ (i.e. constant) for every t in absence of endogenous labor supply. Hence, it is easy to determine, for each t ,

$$\begin{aligned}\hat{w}_t &= F_H(\hat{K}_t, H), \\ \hat{r}_t &= F_K(\hat{K}_t, H) - \delta, \\ \hat{t}_t &= \tau_t \hat{w}_t H,\end{aligned}$$

which are all the elements we need in the budget constraint of the household to solve the household problem at time t .¹

4. Since we know that $c_T(a, \varepsilon) = c^{**}(a, \varepsilon)$, $a_{T+1}(a, \varepsilon) = a'^{**}(a, \varepsilon)$ and $V_T(a, \varepsilon) = v^{**}(a, \varepsilon)$, we can solve the household problem *by backward induction*, by iterating either on the Euler equation or on the value function.

– Euler equation

$$\begin{aligned}& u_c \left((1 + \hat{r}_t) a + (1 - \tau_t) \hat{w}_t \varepsilon + \hat{t}_t - a_{t+1}(a, \varepsilon) \right) \\ & \geq \beta (1 + \hat{r}_{t+1}) \sum_{\varepsilon'} \pi(\varepsilon' | \varepsilon) u_c \left((1 + \hat{r}_{t+1}) a_{t+1}(a, \varepsilon) + (1 - \tau_{t+1}) \hat{w}_{t+1} \varepsilon' + \hat{t}_{t+1} - a_{t+2}(a_{t+1}, \varepsilon') \right).\end{aligned}$$

¹In this step, one can equally guess a path for the interest rate or for wages. Then, from the FOC's of the firm, one would recover aggregate capital at each t .

Note that the function $a_{t+2}(\cdot)$ is always known, so we can derive $\{\hat{c}_t(a, \varepsilon)\}_{t=1}^{T-1}$ and $\{\hat{a}_{t+1}(a, \varepsilon)\}_{t=1}^{T-1}$.

– Value function

$$V_t(a, \varepsilon) = \max_{a_{t+1} \geq 0} u((1 + \hat{r}_t)a + (1 - \tau_t)\hat{w}_t\varepsilon + \hat{t}_t - a_{t+1}(a, \varepsilon)) + \beta \sum_{\varepsilon'} \pi(\varepsilon' | \varepsilon) V_{t+1}(a_{t+1}(a, \varepsilon), \varepsilon')$$

Note that the function $V_{t+1}(\cdot)$ is always known, so we can derive $\{\hat{c}_t(a, \varepsilon)\}_{t=1}^{T-1}$, $\{\hat{a}_{t+1}(a, \varepsilon)\}_{t=1}^{T-1}$ and $\{\hat{V}_t(a, \varepsilon)\}_{t=1}^{T-1}$.

5. Given the policy functions, we can reconstruct the sequence of transition functions $\{\hat{Q}_t\}_{t=1}^{T-1}$ according to

$$\hat{Q}_t((a, \varepsilon), \mathcal{A} \times \mathcal{E}) = I_{\{\hat{a}_{t+1}(a, \varepsilon) \in \mathcal{A}\}} \sum_{\varepsilon_{t+1} \in \mathcal{E}} \pi(\varepsilon_{t+1} | \varepsilon)$$

and, since we know that $\hat{\lambda}_0 = \lambda^*$, we can recover the whole sequence of measures $\{\hat{\lambda}_t(a, \varepsilon)\}_{t=1}^{T-1}$ by using the law of motion

$$\hat{\lambda}_{t+1}(\mathcal{A} \times \mathcal{E}) = \int_{A \times E} \hat{Q}_t((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \hat{\lambda}_t(da, d\varepsilon).$$

Lastly, we can calculate the aggregate supply of assets $\hat{A}_{t+1}$ ²

$$\hat{A}_{t+1} = \int_{A \times E} \hat{a}_{t+1}(a, \varepsilon) \hat{\lambda}_t(da, d\varepsilon).$$

6. Check market clearing in the asset market in every period t , i.e. check if the guess of equilibrium capital stocks $\{\hat{K}_t\}_{t=1}^T$ is consistent with aggregate wealth $\{\hat{A}_t\}_{t=1}^T$ that households would accumulate when facing the sequence of prices induced by the proposed sequence of aggregate capital. In other words, choose a convergence criterion ε and check whether

$$\max_{1 \leq t \leq T} |\hat{A}_t - \hat{K}_t| < \varepsilon. \quad (4)$$

Note that if $|\hat{A}_T - K^{**}| < \varepsilon$ is satisfied, we have implicitly also checked that T is large enough.

²Alternatively, we could compute this integral by using simulation techniques. We simulate histories of length T for N workers (say $N = 10,000$) starting from the steady-state distribution at $t = 1$ (distribution that we also obtain by simulation). Note that when computing the optimal consumption and saving choices of each of the N individuals in our sample at time t , we must use the time t decision rules computed in the previous step.

7. If inequality (4) is not satisfied at every t , we need a new guess of the capital stock, for example

$$\hat{K}_t^{new} = s \times \hat{K}_t^{old} + (1 - s) \hat{A}_t,$$

and go back to step 3 with this new guess. $s = 0.5$ is a good starting point.³

The Appendix discusses some results on the monotonicity of transitional dynamics.

1.3 Computing the Welfare Change from the Tax Reform

The crucial question to ask, from a policy perspective, is: how much agents gain/lose from the tax reform? This question takes us deep into welfare analysis.

In the first steady-state, an agent with initial individual state (a, ε) has expected lifetime utility associated with the stationary decision rule $c^*(a, \varepsilon)$ given by

$$V^*(a, \varepsilon) = E_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t^*) | (a_0 = a, \varepsilon_0 = \varepsilon) \right], \quad (5)$$

where the conditional expectation E_0 is taken over histories of the shocks conditional on a time-zero realization of the shock equal to ε (as made clear by the second equality) and conditional to an agent's wealth level equal to a .

Consider now a policy change, like the permanent rise in the labor income tax from τ^* to τ^{**} from before. Define the expected discounted utility of an agent with initial state (a, ε) at date $t = 0$ going through the transition as

$$\tilde{V}_0(a, \varepsilon) = E_0 \left[\sum_{t=0}^{\infty} \beta^t u(\tilde{c}_t) | (a_0 = a, \varepsilon_0 = \varepsilon) \right],$$

where $\tilde{V}$ differs from V^* because it is computed using the sequence of decision rules $\{\tilde{c}_t(a, \varepsilon), \tilde{a}_{t+1}(a, \varepsilon)\}_{t=0}^{\infty}$ along the transition path.

How to compute discounted utilities V^* and $\tilde{V}_0$? If we solved the problem by value function iteration, then we already have the sequence of value functions $\tilde{V}_t$. Alternatively, if we solved the problem by iterating on the Euler equation, we can compute discounted utilities V^* and V^{**} by simulating S histories of an agent with initial conditions (a, ε) ,

³In practice, depending on the problem, you might find it convenient to choose a smaller s . The cost is that it slows down the algorithm, the advantage is that it might get the algorithm to converge when a larger s will not.

computing discounted utility for every history and then averaging across the S simulations. $\tilde{V}_0$ can be computed by backward induction, as above.

Conditional welfare change— The first question we can ask is: how much would an agent with initial state (a, ε) gain, in percentage terms of lifetime consumption, if she went through the transition induced by the policy reform, compared to the no-reform scenario where she lives in the initial steady-state forever? Precisely, we ask: “how much do we need to change consumption of the agent in the stationary equilibrium so that she would be indifferent between living through the tax reform and living in the pre-reform economy?”. So, welfare changes are expressed in terms of *consumption-equivalent variation*.

The answer to this question is a function $\omega(a, \varepsilon)$ that solves the equation

$$E_0 \sum_{t=0}^{\infty} \beta^t u((1 + \omega(a, \varepsilon)) c_t^*) = E_0 \sum_{t=0}^{\infty} \beta^t u(\tilde{c}_t).$$

If $\omega(a, \varepsilon) > 0$ then the agent is better off under the reform. If instead $\omega(a, \varepsilon) < 0$, then the agent is better off in the no-reform scenario.

For the case of power-utility this welfare calculation is really easy to make. When $u(c) = c^{1-\sigma}$, we can exploit the homogeneity of the utility function and the equation above becomes

$$(1 + \omega(a, \varepsilon))^{1-\sigma} V^*(a, \varepsilon) = \tilde{V}_0(a, \varepsilon) \Rightarrow \omega(a, \varepsilon) = \left(\frac{\tilde{V}_0(a, \varepsilon)}{V^*(a, \varepsilon)} \right)^{\frac{1}{1-\sigma}} - 1. \quad (6)$$

This welfare change is called *conditional welfare change*, because it is computed for an individual that is in a particular state (a, ε) . Thus, we can compute the welfare change for the rich household, the poor household, the productive household, the unproductive household, and so on. Moreover, we can compute the entire distribution of welfare changes and study whether the reform would be politically feasible, e.g. whether the majority of agents have positive welfare gains, hence they would support the reform.

Utilitarian social welfare change— The second type of welfare calculation is based on a Benthamian social welfare function that puts equal weight to every household in the initial steady-state (which is also the initial period of the reform), i.e. it uses the weighting criterion $\lambda^*(a, \varepsilon)$.⁴ The solution to this welfare calculation is *one number only*, ω^U ,

⁴Benthamian = of or relating to Jeremy Bentham (1748–1832), British philosopher, jurist, and social reformer, regarded as the founder of modern utilitarianism.

representing the unique percentage change one would have to apply to the consumption of every agent in the no-reform stationary equilibrium so that the social welfare function in the no-reform equilibrium is the same as the social welfare function obtained if the economy goes through the transition induced by a policy reform.

For the power utility case, the solution to this welfare calculation is

$$\omega^U = \left(\frac{\int_{A \times E} \tilde{V}_0(a, \varepsilon) \lambda^*(da, d\varepsilon)}{\int_{A \times E} V^*(a, \varepsilon) \lambda^*(da, d\varepsilon)} \right)^{\frac{1}{1-\sigma}} - 1.$$

So, ω^U computes the welfare change for “society”, where every agent in society is given equal weight: some will lose, some will gain and we average across everyone with equal weights.

An alternative interpretation of this welfare criterion is that of an *ex-ante* welfare gain, or welfare gain *under the veil of ignorance*. In other words, $\int_{A \times E} \tilde{V}_0(a, \varepsilon) \lambda^*(da, d\varepsilon)$ represents the expected discounted utility of a newborn agent who is dropped at random in the first steady-state without knowing at which point in the distribution she will be, i.e. under the veil of ignorance.

1.4 A Welfare Change Decomposition

Sources of changes in social welfare— The typical utilitarian equal-weight social welfare function

$$\int_{A \times E} V(a, \varepsilon) \lambda(da, d\varepsilon) = \int_{A \times E} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \lambda(da, d\varepsilon) \quad (7)$$

can increase for three reasons:

1. If average consumption $E(c_t)$ increases, since utility is monotone. It is easy to see that in presence of additional resources, at least one individual can be made better off, while the others receive the same utility. We can call this effect, the *level effect*. For example, by reallocating resources more efficiently a tax reform can increase average consumption.
2. If the uncertainty/volatility of each individual consumption path $\{c_t\}_{t=0}^{\infty}$ is reduced, since agents are risk averse. We can call this the *uncertainty effect*. For example, by redistributing from the lucky to the unlucky, the tax reform can provide additional insurance.

3. If inequality across individuals at any point in time is reduced, since the value function is concave. This is easily seen by just applying Jensen's inequality to the left hand side of (7). We can call this effect, the *egalitarian effect*. Note the key difference between 2) and 3): even when there is no uncertainty in consumption sequences, a policy that redistributes initial wealth more equally across agents would achieve a welfare gain, under this social welfare function. This makes the social welfare function not a fully desirable criterion when studying the welfare implication of a policy reform because it *mixes concern for risk/uncertainty with concern for interpersonal equality*.

Conditional welfare is preferable— For this reason, conditional welfare is a somewhat more satisfactory welfare criterion because only the level and uncertainty effect play a role. We now show that, starting from the conditional welfare criterion, the welfare change $\omega(a, \varepsilon)$ can be decomposed additively into 1 and 2. This is a useful result because, when evaluating the welfare implication of a policy reform, one can compute separately the welfare change due to the fact that the policy is 1) generating more/less aggregate consumption, and 2) increasing/decreasing consumption insurance.

Decomposition— For simplicity, we focus on a welfare comparison between steady-states, but the methodology can be extended to transitions. Let the welfare change between economy A (e.g. the initial steady state) and economy B (e.g. the final steady-state) be the ω that solves

$$E_0 \sum_{t=0}^{\infty} \beta^t u((1 + \omega) c_t^A) = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^B), \quad (8)$$

where E_0 is the expectation taken at date $t = 0$ conditional on an initial value for the pair (a, ε) . To simplify the notation, we have omitted the dependence of ω from the pair (a, ε) .

Let C^j denote the average consumption in economy $j = A, B$, i.e.

$$C^j = \int_{A \times E} c^j(a, \varepsilon) \lambda^j(da, d\varepsilon), \text{ with } j = A, B.$$

Then the *welfare gain of increased consumption levels* between A and B ω^{lev} is defined by

$$(1 + \omega^{lev}) C^A \equiv C^B. \quad (9)$$

Next, let the certainty equivalent consumption bundle be defined by $\bar{C}^j$ that solves

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^j) = \sum_{t=0}^{\infty} \beta^t u(\bar{C}^j). \quad (10)$$

Then, we can define the cost of uncertainty p^j as

$$\sum_{t=0}^{\infty} \beta^t u((1-p^j) C^j) \equiv \sum_{t=0}^{\infty} \beta^t u(\bar{C}^j), \quad (11)$$

which is the fraction of average consumption that an individual in economy j would be willing to give up to avoid all the risk associated to productivity fluctuations. Note that p^j depends on (a, ε) because $\bar{C}^j$ depends on (a, ε) .

Then, the *welfare gain of reduced uncertainty* between economy A and economy B is defined by

$$(1 + \omega^{unc}) (1 - p^A) \equiv 1 - p^B. \quad (12)$$

We are now ready to state:

Proposition (Floden, 2001): Assume that $u(xc)$ is “homogenous” in the sense that $u(xc) = g(x)u(c)$, then

$$1 + \omega = (1 + \omega^{unc}) (1 + \omega^{lev}) \Rightarrow \omega \simeq \omega^{unc} + \omega^{lev}.$$

Proof: The total welfare change is given by that value for ω that solves (8). Consider the expected utility in economy B , i.e. the R.H.S. of equation (8):

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^B) = \sum_{t=0}^{\infty} \beta^t u(\bar{C}^B) = \sum_{t=0}^{\infty} \beta^t u((1-p^B) C^B) = g(1-p^B) \sum_{t=0}^{\infty} \beta^t u(C^B),$$

where the first equality follows from (10), the second from (11) and the third from the homogeneity assumption. Then,

$$g(1-p^B) \sum_{t=0}^{\infty} \beta^t u(C^B) = g(1-p^B) \sum_{t=0}^{\infty} \beta^t u((1+\omega^{lev}) C^A),$$

where the equality follows from definition (9). Next,

$$\begin{aligned} g(1-p^B) \sum_{t=0}^{\infty} \beta^t u((1+\omega^{lev}) C^A) &= g(1-p^B) g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u(C^A) \\ &= g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u\left(\frac{1-p^B}{1-p^A} (1-p^A) C^A\right) \\ &= g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u((1-p^A) C^A) \end{aligned}$$

The line above follows from the homogeneity assumption. Using definition (11) ,

$$g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u((1-p^A) C^A) = g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u(\bar{C}^A).$$

From the definition of certainty equivalent consumption

$$g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) \sum_{t=0}^{\infty} \beta^t u(\bar{C}^A) = g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^A).$$

And, finally, from homogeneity

$$g\left(\frac{1-p^B}{1-p^A}\right) g(1+\omega^{lev}) E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^A) = E_0 \sum_{t=0}^{\infty} \beta^t u((1+\omega^{lev})(1+\omega^{unc}) c_t^A). \text{ QED}$$

Floden (2001) contains a more general proof of additive decomposition for preferences which also depend on leisure, and for the notion of ex-ante welfare which also includes a third component, what we called the egalitarian effect.

2 Aggregate Uncertainty in Bewley Models

So far we have assumed away aggregate fluctuations (i.e., business cycles) in our description of the incomplete-markets economies à la Bewley-Aiyagari. In previous sections of the course, you studied RBC models within a representative agent framework. In this section, the objective is to combine aggregate and idiosyncratic risk into an equilibrium model.

The good news is that we can use the recursive language to do this. The bad news is that solving exactly for the equilibrium allocations of this economy is *impossible*. The reason is that the measure of agents across states becomes an aggregate state of the economy, since households need to know it to forecast prices. But a measure is an infinitely dimensional object, i.e. a function: how do we keep track of such a large object? The answer is that we will approximate the exact equilibrium and argue that the approximation is very good for standard parameterizations.

Aggregate and Idiosyncratic Risk— We introduce aggregate fluctuations through an aggregate productivity shock z_t that shifts the production function, i.e.

$$Y_t = z_t F(K_t, H_t),$$

and assume that the aggregate shock can take only two values, $z_t \in Z = \{z_b, z_g\}$ with $z_b < z_g$. To keep things simple, we also assume only two values for the individual productivity shock, $\varepsilon_t \in E = \{\varepsilon_b, \varepsilon_g\}$ with $\varepsilon_b < \varepsilon_g$. For example, if $\varepsilon_b = 0$, then it is as if the worker was unemployed for a period.

Let

$$\pi(z', \varepsilon' | z, \varepsilon) = \Pr(z_{t+1} = z', \varepsilon_{t+1} = \varepsilon' | z_t = z, \varepsilon_t = \varepsilon)$$

be the Markov chain that describes the joint evolution of the exogenous states. This notation allows the transition probabilities for ε to depend on z (the dependence of z on ε is also allowed in principle but does not make sense!). For example, one should expect that

$$\pi(z_b, \varepsilon_g | z_b, \varepsilon_b) < \pi(z_g, \varepsilon_g | z_b, \varepsilon_b), \text{ and } \pi(z_b, \varepsilon_g | z_g, \varepsilon_g) < \pi(z_g, \varepsilon_g | z_g, \varepsilon_g)$$

i.e. finding a job is easier if the economy is exiting from a recession and remaining employed is harder if the economy is entering a recession.

State variables— The two individual states are $(a, \varepsilon) \in S$ and the two aggregate states are $(z, \lambda) \in Z \times \Lambda$, where Λ is the space of probability measures on (S, Σ_S) and $\lambda(a, \varepsilon)$ is the measure of households across states. The individual states are directly budget relevant, whereas the aggregate states are needed to compute and forecast prices. Note that, although it seems reasonable that λ is enough to complete the description of the state (i.e. a recursive equilibrium exists), this is not obvious at all and there are counterexamples of economies for which one needs to keep track of a longer history of distributions.⁵

Household Problem— The household problem can be written in recursive form as:

$$\begin{aligned} V(a, \varepsilon; z, \lambda) &= \max_{c, a'} u(c) + \beta \sum_{\varepsilon' \in E, z' \in Z} \pi(z', \varepsilon' | z, \varepsilon) V(a', \varepsilon'; z', \lambda') & (13) \\ &s.t. \\ c + a' &= w(z, \lambda) \varepsilon + R(z, \lambda) a \\ a' &\geq 0 \\ \lambda' &= \Psi(z, \lambda, z') \end{aligned}$$

⁵If you are interested in this, there is a good, albeit fairly technical, paper by Duffie, Geanakoplos, Mas-Colell and McLennan (Econometrica, 1994), which provides sufficient conditions for the existence of a recursive equilibrium.

where $\Psi(z, \lambda, z')$ is the law of motion of the endogenous aggregate state and depends on z' . The dependence on z' is inherited from π since the fraction of agents with $\varepsilon' = \varepsilon_b$ and $\varepsilon' = \varepsilon_g$ next period, given that the current aggregate productivity level is z , depends also on z' .

The key complication is that the value function V depends on λ which is a distribution, i.e. a huge object. Why? To solve their problem, the households need to compute current prices and, most importantly, forecast future prices next period. From the firm's first order condition, prices depend on the realization of aggregate productivity z and aggregate capital K , but aggregate capital depends on how assets are distributed in the population, i.e. on λ .

Let's be more specific about why agents need to know λ . Consider the Euler equation associated to the problem above and use the fact that profit maximization implies that prices depend on the realization of aggregate productivity z and aggregate capital K :

$$u_c(R(z, K)a + w(z, K)\varepsilon - a'(a, \varepsilon; z, K)) \geq \beta E[R(z', K')u_c(R(z', K')a' - a'(a', \varepsilon'; z', K'))].$$

It is clear that to solve for a' , households need to know current prices and prices next period. To compute current prices and to forecast prices next period, the household needs to know K and K' which, in turn, depend on λ because $K = \int_{A \times E} a\lambda(da, d\varepsilon)$ and $K' = \int_{A \times E} a'\lambda(da, d\varepsilon)$. Since λ is a state variable, we need to know its equilibrium law of motion Ψ which is a very large object!!!

A *Recursive Competitive Equilibrium* for this economy is a value function $V : S \times Z \times \Lambda \rightarrow \mathbb{R}$; decision rules for the household $c, a' : S \times Z \times \Lambda \rightarrow \mathbb{R}$; choice functions for the firm $H, K : Z \times \Lambda \rightarrow \mathbb{R}$; pricing functions $r, w : Z \times \Lambda \rightarrow \mathbb{R}$; and, a law of motion $\Psi : Z \times \Lambda \times Z \rightarrow \Lambda$ such that:

1. given the pricing functions $r(z, \lambda)$ and $w(z, \lambda)$, the decision rules a' and c solve the household's problem (13) and V is the associated value function
2. given $r(z, \lambda)$ and $w(z, \lambda)$, the firm chooses optimally its capital K and its labor H , i.e.

$$\begin{aligned} r(z, \lambda) + \delta &= zF_K(K, H) \\ w(z, \lambda) &= zF_H(K, H) \end{aligned} \tag{14}$$

3. markets clear

- labor market: $H = \int_{A \times E} \varepsilon \lambda(da, d\varepsilon)$
- asset market: $K = \int_{A \times E} a \lambda(da, d\varepsilon)$
- goods market:

$$\int_{A \times E} c(a, \varepsilon; z, \lambda) \lambda(da, d\varepsilon) + \int_{A \times E} a'(a, \varepsilon; z, \lambda) \lambda(da, d\varepsilon) = zF(K, H) + (1 - \delta) K$$

4. For every pair (z, z') , the aggregate law of motion Ψ is generated by the exogenous Markov chain π and the policy function a' as follows:

$$\lambda'(\mathcal{A} \times \mathcal{E}) = \Psi_{(\mathcal{A} \times \mathcal{E})}(z, \lambda, z') = \int_{A \times E} Q_{z, z'}((a, \varepsilon), \mathcal{A} \times \mathcal{E}) \lambda(da, d\varepsilon), \quad (15)$$

where $Q_{z, z'}$ is the transition function between two periods where the aggregate shock goes from z to z' and is defined by

$$Q_{z, z'}((a, \varepsilon), \mathcal{A} \times \mathcal{E}) = I_{\{a'(a, \varepsilon; z, \lambda) \in \mathcal{A}\}} \sum_{\varepsilon' \in \mathcal{E}} \pi_{\varepsilon}(\varepsilon' | z, \varepsilon, z'), \quad (16)$$

where I is the indicator function, $a'(a, \varepsilon; z, \lambda)$ is the optimal saving policy, and $\pi_{\varepsilon}(\varepsilon' | z, \varepsilon, z')$ is the conditional transition probability for ε which can be easily derived from π .

2.1 Computation of an Approximate Equilibrium

The state space of the problem of the household is, technically, infinite-dimensional because it contains a distribution. The problem is to find an efficient way to compute the law of motion

$$\lambda' = \Psi(z, \lambda, z').$$

Krusell and Smith (1998) contains the insight that, since we cannot work with an infinitely dimensional distribution, we need to approximate the distribution with a finite-dimensional object. Any distribution can be represented by its entire (in general, infinite) set of moments. Let $\bar{m}$ be a M dimensional vector of the first M moments (mean, variance, skewness, kurtosis,...) of the *wealth distribution*, i.e. the marginal of λ with respect to a . Our new state is exactly the vector $\bar{m} = \{m_1, m_2, \dots, m_M\}$ with law of motion

$$\bar{m}' = \Psi_M(z, \bar{m}) = \begin{cases} m'_1 = \phi_1(z, \bar{m}) \\ \dots \\ m'_M = \phi_M(z, \bar{m}) \end{cases}. \quad (17)$$

Note that we lost the dependence on z' since we are only interested in the wealth distribution. This method is based on the idea that households have *partial information* about λ . They don't know every detail about that measure, but only a set of moments, e.g. its mean, its variance, the Gini coefficient, the share held by the top 5% and so on. Hence, they use these M statistics to approximate the true distribution and form forecasts.

To make this approach operational, one needs to: 1) fix M and 2) specify a functional form for Ψ_M . Krusell and Smith (and this is their main finding) show that one obtains an excellent forecasting rule by simply setting $M = 1$ and by specifying a law of motion of the form:

$$\ln K' = b_z^0 + b_z^1 \ln K,$$

where only the first moment $m^1 = K$ would matter to predict the first moment next period. In particular, the law of motion does not depend on z' since capital is pre-determined.

The new partial-information problem of the agent becomes

$$\begin{aligned} V(a, \varepsilon; z, K) &= \max_{c, a'} u(c) + \beta \sum_{\varepsilon', z'} V(a', \varepsilon'; z', K') \pi(z', \varepsilon' | z, \varepsilon) \\ &\quad s.t. \\ c + a' &= w(z, K) \varepsilon + R(z, K) a \\ a' &\geq 0 \\ \ln K' &= b_z^0 + b_z^1 \ln K. \end{aligned} \tag{18}$$

Note that this state space is definitely manageable: we collapsed an infinitely dimensional distribution into one number. Now, we also know how to solve this problem. It will be a fixed-point algorithm over the law of motion (i.e., a function) for K : recall that in equilibrium the law of motion used by the agents has to be consistent with the aggregation of the optimal individual decisions (“aggregate consistency” of rational expectation equilibrium).⁶

⁶Recall that to solve for the stationary equilibrium, since the distribution is time-invariant, we guess only one value for the capital stock (or the interest rate). To compute the equilibrium transitional dynamics, we guess a deterministic sequence of capital stocks. With aggregate uncertainty, we need to guess a law of motion for the aggregate capital stock.

2.1.1 Algorithm

The algorithm to solve this problem (and the associated equilibrium) is the following:

1. Guess the coefficients of the law of motion $\{b_z^0, b_z^1\}$
2. Solve the household problem and obtain the decision rules $a'(a, \varepsilon; z, K), c(a, \varepsilon; z, K)$.
Note that with the law of motion for K in hand, we have all we need to solve for decision rules. For example, if we iterate on the Euler equation, then we need to compute the rule $a'(a, \varepsilon; z, K)$ that solves

$$u_c(R(z, K)a + w(z, K)\varepsilon - a'(a, \varepsilon; z, K)) \geq \beta E \{R(z', K') u_c(R(z', K')a' + w(z', K')\varepsilon' - a'(a', \varepsilon'; z', K'))\}.$$

Thus we can use standard methods to obtain $a'(\cdot)$. Note that pricing functions $r(z, K)$ and $w(z, K)$ must satisfy the firm's optimality condition.

3. Simulate the economy for N individuals and T periods. For example, $N = 10,000$ and $T = 2,000$. Draw first a random sequence for the aggregate shocks. Next one for the individual productivity shocks for each $i = 1, \dots, N$, conditional on the time-path for the aggregate shocks. Use the decision rules to generate sequences of asset holdings $\{a_t^i\}_{t=1, i=1}^{T, N}$ and in each period compute the average capital stock

$$A_t = \frac{1}{N} \sum_{i=1}^N a_t^i.$$

4. Discard the first T^0 periods (e.g. $T^0 = 500$) to avoid dependence from the initial conditions. Using the remaining sequence, run the regression

$$\ln A_{t+1} = \beta_z^0 + \beta_z^1 \ln A_t \tag{19}$$

and estimate the coefficients (β_z^0, β_z^1) .

5. If $(\beta_z^0, \beta_z^1) \neq (b_z^0, b_z^1)$, then try a new guess and go back to step 1. If the two pairs are equal for each $z \in \{z_g, z_b\}$, then it means that the approximate law of motion used by the agents is consistent with the one generated in equilibrium by aggregating individual choices. Notice that, once you reached the fixed point, you are sure that

market clearing in the asset market holds: next period prices are determined by K' induced by the law of motion and this law of motion induces individual choices a' that aggregate into $A' = K'$.

6. Recall that this equilibrium computation is approximate: we still need to verify how good this approximation is to the fully rational-expectation equilibrium. For this purpose, compute a measure the fit of the regression in step 4), for example by using R^2 . Next, try augmenting the state space with another moment, for example using $m^2 = E(a_i^2)$. Repeat steps 1)-5) until convergence. If the R^2 of the new equation (19) has improved significantly, keep adding moments until R^2 is large and does not respond to addition of new explanatory moments. Otherwise, stop: it means that additional moments do not add new useful information in forecasting prices.

2.2 A Near-Aggregation Result in the Krusell-Smith Economy

Krusell and Smith's main finding is that a law of motion based only on the mean, i.e.,

$$\ln K' = \begin{cases} 0.095 + 0.962 \ln K, & \text{for } z = z_g \\ 0.085 + 0.965 \ln K, & \text{for } z = z_b \end{cases}$$

delivers an $R^2 = 0.999998$ which means that the agents with this simple forecasting rule make very small errors, for example the maximal error in forecasting the interest rate 25 years into the future is around 0.1%. This result is called *near-aggregation* in the sense that in equilibrium, the evolution of aggregate quantities and prices does not depend on the distribution but, approximately, depends only on the aggregate shock and aggregate capital. Hence, it is almost like in a complete-markets economy, where aggregation of heterogenous individuals holds perfectly.

What is the intuition for the fact that keeping track of the mean of the distribution of assets is enough? Recall that if policy functions are linear, i.e.,

$$a'(a, \varepsilon, z, \lambda) = b_z^0 + b_z^1 a + b_z^2 \varepsilon,$$

then

$$K' = \int_{A \times E} a'(a, \varepsilon, z, \lambda) d\lambda = b_z^0 + b_z^1 K + b_z^2 H_z = \tilde{b}_z^0 + b_z^1 K$$

which would explain why the mean is a sufficient statistic. But saving functions are in general, not linear with uninsurable idiosyncratic shocks. They're exactly linear only

with complete markets (recall the exact aggregation result of the Chatterjee economy with homothetic preferences?), where we showed that the distribution does not affect the dynamics of aggregate variables.

So, why do we get *near-aggregation* in practice? For three reasons. First, the saving functions $a'(a, \varepsilon, z, \lambda)$ for this class of problems usually display lots of curvature for low levels of ε and low levels of assets a , but beyond this region they're *almost linear*. Second, the agents with this high curvature are few and have low wealth, so they matter very little in determining aggregate wealth. What matters for the determination of the aggregate capital stock are the ones who hold a lot of capital, i.e., the rich, not the poor! Third, aggregate productivity shocks move the asset distribution only very slightly, and the mass of the distribution is always where the saving functions are linear.

But why do agents have nearly linear saving functions, i.e. a constant marginal propensity to save out of wealth, for a very wide range of the asset space? After all, if they save for precautionary reasons (as they do in these economies) they should do so more when they hold few assets and less when they hold large assets, so the saving function should be nonlinear. The answer is that most of the consumers in this economy can smooth consumption very effectively through self-insurance, by accumulating a relatively small amount of wealth. Thus their saving behavior is guided mostly by their intertemporal motive rather than their insurance motive, like in complete markets.

To understand why the risk-free asset is such a good vehicle of self-insurance in the Krusell-Smith model, we discuss here two theoretical results in the literature that can help explain it.

First, Yaari (1976) analyzed the optimal consumption path of a perfectly patient household ($\beta = 1$) with general concave preferences (hence with prudence) who lives for T periods, faces *iid* endowment shocks and saves and borrows at rate $r = 0$. Yaari shows that as $T \rightarrow \infty$, the optimal consumption plan converges to that of a consumer who eats a constant fraction of his wealth every period. In this sense, these households behave like certainty-equivalent consumers who are not concerned about future risk. Recall that an agent with quadratic preferences has consumption determined by

$$c_t = \frac{r}{1+r} \left[a_t + E_0 \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j y_{t+j} \right] = \frac{r}{1+r} a_t + H(\mathbf{y}, r),$$

where $\mathbf{y}$ is its whole future income sequence. His assets next period are determined by

$$\begin{aligned} a_{t+1} &= (1+r)(a_t - c_t + y_t) \\ &= (1+r) \left[a_t - \frac{r}{1+r} a_t + H(\mathbf{y}, r) \right] + (1+r) y_t \\ &= a_t + \tilde{H}(\mathbf{y}, r) \end{aligned}$$

and note that the coefficient on past wealth is exactly one, which is very close to the one computed by Krusell-Smith.

Second, Levine and Zame (2001) analyze an economy populated by infinitely-lived consumers with standard preferences satisfying $u''' > 0$ who face stationary individual endowment shocks (i.e., not random-walk) and trade a risk-free asset in zero net supply. They prove that, as $\beta \rightarrow 1$ and the individuals become perfectly patient, “market incompleteness will not matter” in the sense that the welfare of the optimal consumption plan in this economy tends to the welfare of a complete markets economy where every agent consumes her average endowment every period. In other words, a great deal of risk-sharing may take place even in absence of a complicated structure of financial markets.⁷

Finally, at this point it is not surprising that Krusell and Smith find that the cyclical properties of aggregates in their model economy (i.e. volatility of output, consumption, investment, cross-correlations, etc...) are very similar to those of the standard representative agent model.

References

- [1] Domeij, David and Jonathan Heathcote (2003), On the Distributional Effects of Reducing Capital Taxes, *International Economic Review*
- [2] Floden, Martin (2001), The Effectiveness of Government Debt and Transfers as Insurance, *Journal of Monetary Economics*
- [3] Krusell P., and A. Smith (1998);. “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, vol. 106(5), 867-896.

⁷Note however, that Levine and Zame show that their result holds in the presence of aggregate uncertainty only if markets are complete with respect to aggregate risk. In the Krusell-Smith economy there is no insurance against aggregate risk, but aggregate fluctuations are quantitatively small.

- [4] Levine, D., and W. Zame (2001); “Does Market Incompleteness Matter?,” *Econometrica*, vol. 70(5), 1085-1839.
- [5] Yaari, M. (1976); “A Law of Large Numbers in the Theory of Consumer’s Choice under Uncertainty,” *Journal of Economic Theory*, 12, 202-217.

Appendices

Results on the Monotonicity of Transitional Dynamics

Consider first an economy with heterogeneity in individual endowments and a borrowing constraint, but no idiosyncratic risk. Then, the Euler equation for every agent in the economy is

$$u'(c_t) \geq \beta R_{t+1} u'(c_{t+1}),$$

where there is no expectation on the RHS because of the lack of risk. Let $R_{t+1} = 1 + f'(K_{t+1}) - \delta \equiv F'(K_{t+1})$. Assume that the aggregate capital stock is below the steady-state level, so that

$$K_t < K^* \Leftrightarrow \beta F'(K_t) > \beta F'(K^*) \Leftrightarrow \beta R_{t+1} > \beta R^* = 1,$$

where the last step uses the fact that in this economy in steady-state the borrowing constraints do not bind and therefore $\beta R^* = 1$.⁸ Then, we have

$$u'(c_t) \geq \beta R_{t+1} u'(c_{t+1}) > u'(c_{t+1})$$

which implies that $c_{t+1} > c_t$. Consumption grows for every agent in the economy, hence aggregate consumption grows. One can also show that the capital stock is increasing monotonically towards K^* . For more details, see Hernandez (1991).

Consider now the neoclassical growth model with idiosyncratic risk (and borrowing constraints). Assume $u' > 0, u'' < 0, u''' > 0$ and the Inada condition $\lim_{c \rightarrow \infty} u'(c) = 0$. Huggett (1997) proves the following theorem.

Theorem: If $\beta F'(K_t) > 1$ and $K_t > 0$, then $K_{t+1} > K_t > K_{t-1}$.

Recall that the steady state of this model features $\beta R^* = \beta F'(K^*) < 1$, so the Theorem says that, when the capital stock is sufficiently below the steady-state, the convergence will be monotonic. The proof is by contradiction. Assume that $\{K_t\}$ is decreasing. Then, the sequence $\{\beta R_\tau\}_{\tau=t}^\infty$ will be increasing and always above 1 from date t onward. The Euler equation for the individual is

$$u'(c_t(a_t, \varepsilon_t)) \geq \beta R_{t+1} E_t[u'(c_{t+1}(g_t(a_t, \varepsilon_t), \varepsilon_{t+1}))].$$

⁸See Hernandez (1991) for the proof.

Integrate both sides of the expression wrt λ_t and obtain

$$\int_{A \times E} u'(c_t(a_t, \varepsilon_t)) \lambda_t(da_t, d\varepsilon_t) \geq \beta R_{t+1} \int_{A \times E} E_t[u'(c_{t+1}(g_t(a_t, \varepsilon_t), \varepsilon_{t+1}))] \lambda_t(da_t, d\varepsilon_t).$$

From Theorem 8.3 in SLP, we know that for a continuous function h :

$$\int (Th)(z) \lambda(dz) = \int h(z') (T^* \lambda)(dz')$$

which implies

$$\int_{A \times E} E_t[u'(c_{t+1}(g_t(a_t, \varepsilon_t), \varepsilon_{t+1}))] \lambda_t(da_t, d\varepsilon_t) = \int_{A \times E} u'(c_{t+1}(a_{t+1}, \varepsilon_{t+1})) \lambda_{t+1}(da_{t+1}, d\varepsilon_{t+1}),$$

since both integrals express the expected value of h next period: it does not matter the order in which the integration is performed. Put it differently, the LHS integrates wrt ε_{t+1} through the E operator and then integrates over all the pairs (a, ε) the function $g_t(a, \varepsilon)$ which is equivalent to integrating over a_{t+1} . So, both integrals are with respect to $(a_{t+1}, \varepsilon_{t+1})$. Therefore we obtain

$$\int_{A \times E} u'(c_t(a_t, \varepsilon_t)) \lambda_t(da_t, d\varepsilon_t) \geq \beta R_{t+1} \int_{A \times E} u'(c_{t+1}(a_{t+1}, \varepsilon_{t+1})) \lambda_{t+1}(da_{t+1}, d\varepsilon_{t+1})$$

And repeating this argument, we get

$$\int_{A \times E} u'(c_t(a_t, \varepsilon_t)) \lambda_t(da_t, d\varepsilon_t) \geq \beta^n (R_{t+1} \cdot R_{t+2} \cdot \dots \cdot R_{t+n}) \int_{A \times E} [u'(c_{t+n}(a_{t+n}, \varepsilon_{t+n}))] \lambda_{t+n}(da_{t+n}, d\varepsilon_{t+n})$$

The first term of the RHS goes to infinity because $\beta R_\tau > 1$ for every $\tau > t$. Therefore, $\int_{A \times E} [u'(c_{t+n}(a_{t+n}, \varepsilon_{t+n}))] \lambda_{t+n}(da_{t+n}, d\varepsilon_{t+n})$ must converge to zero because the LHS is finite. Given any $\varepsilon > 0$, let n such that

$$\begin{aligned} \varepsilon > \int_{A \times E} [u'(c_{t+n}(a_{t+n}, \varepsilon_{t+n}))] \lambda_{t+n}(da_{t+n}, d\varepsilon_{t+n}) &> u' \left(\int_{A \times E} c_{t+n}(a_{t+n}, \varepsilon_{t+n}) \lambda_{t+n}(da_{t+n}, d\varepsilon_{t+n}) \right) \\ &= u'(C_{t+n}) \end{aligned}$$

where the second inequality requires $u''' > 0$ and Jensen's inequality. This second inequality implies aggregate consumption C_{t+n} goes to infinity by the Inada condition, which violates the resource constraint, since the aggregate capital stock is assumed to be a decreasing sequence. This cannot be an equilibrium, and we found a contradiction.

1 Industry equilibrium

We develop a model of entry, exit and firm dynamics. The model is based on Hopenhayn (1992), and Hopenhayn and Rogerson (1993). The building blocks of the model are as follows:

- The industry is competitive and produces a homogeneous good. The industry is “small”, so it takes the wage and the interest rate as given. The equilibrium of the industry determines the price and quantity of the good and the amount of labor hired in the industry, but not the wage which is treated as a parameter in our analysis below.
- The model is not fully general equilibrium. Instead of modelling a household sector explicitly, we’ll assume that there is an exogenously given demand function for the good produced in the industry.
- Plants in the industry operate a decreasing returns to scale technology and are subject to productivity shocks. They hire labor as their only input. They pay a fixed operating cost.
- Every period incumbent firms choose whether to exit the market. There is free entry into the industry, subject to paying a fixed entry cost.

Jovanovic (1982) is another seminal paper on industry equilibrium which characterizes the evolution of an industry where costs are random and different among firms. Firms all have the same prior belief about costs when they enter the industry, and learn over time about their own true cost.

Applications of the model: One can use the model to study, for example:

- *Plant turnover* (in equilibrium some plants die, others enter) and *job turnover* from the growing firms (those with good shocks) to shrinking firms (those with bad shocks).
- The size distribution of firms: in the model, the large firms are those who had long sequences of good shocks.

- Effects of policies [e.g. a wage subsidy, a profit tax, a firing tax] on the size distribution, plant turnover, average profits, average productivity, etc...
- The diffusion of technologies across plants. If, in the model, the technology diffuses from young/small plants to old/large plants then we can replicate "S-shaped" diffusion curves.
- The role of financing constraints for firm's growth.
- Trade dynamics and export decisions of firms.

1.1 The Economy

Household sector: We represent the household sector simply through a demand function $D(p)$ where p is the price of the good produced by the plants in the industry and $D' < 0$.

Plant-level production technology: Each plant produces the homogenous good with technology

$$y = zf(n),$$

where $f' > 0$, $f'' < 0$ and $f(0) = 0$. The term $z \in Z$ denotes the plant-level idiosyncratic productivity shock which follows the continuous process $\Gamma(z'|z) = \Pr\{z_{t+1} = z' | z_t = z\}$.

Problem of an incumbent plant: Incumbent plants (i.e., those producing in the current period) incur in the per-period fixed cost ϕ in order to operate (e.g., they hire one unit of managerial time every period at cost ϕ). They hire labor at the wage w .

Let the profits of the plant be denoted by $\pi(z)$. A plant that takes $\{p, w\}$ as given, solves:

$$\pi(z) = \max_n \{pzf(n) - wn - \phi\}. \quad (1)$$

Note that the profit function is increasing in z . Note also that $w = pzf_n(n)$. Competitive labor markets means that in equilibrium all plants face the same wage, so the more productive plants hire more labor. For example, suppose $f(n) = n^\alpha$. Then

$$pz\alpha n^{\alpha-1} = w \Rightarrow n(z; p) = \left(\frac{zp\alpha}{w}\right)^{\frac{1}{1-\alpha}}.$$

Therefore, take two plants with productivity z_i and z_j . The ratio between their size will be directly proportional to the productivity ratio, i.e.

$$\frac{n(z_j; p)}{n(z_i; p)} = \left(\frac{z_j}{z_i}\right)^{\frac{1}{1-\alpha}}.$$

So, once the decreasing return parameter α is known, by observing the size distribution one can fully recover the productivity distribution.

Exit decision: At the beginning of every period, before realizing the productivity shock z , the plant decides to exit. The value of exit is normalized to zero, i.e. the plant has no “scrap value”. The value of an incumbent plant is:

$$v(z; p) = \pi(z; p) + \frac{1}{1+r} \max \left\{ \int_Z v(z'; p) \Gamma(z'|z) dz', 0 \right\} \quad (2)$$

where the first argument of the max operator is the expected continuation value of the incumbent firm. Since profits are increasing in z and Γ is monotone, the value function v is also increasing in z . Let $\chi(z; p) = \{0, 1\}$ be the exit decision. The exit decision involves a reservation rule. There exists a threshold productivity level $z^*(p)$ such that, for all $z < z^*(p)$, the firm will decide to exit. In equilibrium, incumbent firms may incur in negative profits temporarily and keep operating, if the shock is mean reverting.

Problem of an entrant plant: An entrant plant must pay the fixed cost κ for one period (e.g., hire the manager to set-up the plant) and then it draws its initial productivity z' from the distribution $G(z')$. The value of an entrant plant is, therefore,

$$v^e(p) = -\kappa + \frac{1}{1+r} \int_Z v(z'; p) G(z') dz'. \quad (3)$$

The entry decision is simple: a plant should enter as long as $v^e(p) \geq 0$. Free entry of plants will guarantee that, in equilibrium, $v^e(p) = 0$.¹ Let m denote the measure of entrant plants, i.e. those preparing to produce but not yet producing, hence not yet hiring labor.

1.2 Equilibrium

A *stationary recursive competitive equilibrium* for this industry is a list of: plants' decision rules n, χ value functions v, v^e , price p , an invariant measure of incumbent firms λ and a measure of entrant plants m , such that:

- Given p , $n(z; p)$ solves the static hiring decision (1).

¹Depending on the parameter values, there may be equilibria where $v^e(p) < 0$, so there is no entry and no exit.

- Given p , $\chi(z; p)$ solves the exit decision (2) of the incumbent firm, and $v(z, p)$ is the associated value function
- Free entry of firms implies that $v^e(p) = 0$, i.e., at the equilibrium price p

$$\kappa = \frac{1}{1+r} \int_Z v(z'; p) G(z') dz'$$

- The good market clears

$$D(p) = \int_Z z f(n(z; p)) \lambda(dz; p) \quad (4)$$

- The invariant measure of incumbent plants solves

$$\lambda(z'; p) = \int_Z \Gamma(z'|z) [1 - \chi(z; p)] \lambda(dz; p) + m G(z'). \quad (5)$$

Note that the measure λ is linearly homogenous in m . To see this, suppose that Z is a discrete set with I values, then G is a vector and λ is a vector that satisfies

$$\lambda_{(I \times 1)} = \Omega_{(I \times I)(I \times 1)} \lambda_{(I \times 1)} + m_{(I \times 1)} G$$

where Ω is the transition matrix induced by the law of motion Γ and the exit decision χ . The above linear system of equations has solution

$$\lambda = m [(I - \Omega)^{-1} G]$$

which shows clearly that λ is homogeneous in m .

1.3 Solution method

The key difference with respect to the computation of the equilibrium in Bewley models with heterogeneous households is that in models of industry equilibrium, we also need to determine the *number of plants*, which is endogenous. A fixed-point algorithm to solve for the equilibrium is as follows.

1. Guess a price p_0 and solve the static hiring problem (1) for the hiring decision $n(z; p_0)$. Let

$$\pi(z; p_0) = p_0 z f(n(z; p_0)) - w n(z; p_0) - \phi$$

2. Given the profit function $\pi(z; p_0)$, one can look for the fixed point of the Bellman equation

$$v(z; p_0) = \pi(z; p_0) + \frac{1}{1+r} \max \left\{ \int_Z v(z'; p_0) \Gamma(z'|z) dz', 0 \right\}$$

and from this step, obtain $\chi(z; p_0), v(z; p_0)$. Note that $v(z; p_0)$ is increasing in p_0 since the flow profits are increasing in p_0 .

3. Verify if the price p_0 satisfies the free-entry condition

$$\kappa = \frac{1}{1+r} \int_Z v(z'; p_0) G(z') dz'$$

If, for example, the equation above holds with the $>$ sign, then guess a new value $p_1 > p_0$. Go back to step 1) with the new guess. Continue until we find a price that satisfies the free entry condition. Now that we solved for p , all we need to determine is the measure of incumbents λ and the mass of entrants m .

4. Exploit the linear homogeneity of λ in m . Guess a value of entrant plants m_0 and from (5) compute, either by simulation or using the transition function, the invariant measure λ .
5. Compute total sales of the industry, the RHS of equation (4) and verify that, at the equilibrium price found in step 3), total sales equal the exogenously given aggregate demand. If, for example, total sales are below demand, update your guess of entrants to $m_1 > m_0$ and go back to step 4). However, now step 4) is much simpler because we don't need to recompute λ , we just rescale the invariant measure already obtained.

1.4 An industry with firing costs

Hopenhayn and Rogerson (1993) study the impact of firing restrictions on employment turnover and the average productivity of the industry. Suppose, for example, that the government imposes *firing costs* that can be summarized by the function

$$g(n, n_{-1}) = \begin{cases} \chi(n_{-1} - n) & \text{if } n < n_{-1} \\ 0 & \text{if } n \geq n_{-1} \end{cases}$$

i.e., the government imposes a severance payment of size χ to the firm for every workers who is laid off.

The key novelty, with respect to the previous model, is that the employment choice is dynamic and past employment n_{-1} is a *state variable*. Plants need to keep track of their past employment to calculate the firing cost $g(\cdot)$ associated to their employment decision.

The hiring problem becomes similar to the investment problem (with or without adjustment costs) and the value function of the incumbent is:

$$v(z, n_{-1}; p) = \max_n \{ pz f(n) - \omega n - \phi - g(n, n_{-1}) + \frac{1}{1+r} \max \left\{ \int_Z v(z', n; p) \Gamma(z'|z) dz', -g(0, n) \right\} \}$$

Results: A firing cost will reduce labor reallocation from the low-productivity firms (which should be shrinking by shedding workers) towards the high-productivity firms (which should be expanding by hiring workers). It also prevents inefficient firms from exiting because of the large exit cost associated to firing the entire workforce. Overall, it is easy to see that this policy reduces average productivity of the industry and labor turnover.

References

- [1] Hopenhayn, Hugo (1992); Entry, Exit and Firm Dynamics in Long Run Industry equilibrium, *Econometrica*
- [2] Hopenhayn, Hugo and Richard Rogerson (1993); Job Turnover and Policy Evaluation: A General Equilibrium Analysis, *Journal of Political Economy*
- [3] Jovanovic, Boyan (1982); Selection and Evolution of Industry, *Econometrica*